北京化工大学

博士学位论文

题　　目　Mathematical Modeling and Experimental Validation of Processing Parameters for Enhanced Resin Impregnation in Thermoplastic Composites

专　　业　材料科学与工程

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Mathematical Modeling and Experimental Validation of Processing Parameters for Enhanced Resin Impregnation in Thermoplastic Composites

ABSTRACT

Thermoplastic composites are increasingly used in industries such as aerospace, and automotive due to their superior mechanical properties, thermal stability and recyclability. These materials offer a light weight, environmentally friendly alternative as compared to thermosetting composites with the added advantage of being easier to recycle and reuse. However the production of high thermoplastic composites remains challenging. Achieving optimal resin impregnation and fiber-matrix bonding is critical for ensuring composite strength, durability and performance. These parameters are significantly influenced by key processing factors such as mold temperature and fiber velocity.

The impregnation process involves the infiltration of resin into the fiber bundles, which is controlled by the flow dynamics of molten resin. Parameters like temperature affect resin viscosity, while fiber velocity determines the contact time between resin and fibers. Excessive fiber velocity can lead to incomplete impregnation and void contents compromising composite quality, on the other hand, suboptimal temperature may result in high resin viscosity reducing resin flow and penetration into fiber bundles. Models such as Darcy law are limited in capturing the non-Newtonian behavior of thermoplastic melts including shear thinning and viscoelasticity. To address these limitations this research work employs a combination of advanced mathematical modeling and experimental validation, focusing on basalt fiber-polypropylene (BFPP), glass fiber-polypropylene (GFPP) and carbon fiber-polyamide 66 (CFPA66) composites.

The First study focused on basalt fiber polypropylene (BFPP) composites, integrating a mathematical model to predict resin impregnation quality and fiber fracture probability. The model integrates Darcy's law for the resin flow through fiber bundle, the Reynolds equation for individual fiber pressure distribution, the Weibull Distribution (WD) for fracture probability estimation, Cohesive Zone Model (CZM) for interfacial bonding and Griffith's criterion for crack propagation. Experimentally, basalt fiber polypropylene (BFPP) composites samples were prepared using twin-screw extruder with mold temperatures of 240 °C, 250 °C and 260 °C with varying fiber velocities of 0.5, 0.8, 1.0 and 1.5 m/min. Scanning electron microscopy (SEM) and thermogravimetric analysis (TGA) validate the findings. The results showed that the optimal impregnation and minimal fiber fracture occurred at 260 °C and 0.5 m/min. The mathematical model align with experimental results, providing a robust framework for optimizing BFPP processing conditions.

The second study develop a mathematical model to analyze the effects of temperature and pressure on resin viscosity, fiber tension and impregnation quality in glass fiber-reinforced polypropylene (GFPP) composites. The model integrated Arrhenius based viscosity equation, Reynolds equation, Darcy law and Oldroyed –B viscoelastic model to predict resin flow, pressure distribution and fiber tension dynamics during melt impregnation. Experimentally, GFPP composites were fabricated using a twin-screw melt impregnation process with mold temperature ranging from 220 °C to 250 °C and fiber velocity of 0.7 m/min. Scanning electron microscopy, thermogravimetric analysis and differential scanning calorimetry confirmed that higher temperature (250 °C) significantly reduces resin viscosity, improving resin penetration, fiber wetting and fiber-matrix bonding. Result showed that temperature-dependent viscosity reduction enhances pressure distribution and lower fiber tension optimizing impregnation quality. MATLAB simulation validated the model with an error margin of 2.5% -5.0% between theoretical and experimental results. Optimal processing temperature (250 °C) improved mechanical performance, thermal stability and crystallinity making GFPP composites suitable for automotive and aerospace applications. The third study, investigated the impact of fiber velocity (0.5, 0.8, 1, and 1.5 m/min) on pressure distribution in wedge area of melt impregnation at a constant temperature of 250 °C using of glass fiber polypropylene composite (GFPP), to validate the experimental results a nonlinear Darcy law flow mathematical model was developed, to calculate the pressure in wedge area at different fiber velocity MATLAB was used for simulation. According to MATLAB simulation and SEM analysis show lower fiber velocity results in better impregnation quality with minimum void content and lower pressure in wedge area compared to higher velocities due to low velocity allowing more time for the resin to impregnate onto the fiber. This model-based approach provides insights for optimizing melt impregnation parameters and supporting the production of high-quality thermoplastic composites.

The final study addressed the limitation of Darcy law inadequately describe the non-Newtonian behavior of thermoplastic melts including shear thinning and viscoelasticity limiting their ability to optimize impregnation processes. This study develops a comprehensive multi-physics model that integrates Darcy law with Carreau-Yasuda viscosity model to predict resin flow dynamics, pressure gradient, and impregnation efficiency. Polyamide 66 (PA66) reinforced with carbon fibers was used to validate the model experimentally under varying mold temperatures (reinforced carbon fibers was used to validate the model experimentally under varying temperatures (320 °C and 330 °C) and fiber velocities (1.5 m/min and 2 m/min). The results shows that the optimal impregnation occurs at 330 °C and 1.5 m/min, where reduced viscosity enhances resin flow, minimizes voids contents and improves carbon fiber-polyamide 66 (CFPA66) bonding. The experimental results validate with mathematical model predictions.

KEY WORDS: Thermoplastic composites; Void content; Mold temperature; Fiber velocity; Mathematical modeling.

热塑性复合材料中树脂浸渍优化的工艺参数数学建模与实验验证研究

摘要

热塑性复合材料因其优越的机械性能、热稳定性和可回收性，越来越多地应用于航空航天和汽车等行业。这些材料相比热固性复合材料提供了更轻便、环保的替代方案，且具有更易回收和再利用的优势。然而，生产高性能热塑性复合材料仍面临挑战。实现最佳的树脂浸渍和纤维-基体结合对于确保复合材料的强度、耐久性和性能至关重。这些参数受到模具温度和纤维牵引速度等关键加工因素的显著影响。浸渍过程涉及树脂渗透到纤维束中，这一过程由熔融树脂的流动过程决定。温度等参数会影响树脂的粘度，而纤维牵引速度则决定树脂与纤维之间的接触时间。过高的纤维牵引速度可能导致浸渍不完全和空隙含量增加，从而影响复合材料的质量；另一方面，温度不足可能导致树脂粘度过高，从而降低树脂的流动性和渗透性。像达西定律这样的模型在捕捉热塑性熔体的非牛顿行为（包括剪切稀化和粘弹性）方面有限。为了解决这些局限，本研究结合了先进的数学建模和实验验证，重点研究了玄武岩纤维-聚丙烯（BFPP）、玻璃纤维-聚丙烯（GFPP）和碳纤维-聚酰胺 66（CFPA66）复合材料。

第一项研究集中于玄武岩纤维聚丙烯（BFPP）复合材料，结合数学模型预测树脂浸渍质量和纤维断裂概率。该模型结合了达西定律用于计算树脂浸渍纤维束的深度，雷诺方程用于计算单根纤维的压力分布，韦布尔分布（WD）用于断裂概率估算，粘结区模型（CZM）用于界面

结合，格里菲斯准则用于裂纹扩展。实验中，使用双螺杆挤出机制备了BFPP复合材料样品，模具温度分别为240 °C、250 °C和260 °C，纤维牵引速度分别为0.5、0.8、1.0和1.5 m/mi。扫描电子显微镜（SEM）和热重分析（TGA）验证了这些结果。结果显示，最佳浸渍和最小的纤维断裂发生在260 °C和0.5 m/min时。数学模型与实验结果一致，为优化BFPP加工条件提供了坚实的框架。

第二项研究建立了一个数学模型，分析温度和压力对玻璃纤维增强聚丙烯（GFPP）复合材料中树脂粘度、纤维张力和浸渍质量的影响。该模型结合了基于阿伦尼乌斯方程的粘度公式、雷诺方程、达西定律和Oldroyd-B粘弹性模型，以预测熔融浸渍过程中树脂流动、压力分布和纤维张力的变化。在实验过程中，研究人员采用双螺杆熔融浸渍工艺制备GFPP复合材料，模具温度范围为220 °C至250 °C，纤维运行速度为0.7 m/min。扫描电子显微镜、热重分析和差示扫描量热法 分析表明，较高温度（250 °C）可显著降低树脂粘度，从而增强树脂渗透性，提高纤维润湿效果和纤维-基体结合强度。研究结果显示，温度升高会降低树脂粘度，使压力分布更均匀，同时减少纤维张力，从而优化浸渍质量。MATLAB仿真验证了该模型，理论预测值与实验数据的误差范围在2.5%–5.0%之间。最优加工温度（250 °C）可显著提升GFPP复合材料的力学性能、热稳定性和结晶度，使其更适用于汽车和航空航天领域。

第三项研究调查了速度（0.5、0.8、1和1.5 m/min）对熔体浸渍楔形区域压力分布的影响，温度保持在250 °C，使用玻璃纤维聚丙烯复合材料（GFPP）。为了验证实验结果，开发了一个非线性达西定律流动数学模型，通过该模型计算不同纤维牵引速度下楔形区域的压力，使用MATLAB进行模拟。根据MATLAB模拟和扫描电子显微镜（SEM）分析，与较低的纤维牵引速度相比，较高的速度，会使得浸渍质量更好，

空隙含量最小，并且楔形区域的压力较低，因为低牵引速度使得树脂有更多时间渗透到纤维中。这种基于模型的方法为优化熔体浸渍参数提供了参考，并支持高质量热塑性复合材料的生产。

最后一项研究解决了达西定律在描述热塑性熔体的非牛顿行为（包括剪切稀化和粘弹性）方面的局限性，这限制了其优化浸渍过程的能力。本研究开发了一个综合的多物理场模型，将达西定律与 Carreau-Yasuda 粘度模型结合，用于预测树脂流动动力学、压力梯度和浸渍效率。使用碳纤维增强聚酰胺 66（PA66）来验证该模型的实验结果，验证过程中温度变化为 320 °C 和 330 °C，纤维速度为 1.5 m/min 和 2 m/min。结果表明，最佳浸渍发生在 330 °C 和 1.5 m/min 时，降低的粘度促进了树脂流动，最小化了空隙含量，并改善了碳纤维-聚酰胺 66（CFPA66）的结合。实验结果与数学模型预测相符。

关键词：热塑性复合材料；熔体浸渍；空隙含量；树脂浸渍；模具温度；纤维速度；非牛顿行为；数学建模

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List of Abbreviations

GF	Glass fiber
CF	Carbon fiber
BF	Basalt fiber
PP	Polypropylene
PA66	Polyamide 66
SEM	Scanning electron microscopy
TGA	Thermogravimetric analysis
DSC	Differential scanning calorimetric
d_f	Fiber diameter
ε_0	Porosity
δ	Melt film thickness
k_0	Carman kozeny co efficient
°C	Centigrade
q	Flow rate
∇p	Pressure gradient
V	Velocity
k	Permeability
L	Length
A	Cross-sectional area of the fiber bundle
$\sigma_r (r)$	Radial stress
$\sigma_\theta (r)$	Hoop stress
r_i	Inner radius
r_o	Outer radius
F_z	Axial force applied along the length of the fiber
σ_{total}	Total stress

σ_n	Normal stress across the interface,
σ_{max}	Maximum stress before failure
δ_n	Normal displacement across the interface,
δ_0	Critical displacement at which interface failure begins.
τ_n	Tangential shear stress
τ_{max}	Maximum shear stress
G_c	Critical energy release rate for interface failure.
e	Young's modulus
CZM	Cohesive zone model
P_f	Fracture rate
$\eta(T)$	Resin viscosity at temperature
Ea	Activation energy
η_0	Reference viscosity
R	Universal gas constant
φ	Void content
T_d	Theoretical density of the composite
M_d	Measured density of the composite.
λ_1	Relaxation time
λ_2	Retardation time
D	Rate of deformation tensor
h	Film thickness
h_0	Initial film thickness
P	Pressure distribution
η	Viscosity of resin
ρ	Density
Cp	Specific heat capacity
k	Thermal conductivity

η_{eff}	Effective viscosity
Φ_f	Fiber volume fraction
μ_0	Zero-shear viscosity
μ_∞	Infinite-shear viscosity

Chapter 1 Introduction

1.1 Introduction

Composite materials are combinations of two or more materials with different properties to create a new material with synergetic performance [1, 2]. The term most commonly refers to fiber reinforcement, typically involving carbon or glass fibers, encased in a polymer matrix [3, 4]. Composite materials often boast higher specific strength compared to common engineering metals of similar specific stiffness. In essence, this means that a composite can achieve comparable strength to metal while significantly reducing the part's weight [5-9].

Processes that achieve high fiber alignment, high fiber content, and low void content, such as pultrusion, can result in composites with superior mechanical properties along the fiber direction [10]. However, these processes typically require long cycle times, extensive energy use, or both, to achieve the high compaction pressures necessary to efficiently consolidate the composites.

The development of composite technology utilizing fiber-reinforced polymeric materials is one of the most significant advancements in recent times. Extensive research conducted in the field of composites over the past few years has had a profound impact on meeting the demands of the structural engineering community and addressing their requirements effectively [11, 12]. The development of polymers capable of withstanding high service temperatures, along with the production of high-performance composites utilizing these polymers, addresses various challenges in the aerospace, marine, and nuclear industries. In aerospace applications, the majority of engineering composite materials consist of continuous carbon or glass fibers reinforcing a polymeric matrix [13]. In recent decades, composite materials have been widely used in various applications due to their outstanding characteristics, including higher specific stiffness, greater energy absorption, design flexibility, thermal stability, lightweight properties, higher fatigue limits, and longer service life [14].

1.2 Comparison between Thermoplastic and Thermoset Composite Material

Polymer matrices can be classified into thermoplastic and thermosetting materials. Cured thermosetting polymers do not melt when heated; however, they can undergo pyrolysis at high temperatures. For example, orthophthalic resin undergoes pyrolysis at 300 °C, resulting in 82.6% solid residues, 6.1% gas yields, and 9.7% liquid yields (mainly aromatic compounds) by weight [15]. In contrast, thermoplastic polymers melt at specific temperatures, typically between 150 °C and 250 °C above their glass transition temperature. For instance, the melting point of polypropylene ranges from 160 °C to 175 °C [16]. Thermoplastic polymers harden again upon cooling. The most widely used thermoplastics include polyethylene, polypropylene (PP), polyvinyl chloride (PVC), and polystyrene [17].

Thermoplastic matrices offer advantages in recyclability due to their ability to be remolded [18], as well as high toughness and significant resistance to micro-cracking [19]. However, the melting point of thermoplastic polymers limits their application temperature range. Additionally, the mechanical properties, such as tensile modulus and strength, of thermoplastic polymers are generally lower than those of thermosetting polymers [20]. This is because the linear molecular chain of thermoplastic polymers as shown in Figure 1-1 (a) can be easily stretched under load. In contrast, thermosetting polymers such as epoxy, polyester, and vinyl ester have a more stable molecular structure with crosslinks (as shown in Figure 1-1 (b), making them capable of withstanding higher loads and temperatures compared to thermoplastic polymers [21].

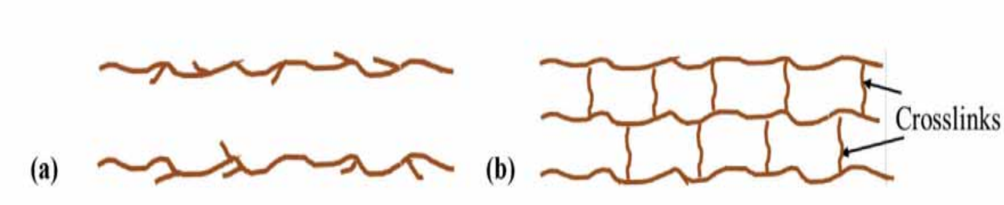

Figure 1-1 structure of (a) thermoplastic polymer and (b) thermosetting polymer

It is well known that the manufacturing process of thermoplastic composites is more energy-intensive due to the high temperatures and pressures required to melt the resin and impregnate the fibers with the matrix. Despite these efforts, the impregnation quality of thermoplastic matrices is generally inferior to that of thermosetting matrices [22].

Table 1-1 Comparison between Thermoplastic and Thermoset [26-29]

Property/Characteristic	**Thermoplastic Composites**	**Thermoset Composites**
Heat Behavior	Can be reheated, reshaped, and cooled multiple times without altering chemical structure	Undergoes irreversible chemical change when heated, cannot be remolded or reheated
Molecular Structure	Linear or branched polymers with no cross-linking	Cross-linked polymers forming a three-dimensional network
Recyclability	Recyclable and eco-friendly	Non-recyclable, cannot be remolded
Impact Resistance	High impact resistance	Generally lower impact resistance, can be brittle
Thermal Stability	Lower thermal stability, may soften at high temperatures	High thermal stability, retains strength and shape at elevated temperatures
Chemical Resistance	Good chemical resistance, but varies with type of resin	Excellent chemical resistance
Processing Time	Short processing times, suitable for high-volume production	Longer processing times due to curing process
Cost	Generally higher material costs	Lower material costs, but higher tooling costs
Applications	Aerospace interiors, automotive parts, medical devices, food processing	Electrical housings, automotive components, industrial machinery, household appliances
Examples	ABS, PVC, Polyamide Polypropylene, Polycarbonate	Epoxy, Polyester, Polyurethane, Melamine, Silicone
Advantages	Re-moldable, recyclable, good impact resistance, better aesthetic finishing	High-temperature resistance, good chemical resistance, excellent dimensional stability
Disadvantages	Can degrade under UV exposure, may soften with heat, higher cost	Brittle, non-recyclable, poor thermal conductivity

Thermoplastic composites offer several advantages, including excellent specific strength, specific modulus, high impact toughness, high production efficiency, low manufacturing cost, and environmentally friendly recycling [23]. Consequently, their market share has surpassed that of thermosetting composites, as shown in Table 1-1. Thermoplastic composites are widely used in the automotive industry, home appliances, aviation industry, and reinforced thermoplastic pipes [24].

The use of thermoplastic composites (TPC) in high-performance structural parts and assemblies has grown significantly in recent years. The global market for TPC is projected to reach $16.3 billion by 2023 [25].

1.3 Thermoplastic Composite Manufacturing Techniques

The preparation of fiber-reinforced thermoplastic resin prepregs involves the impregnation of fiber bundles with thermoplastic resins. This process is particularly challenging due to the high viscosity of thermoplastic resins, which impedes their flow between fibers, often resulting in poor impregnation quality in the final prepreg.

Currently, the main impregnation processes for preparing prepregs include the solution impregnation method [30, 31], melt impregnation method [31], powder impregnation method [32], fiber blending method [33, 34], Thin film lamination method [35, 36], and melt impregnation method [37, 38].

1.4 Mechanism of Thermoplastic Melt impregnation Machine

There are six main components of a melt impregnation machine, discussed below:

1.4.1 Impregnation Mold

The impregnation mold is crucial for ensuring that the carbon fiber bundles are fully impregnated with the molten polymer. The mold typically includes a series of parallel pins through which the fiber bundle passes, allowing the molten polymer to penetrate the fibers due to tension as the bundle is pulled through the device [39].

1.4.2 Twin-Screw Extruder

A twin-screw extruder is used to melt the polymer and feed it into the impregnation mold. The extruder must maintain a consistent melt flow and temperature to ensure uniform impregnation [39].

1.4.3 Tension and Traction Devices

Tension rollers and traction devices are essential for controlling fiber tension and line speed. These devices help maintain the necessary pressure between the fiber bundle and the pins, enhancing the degree of impregnation [39, 40].

1.4.4 Heating Elements

Heating rods or other heating elements are used to maintain the temperature of the molten polymer and the impregnation mold. This is critical for reducing the polymer's viscosity and improving impregnation efficiency.

1.4.5 Dispersion and Elimination Devices

Dispersion devices help spread the fibers evenly before they enter the impregnation mold, while electrostatic elimination devices prevent static buildup that could interfere with the impregnation process [39].

1.4.6 Impregnation Wheel

An impregnation wheel can combine the impregnation process with other manufacturing techniques like filament winding. This wheel allows for continuous impregnation and winding, making the process more efficient and reducing the need for intermediate steps [40].

1.5 Process Parameters

The melt impregnation machine operates based on four key process parameters, described below;

1.5.1 Temperature

The temperature of the polymer melt significantly affects the degree of impregnation. Higher temperatures reduce the polymer's viscosity, allowing it to penetrate the fiber bundles more effectively. For example, increasing the temperature from 180 °C to 230 °C can improve the impregnation degree by 20% [39].

1.5.2 Pressure

The pressure applied to the molten polymer is another critical factor. Higher pressure enhances the penetration of the polymer into the fiber bundles. This can be achieved by adjusting the tension on the fiber bundle and the pressure in the extruder [39, 40].

1.5.3 Line Speed

The speed at which the fiber bundle is pulled through the impregnation mold affects the duration of impregnation. A lower line speed allows more time for the polymer to penetrate the fibers, resulting in a higher degree of impregnation. However, in practical applications, a balance must be struck between line speed, production efficiency, and product performance [39].

1.5.4 Fiber Tension

Increasing the tension on the fiber bundle enhances the degree of impregnation by increasing the pressure between the fibers and the pins. This helps achieve a more uniform and thorough impregnation [39, 40].

1.6 Mathematical Modeling of Fiber-Reinforced Thermoplastic Composites Manufacturing

1.6.1 Advances in Impregnation Theory for the Melt impregnation Process

The impregnation stage is crucial in the preparation of continuous fiber-reinforced thermoplastic composites. It serves as the core technology in this process. Therefore, establishing a theoretical model for this stage is essential to guide predictive experiments. Most scholars base their theoretical models on the following principles;

Darcy's law [41], lubrication approximation theory, and Reynolds equation [42, 43], wall slip wetting theory [44], capillary wetting theory [45], etc. Darcy's Law is frequently used to study the movement of fluids in porous materials, making it one of the most widely used equations in this field. It was originally developed by Darcy to investigate the relationship between flow rate, flow path length, and the cross-sectional area of water flow within a porous medium.

Numerous researchers have suggested models for infiltration based on this definition of Darcy's Law. These basic models involve applying Darcy's Law to calculate the duration of infiltration [46].

Darcy's law explains how a viscous fluid moves through a porous material. The equation includes several variables: "q", which represents the flow rate of the fluid through the fibers; "k", the permeability constant of the fibers; "μ", the fluid's viscosity; and "Δp", the pressure drop across the fibers [47, 48].

In models that are based on Darcy's Law, the relationship between the time required for infiltration and various factors such as pressure, permeability, viscosity, and infiltration length is examined. It has been observed that the infiltration time shows a linear correlation with the applied pressure, permeability, and viscosity, while it shows a nonlinear (squared) correlation with the infiltration length. Therefore, reducing the infiltration length can have a significant impact on decreasing the infiltration time. However, when dealing with highly compressible preforms, these variables are interconnected, and simply reducing the infiltration length may not always be the most effective approach to reducing the infiltration time [49, 50]. The volume fraction of the fibers determines the properties of the final products. This fraction is influenced by the impregnation of the fibers. Moreover, the strength of the material is also affected by the volume fraction and impregnation of the fibers [51]. This law helps us understand impregnation efficiency and allows us to optimize process parameters for better fiber wetting and reduced voids [52].

The Reynolds equation, commonly used in lubrication theory, has been modified to simulate resin flow in the melt impregnation process. By taking into account the pressure distribution in the resin film located between the fibers, this equation enables the accurate prediction of resin distribution and flow characteristics under different process conditions [53].

Gennaro et al. [54, 55], and Carman et al. [54, 55], uses a parameterization similar to the Kozeny-Carman equation. This provides a clearer explanation of the power-law characteristics displayed by the melt Bates et al. [56], conducted experiments in order to establish an empirical equation for impregnation pressure Sharma et al. [57], used the lattice Boltzmann method to study how the die affects melt pressure in the pultrusion process Grouve et al. [58], developed an impregnation model by combining Darcy's law and the Navier-Stokes equation to solve for melt pressure, Yuan Man [59], concluded the impact of these parameters on the average impregnation pressure and conducted a comprehensive analysis of their influence. Li et al. [60], simplified the Reynolds equation by considering the upper and lower surfaces of the fiber bundle as flat plates, developed an impregnation model with zero flow drag at the outlet, and investigated the relationship between pressure, process parameters, and mold structure.

Table 1-2 Comparison of the advantages and disadvantages of various impregnation models

Model	**Fundamental**	**Scope of application**	**Advantage**	**Disadvantages**
Darcy's Law	$\mu = \frac{PK}{n}$	Pressure gradient Micro scale Flow	Simple formula High calculation accuracy	Empirical formula fails to explain the cause of penetration of theoretical level
Reynolds equation	Reynolds transport equation	Roller system, curved channel mold	The wedge area pressure can be calculated	The solution is complicated and cannot be solved directly
Wall slip wetting theory	$\mu = \frac{\lvert S \rvert \ tan\theta K}{3ln}$	Thin film coating	Explaining microfluidic behavior	Calculation without pressure gradient
Capillary Wetting Theory	Surface Tension	Liquid Molding Resin Transfer Molding	Evaluating the effect of surface tension on impregnation	Narrow application range
Finite Difference Model	Finite element, Reynolds equation	Roller system, curved channel mold	Calculation of melt pressure distribution	Calculation complexity
Equivalent impulse Model	Reynolds equation Darcy's Law	Roller system, curved channel mold	Simple calculation	The average value is used instead of the actual pressure distribution
Zero flow drag model	Reynolds equation Darcy's Law	Roller system, curved channel mold	Simple calculation High accuracy	

Melt impregnation has become the leading technology for producing prepreg with continuous fiber-reinforced thermoplastic composites. The main reasons for this are its simple process equipment, efficient production cycle, and continuous manufacturing capability [61, 62], The process of melt impregnation, as well as the strategies to increase melt pressure and achieve effective impregnation of high viscosity thermoplastic melt into micron-sized pores of fiber bundles, is a challenging task [56], Impregnation in a composite system occurs when thermoplastic matrices and fiber bundles are combined. This process is facilitated by the high viscosity of the thermoplastic melt, which typically ranges from 500 to 5000 Pa.s [63]. However, if the impregnation quality is poor, it can negatively affect the mechanical properties of the composite [64, 65]. In the extrusion-pultrusion system, enhancing fiber wetting and ensuring sufficient impregnation can be achieved through the design of a pin-assisted die and a mechanism that allows for fiber movement in the plastic melt. The impregnation quality in a pin-assisted impregnation die depends on various parameters, including the melting temperature and pressure of the plastic, the number and dimensions of the pins, the layout of the pins, the tension of the fiber, and the speed at which the fiber is pulled [66]. During impregnation, it is crucial to laterally stretch out the fiber bundles to align each strand, avoid overlap, and achieve uniform wetting by the matrix. Mechanical fiber stretching can be accomplished by passing the fiber through spreader pins attached to the die. The misalignment of the pins causes the fiber bundle to spread widely as it passes through. The curvature radius of the pins and the height difference between them affect the stretching of the fibers [67]. Table 1-2 provides a comparison and analysis of the advantages and disadvantages of the existing impregnation theories mentioned above.

1.6.2 Theoretical and Experimental studies on Thermoplastic Melt impregnation

The wedge areas formed between fibers and pins have been recognized as critical zones for establishing the pressure needed to drive resin melt into the fiber bundle. Consequently, these wedge areas have become a focal point for research on melt impregnation in recent years. Numerous studies have aimed to calculate the pressure in these wedge areas using theoretical models and to establish impregnation degree models based on Darcy's Law to evaluate fiber bundle impregnation quality [68-70].

Traditional processes for manufacturing continuous fiber-reinforced thermoplastic composites include powder impregnation, solvent impregnation, and melt impregnation. Among these methods, melt impregnation has received considerable attention. During this process, fiber bundles are pulled through guide pins submerged in molten thermoplastic.

Due to the high viscosity of the molten thermoplastic, complete penetration of the polymer into the fiber bundle is challenging. To address this issue, many studies have focused on reducing viscosity to achieve better impregnation [71], In the preparation of thermoplastic composite materials through melt impregnation, challenges such as difficulty in impregnation and accurately describing the process arise due to the high viscosity of the thermoplastic resin [72], Pre-impregnated thermoplastic composites are usually produced by two methods. The first involves passing the fibers through a melted polymer solution, creating a pre-impregnated material with some rigidity and consistent polymer distribution if done correctly. The second method involves intimate contact between fibers and polymer in powder form, resulting in a very cost-effective manufacturing process [73], Impregnating fiber bundles with thermoplastic polymer is slow, primarily due to the inherent high viscosity of thermoplastic polymers. While heating reduces viscosity, the generally low thermal conductivity limits the heating rate. In contrast, thermoset impregnation is much easier due to significantly lower viscosity. However, thermoplastic polymers offer advantages such as high environmental resistance, high impact strength, low product cycles, high recyclability, and reform ability [74]. Common film-type prepregs are manufactured continuously using rollers or belts. Considerable research has focused on reducing manufacturing time without compromising properties to lower costs. The manufacturing process for film-type prepregs must consider melting the film without thermal degradation and impregnating the reinforcing structure. Important factors include the onset of melting, rheological behavior at melt processing temperatures, and the time required for impregnation [75].

Ren et al. [76], designed a mathematical model that combines the Reynolds equation and Darcy's law to calculate the pressure distribution in the wedge area. This model aids in predicting the effects of processing parameters on the impregnation degree of the fiber bundle. The model's effectiveness was confirmed through satisfactory results, demonstrating its capacity to predict the influence of processing parameters on impregnation. The findings indicate that increasing the processing temperature, the number and radius of pins, or decreasing the pulling speed and the center distance of pins, improves the impregnation degree of the fiber bundle. This provides a potential solution to the challenges posed by high-viscosity melts in melt impregnation and offers ways to optimize the impregnation process.

Ret et al. [77], investigated the effects of process variables such as roving pulling speed, melt temperature, and the number of pins on fiber fracture during the processing of thermoplastic composites reinforced with continuous glass fibers. They developed a mathematical model for the impregnation process to predict the fiber fracture rate, using the Weibull intensity distribution function to explain the results. Their observations revealed that fiber fracture is caused by viscous shear stress on the fiber bundle within the melt impregnation mold during pulling.

Song et al. [78], analyzed the impregnation process in preparing long glass fiber-reinforced polypropylene composites. They developed a theoretical model for the impregnation of fibers with molten thermoplastic resin, highlighting how various processing conditions, melt viscosity, and fiber structure influence the time required for complete fiber saturation.

Chen et al. [79], investigated the effects of mold structure on the properties of continuous glass fiber-reinforced polypropylene prepreg tape. Using two different self-designed melt impregnation molds, the study tested their impact on porosity, fiber fracture rate, interface morphology, fiber dispersion uniformity, and tensile strength. The research established fiber impregnation and fracture models to predict these properties and concluded that a helical mold with a multi-wedge structure is more effective in reducing porosity and improving fiber dispersion compared to a wavy mold, which shows better tensile properties due to its larger fillet radius and fewer wedge zones.

Zhang et al. [80], emphasized that the impregnating die is the key component of melt impregnating equipment and plays a decisive role in impregnation. They established an impregnation model for the slot die and studied the influences of die design parameters. Their observations indicate that impregnation can be improved by appropriately increasing the number of crests, having a longer runner characteristic length (L_0), a smaller crest rounder (R), a smaller crest angle (β), and a smaller runner gap (H_0). They also found that the tension of fiber bundles provides the major driving force for polymer melt impregnation, with higher tension leading to better impregnation.

Yifei et al. [81], investigated the effects of fiber bundle tension and impregnation pressure variation on the permeation rate in the melt impregnation process with high-viscosity resin. They observed that higher fiber bundle tension correlated with reduced permeability, while increased impregnation pressure resulted in greater permeability.

Jie et al. [82], studied fiber fracture in continuous glass fiber-reinforced polypropylene prepreg tapes during melt impregnation. They established a mathematical model of fiber fracture using the Weibull distribution function to predict the fiber fracture rate during the prepreg tape production process. The study found that fiber fracture was mainly influenced by the resin melt's effect on the fiber bundle in the mold and the friction between the fiber and the equipment. By appropriately increasing the mold temperature, reducing the fiber traction speed, and increasing the mold gap, fiber breakage was effectively reduced, and process stability was improved.

Chandler et al. [83], identified the wedge areas formed between fibers and pins as critical for establishing the pressure needed for resin melt impregnation into fiber bundles. They divided the wedge area into four regions: the wedge entrance area, the impregnation zone, the contact area, and the leaving area. Using a simplified Reynolds equation, they developed a calculation to determine the tension in the fiber bundle, concluding that the melt pressure in the wedge area is approximately equal to the tension of the fiber bundle divided by the pin radius.

1.6.2.1 Impregnation

Hampered by the high viscosity of thermoplastics, several mathematical models have been developed to determine optimal manufacturing conditions and to explore the relationship between the degree of impregnation and process parameters. Most of these models describe the nonreactive thermoplastic pultrusion with commingled yarns and are based on the following assumptions:

- Reinforcing fibers are represented by separate groups (agglomerations) within the thermoplastic melt.
- These groups have either an elliptical or circular cross-section.
- Fibers are uniformly impregnated throughout the bulk of the product.

Darcy’s law is a well-established model for calculating the resin flow through a fiber matrix in composite manufacturing [84].

Most composite infiltration models are based on Darcy's Law, which in one dimension (1D) is expressed as follows:

$$\frac{Q}{A} = -\frac{K}{\mu}\frac{\Delta P}{L} \tag{1-1}$$

Whereas “Q” is the volumetric flow rate, “A” is the cross-sectional area through which the fluid flows, “k” is the permeability, “μ” is the fluid dynamic viscosity, and “ΔP” is the pressure drop over length “L”.

Taking into account the fiber volume fraction and the local velocity of the fibers and thermoplastic melt, derived the following expression for Darcy's Law [85]:

$$\left(1-V_f\right)\left(U_l-U_s\right)=-\frac{K}{\mu}\nabla p \qquad (1-2)$$

Whereas “V_f” fiber volume fraction, “U_l” and “U_s” local speeds of thermoplastic melt and fiber, respectively.

Since permeability “K” is neither constant nor uniform in all directions, modeling the impregnation process is challenging. To calculate permeability in the direction parallel to fiber orientation, the Kozeny–Carman equation 1-3, is employed [86].

$$K\parallel = {r^2}_f \cdot \frac{\left(1-V_f\right)^3}{4\cdot K_0\cdot {V^2}_f} \qquad (1-3)$$

Whereas “r_f” fiber radius, “V_f” instantaneous fiber volume fraction depending on the pressure, and “k_0” the permeability constant.

To determine the permeability of fibers in the transverse direction, the equation 1-4, proposed by is used [87].

$$K\perp = \frac{{r^2}_f\left(\sqrt{\frac{V_a}{V_f}}-1\right)^3}{4K_0\left(\frac{V_a}{V_f}+1\right)} \qquad (1-4)$$

Whereas “V_a”maximum possible fiber volume fraction and “k_0” he permeability constant.

When modeling the impregnation process during pultrusion, it is essential to consider the movement of the matrix along the fibers.

The micro model describes the impregnation of fibers in the transverse direction, while the macro model addresses the lengthwise flow of the matrix. The macro model is based on Darcy's Law and the Kozeny–Carman equation. Subsequently, the model was enhanced with mass conservation equations in a cylindrical coordinate system, equation 1-5, for the matrix and equation 1-6, for the fiber reinforcement [88].

$$\frac{\partial}{\partial t}\left(1-V_f\right)+\frac{1}{\mathrm{r}}\frac{\partial}{\partial r}\left(\left(1-V_f\right)rU_l\right.=0 \qquad (1-5)$$

$$\frac{\partial}{\partial t}V_f+\frac{1}{\mathrm{r}}\frac{\partial}{\partial r}\left(\left(V_f r U_s\right)\right.=0 \qquad (1-6)$$

Whereas "r" matrix propagation front

Koubaa et al. [89, 90], developed a pultrusion impregnation model based on the Navier–Stokes equation, expressed as follows in a cylindrical coordinate system.

$$\frac{\partial \mathrm{p}}{\partial z}=\frac{\mu}{\mathrm{r}}\frac{\partial}{\partial r}\left(r\,\frac{\partial \mathrm{Uz}}{\partial r}\right) \qquad (1-7)$$

Whereas "z" coordinate axis coinciding with the pulling direction and "uz" fluid motion speed along the longitudinal axis.

The following boundary conditions are imposed: zero fluid motion speed outside the fibers and equality of fluid motion speed inside the fibers to the pulling speed. The model proposed by Gibson et al. [91], considers the influence of capillary force. Sala et al. [92], conducted numerical and experimental studies, proposing a model that utilizes both Newtonian and power-law relationships to predict the outcomes of the impregnation process. Haffner et al. [93], developed a mathematical model describing the microscopic flow of resin, which accounts for different fiber arrangements, reinforcement volume fraction, and impregnation time. Miller et al. [94], proposed an impregnation model for composite materials based on tow pregs. Bechtold et al. [95], proposed two different methods to model the impregnation process in thermoplastic pultrusion with braided commingled yarns, Koubaa et al. [96], studied the impregnation of a single glass-fiber bundle, proposing a model based on the Young–Laplace law that accounts for capillary force influence, Ngo et al. [97], proposed a model for thermoplastic pultrusion with carbon fiber-reinforced prepreg, which accounts for a multi scale 3D impregnation die.

1.6.2.2 Temperature Distribution

The thermal model allows for the determination of temperature distribution across the heated die and the degree of crystallinity in a thermoplastic polymer. All developed models are based on the heat transfer equation, which has been adapted for the pultrusion process.

This adaptation introduces a pulling speed term on the left side of the following equation 1-8 [98].

$$\rho\, CV \frac{\partial \mathrm{T}}{\partial x} = \frac{\partial}{\mathrm{z}}\left(K \frac{\partial \mathrm{T}}{\partial z} + Q \right) \qquad (1-8)$$

Whereas “ρ” specific density, “C” specific heat capacity, “V” pulling speed, “x” coordinate axis parallel to the die block axis, “z” coordinate axis perpendicular to the die block axis, “T” temperature, and “Q” energy released during crystallization.

Most polymers are amorphous and do not form a crystal lattice [98], which generally allows the “Q” variable to be disregarded in most cases.

Using the model proposed by Åström, et al. and Babeau, et al. [99-101], conducted experimental tests and numerical simulations, obtaining results that closely match the analytical model.

Carlsson et al. [102], proposed the following expression for the energy released during the crystallization of polypropylene.

$$\mathrm{Q} = \mathrm{m} \frac{\partial \alpha}{\partial t} \mathrm{H} \qquad (1-9)$$

Whereas “m” mass fraction of a polymer matrix, “α” degree of crystallinity, and “H” theoretical ultimate heat of crystallization at 100% crystallinity.

The crystallization process can be divided into two stages. The first stage involves the formation of primary nuclei, while the second stage involves the growth of crystals on these nuclei. Therefore, the derivative of the degree of crystallinity can be expressed as shown in equation 1-10 [102].

$$\frac{\partial \alpha}{\partial t}(\mathrm{T}, \alpha) = (f1(\mathrm{T}) + f2(\mathrm{T})\alpha)(1 - \alpha) \qquad (1-10)$$

Whereas the “f_1” function accounts for formation of primary nuclei and “f_2” accounts for further growth of the crystal.

Expressions for f_1 and f_2 were proposed by Malkin [103].

1.6.2.3 Pressure and Pulling Force

Åström [100], proposed a model that describes the distribution of pressure over the heated die and the pulling force model, utilized the integral relationship between the pulling force and the drag per unit area, as expressed in equation 1-11.

$$ftot\ (x) = (\ 1 - Q\ (x)\ f\mathrm{v}\ (\mathrm{x}) + fc(x) + \Omega(\mathrm{x}) + ff\ (x) \qquad (1-11)$$

Whereas "Ω(x)" the part of a composite subjected to pressure, "fv(x)" viscous drag for Carreau fluid, "fc(x") compaction resistance resulting from fibers compaction in the tapered portion of the heated die block, and "f f(x)" friction resistance.

The model uses the Carreau model and considers the nonlinear nature of thermoplastic melt viscosity [100];

$$\eta_a = \eta_0\ (1 + (\lambda\gamma)^2)\frac{n-1}{2} \qquad (1-\ 12)$$

Whereas "η_0" zero-shear rate viscosity, "γ" shear rate, "indicates the shear rate at which shear thinning effects become significant, and the dimensionless constant "n" describes the degree of shear thinning.

Minchenkov et al. [104, 105], Lee et al. [104, 105], proposed a numerical model to predict pressure, pulling force, temperature, and crystallinity. Simultaneously, Åström and Pipes [106, 107], developed a model to predict pulling resistance from the die, along with the temperature and pressure distribution within a composite. Parasnis et al. [108], studied the influence of viscosity and shear load on the pulling force using a finite element model and compared the calculated pulling force values with experimental data. They reported a discrepancy between experimental and calculated data that appeared after a certain pulling speed was reached. While the model predicted an exponential increase in pulling force, the experiments showed that the increase in pulling force occurred only until a certain pulling speed value was reached, after which the pulling force remained unchanged. Blaurock et al. [109], proposed a method for predicting pulling force and compared it with experimental data. Roy et al. [110], analyzed the relationship between pulling force and viscous characteristics. Bates et al. [42, 56, 111], hypothesized that the roving tension is equal to the inlet tension on the roving before it enters the process, viscous shear stresses occur between the roving and the pin surface, and solid-solid friction exists between roving monofilaments. They believed that the maximum pressure in the wedge area was equal to the roving tension divided by the fiber bundle broadening of the product and the radius of the pin. The empirical formula was given as follows equation 1-13.

$$P = f.\frac{\mathrm{T}}{WR} \tag{1-13}$$

Whereas “T’’ is the tension, “W” is the fiber bundle broadening and “R’’ is the radius of the pin

Bates et al. [111], used an empirical formula combined with Darcy’s Law to calculate the impregnation degree of fiber bundles. They found that the impregnation degree improves by increasing the radius and number of pins or by decreasing the pulling speed. Seo et al. [68], discovered that increasing the temperature and impregnation time also enhances impregnation.

Peltonen et al. [69], utilized a self-made crosshead impregnation mold to prepare two types of prepregs, granular and strip-shaped, and investigated factors affecting impregnation degree. They found that extending impregnation time, increasing polymer resin melt temperature, or selecting a low-viscosity resin effectively improves impregnation degree. Bijsterbosch et al. [112], confirmed that Darcy's law can describe the impregnation process by varying fiber bundle residence time in the mold and polyamide melt viscosity when preparing glass fiber-reinforced polyamide composites.

Gebart et al. [47], later proposed a new model to express fiber bundle permeability, effectively predicting the influence of continuous fiber bundles on impregnation , Bates et al. [56, 111], proposed an empirical expression to predict resin melt pressure in the mold, linking pressure in the impregnation mold to fiber bundle broadening, tension, impregnation roller diameter, and friction force on the fiber bundle. Yang et al. [113], employed the finite difference method to calculate melt pressure distribution in the wedge-shaped zone inside the impregnation mold based on Reynolds equation, analyzing the impact of various parameters on melt pressure in the wedge zone. Yuan et al. [114], simplified the Reynolds equation using the concept of equivalent impulse and sliding bearing theory, deriving an analytical equation to describe melt pressure in the wedge zone and analyzing process parameters' influence on average pressure.

Tipboonsri et al. [115], investigated three factors: filling ratio, molding temperature, and pulling speed, analyzing resin impregnation in LFTP using a microscope. They observed that un-impregnation decreased with increased filling ratio and molding temperature but increased with pulling speed. The void content was minimally affected by molding temperature and pulling speed. Statistical analysis identified the optimal pultrusion process conditions for LFTP: a filling ratio of 104.73%, a molding temperature of 230 °C, and a pulling speed of 10 cm/min, resulting in the lowest un-impregnation value of 5.85%. Gaymans et al. [116], found that increasing pin diameter reduced the impregnation rate, as fiber tension on the pins varied with pulling speed. Increasing fiber tension decreased permeability.

Marissen et al. [117], similarly developed an impregnation system for continuous CFR-PA66 using parallel pins. Nygard et al. [118], compared different melt impregnation methods, finding the radial slot method provided the best overall impregnation efficiency. Impregnation quality depends on contact time between the fiber bundle and the impregnation bar; higher speeds increase pulling force, potentially exceeding fiber bundle strength and reducing impregnation quality. While these methods improve impregnation, their complexity limits mass production applicability, focusing solely on impregnation without considering other quality indicators like fiber-matrix interactions.

Tang et al. [119], developed a theoretical model for impregnating continuous fiber-reinforced polypropylene composites, integrating fiber diameter, resin viscosity, impregnation pin number, pulling speed, and processing temperature. Their model effectively predicted impregnation degree, with experimental validation showing accuracy. Porosity measurements provided a straightforward method for evaluating impregnation quality.

1.7 Fiber Reinforcement

Due to their superior properties, which enable the production of composites with more predictable behavior and enhanced mechanical, thermal, and chemical strength, synthetic fibers such as glass, carbon, and aramid are predominantly used in structural applications [120].

1.7.1 Carbon Fiber

Carbon fibers possess high strength, high stiffness, low density, high chemical and thermal stability, good thermal and electrical conductivity, lightness, and low thermal expansion. These properties make them an exceptionally popular and ideal reinforcement for many structural applications [121, 122]. It is important to note that these properties are particularly superior along the fibers' direction (i.e., carbon fibers exhibit anisotropic behavior), as the carbon atoms tend to naturally align along the reinforcement axis [120].

When combined with polymer matrices, the excellent mechanical properties of carbon fibers are naturally reflected in carbon fiber-reinforced plastics (CFRPs). These composites exhibit attributes such as specific tensile strength, specific stiffness, and corrosion resistance, making them widely employed in various industries. However, the intrinsic surface characteristics of carbon fibers, such as morphological factors and reactive functional groups, make the production of CFRPs a challenging process. Although many manufacturing processes are automated, the impregnation of fibers into the matrix must be quick, uniform, and with minimal void content between the two composite phases. Meeting these requirements is essential to achieve good interfacial adhesion, which ensures high interfacial strength [123].

The use of carbon fibers as reinforcement in polyamide 66 (PA66) composites offers several advantages over glass fiber reinforcement. Carbon fibers provide higher specific strength and stiffness compared to glass fibers, resulting in better mechanical performance at lower fiber loadings. Additionally, carbon fiber-reinforced PA66 exhibits higher tensile and flexural strengths than glass fiber-reinforced grades [124, 125]. This improvement enhances heat dissipation in carbon fiber-reinforced PA66, making it suitable for applications requiring effective heat management, such as electronics housings. Additionally, carbon fiber-reinforced PA66 exhibits lower friction coefficients and wear rates compared to both unreinforced and glass fiber-reinforced PA66 under dry sliding conditions [124]. Carbon fibers provide better dimensional stability and creep resistance to PA66 composites compared to unreinforced PA66, similar to the benefits provided by glass fiber reinforcement [125], While carbon fiber-reinforced PA66 offers superior performance, the higher cost of carbon fibers compared to glass fibers is a significant consideration. This makes glass fiber reinforcement more economical for applications with less stringent requirements [125].

In terms of applications, carbon fiber-reinforced PA66 composites are preferred for uses requiring high mechanical strength, thermal resistance, and wear resistance. These include automotive components (such as housings, connectors, and electronics), as well as sporting goods and recreational equipment [126], Industrial machinery and equipment housings [127].

On the other hand, carbon fiber-reinforced polypropylene (PP) composites are more suitable for applications where weight reduction is crucial and moderate mechanical properties are acceptable. These applications include automotive interior and exterior components, as well as aerospace components [127], Ultimately, the choice between carbon fiber-reinforced PA66 and PP composites depends on the specific requirements of the application, considering factors such as mechanical performance, thermal resistance, chemical resistance, cost, and weight considerations [125-127].

1.7.2 Glass Fiber

Glass fibers are widely used in the reinforcement market because they offer a high performance-to-cost ratio, accounting for nearly 90% of usage. The production process involves melting minerals, specifically silica and oxides, at temperatures exceeding 1370 °C. The molten material is then extruded through tiny holes using gravity, resulting in the production of filaments that are pulled and wound onto bobbins [128]. Glass fibers have excellent adhesion in polymeric matrices, allowing for the production of higher quality composites. They also possess dielectric properties, making them electrically insulating. Additionally, they are cost-effective and have specific mechanical properties [120, 129]. Furthermore, by adjusting their chemical composition, different mechanical properties can be achieved, with the enhanced property identified with a letter [130].

When comparing glass fibers to carbon fibers, the latter have approximately three times higher stiffness and tensile modulus. However, carbon fibers are significantly more expensive, costing around ten to fifteen times more. This high cost explains their predominant use in industries such as aerospace, safety, renewable energy, and sports, where high structural stiffness and low weight are required for demanding applications. Aside from cost, carbon fibers also have drawbacks such as low impact strength and susceptibility to galvanic corrosion [120, 130, 131].

The production of carbon fiber reinforced polymers (CFRPs) is an expensive, time-consuming, complex, and delicate procedure. Similar to glass fibers, carbon fibers undergo a surface treatment to enhance adhesion with the matrix. This treatment involves coating the fibers with a thin polymeric layer to adjust their physical and chemical properties without causing damage. It facilitates the handling of the fibers and protects them during the manufacturing process [120, 123, 132]. Incorporating glass fibers, typically in the range of 10-30% with 30% being the most common, significantly enhances the mechanical strength, rigidity, heat resistance, creep resistance, and fatigue strength of PA66 (polyamide 66). Tensile strength, flexural strength, and elastic modulus increase as the glass fiber content rises, while elongation at break decreases. Glass fiber reinforcement also reduces moisture absorption, molding shrinkage, and dimensional instability of PA66. Glass fiber-reinforced PA66 is widely used in automotive components (housings, covers, gears), industrial machinery, and other areas requiring high mechanical performance and heat resistance [133, 134].

Incorporating glass fibers into PP (polypropylene) improves its tensile strength, tensile modulus, and yield strength. Higher fiber content (up to 20%) leads to improved mechanical properties. However, elongation at break decreases significantly with increasing glass fiber content in PP composites. Glass fiber reinforcement enhances the stiffness and dimensional stability of PP while maintaining its good chemical resistance. Glass fiber-reinforced PP composites are used in automotive interior and exterior components where moderate mechanical properties and lightweight are desired [135].

In summary, the addition of glass fiber reinforcement provides significant improvements in mechanical properties, thermal resistance, and dimensional stability for both PA66 and PP. However, PA66 composites offer superior overall mechanical performance, heat resistance, and chemical resistance compared to PP composites, making them suitable for more demanding applications [133-135].

1.7.3 Basalt Fiber

Basalt fibers have unique properties that make them ideal for various applications. These properties include excellent mechanical strength, being sourced from environmentally friendly materials, corrosion resistance, sound insulation, and cost-effectiveness [136]. Compared to GFRP, BFRP composites have stronger creep properties, with creep fracture stresses reaching 54% of their tensile strength. This makes BFRP composites more suitable for pre stressing and cable applications [137]. Basalt fiber composites reinforced with PP also demonstrate excellent resistance to a wide range of chemicals, solvents, and moisture, enhancing their durability in different environments [138]. Additionally, basalt fiber-PP composites are lightweight due to the relatively low density of PP, making them ideal for weight reduction applications like in the automotive industry [138, 139]. The manufacturing process of basalt fiber-PP composites is facilitated by the ease with which PP can be processed using conventional techniques like injection molding, extrusion, and compression molding [138-140]. Furthermore, the thermal stability of PP complements the excellent thermal resistance of basalt fibers, making the composites suitable for applications requiring heat resistance [138, 139]. Both PP and basalt fibers are relatively inexpensive compared to other polymer matrices and reinforcements like carbon fibers, making basalt fiber-PP composites a cost-effective solution [138, 139]. The mechanical properties of PP are significantly enhanced when basalt fibers are incorporated into the matrix, such as tensile strength, flexural strength, and impact resistance, compared to unreinforced PP [138-140].

Table 1-3 Fiber reinforcement properties in thermoplastic composite material

Parameters	**Carbon Fiber**	**Glass Fiber**	**Basalt Fiber**	**References**
Fiber Properties	High strength, high stiffness, low density	Lower strength and stiffness than carbon fiber	Higher strength than glass fiber, good thermal and chemical resistance	[143-145]
Impregnation Behavior	Difficult due to high viscosity of thermoplastic melts	Easier than carbon fiber due to lower viscosity requirements	Similar to glass fiber	[143, 144, 146]
Void Formation	Higher tendency for void formation due to poor impregnation	Lower tendency for void formation compared to carbon fiber	Lower tendency than carbon fiber	[143, 147]
Interfacial Adhesion	Requires surface treatment or sizing for good adhesion	Good adhesion with many thermoplastics without treatment	Requires surface treatment for good adhesion	[144, 145]
Mechanical Properties	Highest strength and stiffness	Lower than carbon fiber composites	Higher than glass fiber composites, but lower than carbon fiber	[143, 144]
Cost	Highest cost	Lowest cost	Higher cost than glass fiber, but lower than carbon fiber	[144, 145]

The use of coupling agents or surface treatments can improve the adhesion between the basalt fibers and the PP matrix, leading to efficient stress transfer and better mechanical performance [138, 140]. On the other hand, PA66 is known for its high strength, stiffness, and toughness, making it suitable for structural applications when reinforced with basalt fibers. Basalt fiber-PA66 composites exhibit improved tensile, flexural, and impact strengths compared to unreinforced PA66. PA66 also has good thermal stability and can withstand high temperatures, which complements the excellent thermal resistance of basalt fibers. As a result, basalt fiber-PA66 composites are suitable for applications requiring heat resistance. Basalt fibers offer a cost-effective alternative to carbon fibers as reinforcement, and when combined with PA66, they provide a balance of performance and cost for various applications [141]. In comparison, PP is more suitable for lightweight and cost-effective applications, while PA66 is better suited for applications requiring high mechanical strength, thermal resistance, and chemical resistance. PA66 generally exhibits better fiber-matrix adhesion and mechanical performance with basalt fibers compared to PP [140-142]. However, for applications where weight reduction is critical, such as in the automotive industry, basalt fiber-PP composites may be preferred due to their lower density. Table 1-3 provides an overview of the fiber reinforcement properties in thermoplastic composite materials.

1.8 Thermoplastic Matrices

During prepreg production, the choice of thermoplastic matrix is a crucial step. It has a significant impact on both the processing ease and the behavior of the final composite as shown in Table 1-4 [131]. From a mechanical perspective, the density, toughness, ductility, and impact behavior of the selected polymer directly affect the final composite. Especially for continuous fiber composites, high-strength polymers are commonly used to compensate for the anisotropic behavior of these composites, on the other hand, from a chemical standpoint, considerations such as thermoplastic processing behavior and the binding affinity between the fibers and the polymer are important for achieving well-consolidated composites [120]. Overall, the choice of thermoplastic is primarily influenced by its market consumption and the quality of its mechanical performance-to-cost ratio. Some notable polymers in this regard are PA, PP, PE, PET, and PC [131].

1.8.1 Polyamide

Polyamide (PA) is widely used in the automotive industry for injection-molded components, making it a profitable market. Initially, extrusion was the preferred method for producing reinforced thermoplastics. However, the industry later shifted towards short fiber reinforced thermoplastics (SFRTs) instead of long fiber reinforced thermoplastics (LFRTs) for projects. PA is a polymer that enhances the mechanical performance of SFRTs and is therefore commonly used. Additionally, PA has superior mechanical properties and better adhesion to reinforcement fibers compared to materials like PP, allowing for higher load transfer between the matrix and the reinforcement [131]. Due to its strength and durability, PA6 is used in automotive parts, industrial machinery, and consumer goods [148].

1.8.2 Polypropylene

Polypropylene (PP) is a hydrocarbon polymer consisting of 85.7% carbon and 14.3% hydrogen. It is created by polymerizing propylene and is available in different forms, such as fiber, filament, film, and molded plastic. PP is widely used in various industries due to its strength, rigidity, and crystalline structure [149, 150].

PP's versatility allows for its use in a wide range of sectors:

1. Biomedical: PP is used in medical devices, diagnostic equipment, surgical instruments, and packaging due to its biocompatibility and sterilizability.
2. Automotive: PP composites are used in various vehicle components due to their lightweight nature and high strength-to-weight ratio.
3. Aerospace: PP is used in aerospace applications due to its exceptional mechanical properties and resistance to environmental factors.
4. Packaging: PP film is extensively used in packaging, including cigarette package wrapping and snack food packaging.
5. Textiles: PP fibers and filaments are used in the backing structure of tufted carpets, automotive interior trim, and upholstery fabrics.
6. Industrial Equipment: PP is used in the manufacturing of various items, such as kitchen utensils, waste baskets, flower pots, toys, luggage, crates, and gardening equipment [149, 150].

Table 1-4 Properties of Thermoplastic resin [151]

Property	Polypropylene (PP)	Polyamide 6 (PA6)	Polyamide 66 (PA66)	Polyamide 12 (PA12)
Melting Point	160-171 °C (320-340 °F)	220 °C (428 °F)	260 °C (500 °F)	180 °C (356 °F)
Density	0.855-0.946 g/cm³	1.13 g/cm³	1.14 g/cm³	1.01 g/cm³
Tensile Strength	31-38 MPa	70-80 MPa	80-90 MPa	50 MPa
Elongation at Break	100-600%	60-280%	15-300%	>300%
Flexural Modulus	1.1-1.6 GPa	2.8 GPa	2.9 GPa	0.8 GPa
Impact Strength (Notched Izod)	2-8 kJ/m²	5-10 kJ/m²	5-10 kJ/m²	No break
Water Absorption (24 hrs)	<0.1%	1.5-2.5%	1.5-2.5%	0.5%
Thermal Conductivity	0.22 W/m·K	0.24 W/m·K	0.24 W/m·K	0.19 W/m·K

1.9 Problem Statement, Scope of Study, Research Objectives and Innovation Points

1.9.1 Problem Statement

Fiber-reinforced thermoplastic composites such as, basalt fiber-polypropylene (BFPP), glass fiber-polypropylene (GFPP) and carbon fiber-polyamide 66 (CFPA66) are widely used in aerospace, automotive and industrial applications due to their recyclability, lightweight properties and thermal stability. However, manufacturing high quality composites remains a challenge, particularly in optimizing resin impregnation during melt processing. Incomplete impregnation, void contents and weak fiber-matrix adhesion significantly degrade the mechanical and thermal performance of these materials. High mold temperatures are critical for reducing resin viscosity and improving fiber wetting, but excessive temperatures risk thermal degradation of the matrix and fibers as demonstrated by Hu et al. [152], and Alsinani et al. [153]. Conversely, high fiber velocities enhance production rates but reduce resin-fiber interaction time leading to void contents and inadequate impregnation as explained by Zhao et al. [154], and Weidmer et al. [155].

Mathematical model such as Darcy law which define resin flow through porous media, fail to account for the non-Newtonian behavior of thermoplastic melts including shear thinning and viscoelastic effects. Mukherjee et al. [156], emphasized the importance of incorporating these rheological properties into resin flow models, yet existing approaches often simplify these effects, limiting their applicability in complex melt impregnation processes. Moreover, while recent efforts by Tipboosri et al. [157], optimized the pultrusion process for fiber-reinforced composites they did not address the nonlinear viscosity behavior and pressure gradient that influence resin impregnation quality. Experimental studies including those by Steggall et al. [147], have highlighted the critical role of controlling temperature and velocity in reducing void content, yet a disconnect persists between these empirical observations and predictive models.

Despite advancement in experimental characterization techniques such as scanning electron microscopy (SEM) and thermogravimetric analysis (TGA), there remains a lack of integration between these results and theoretical models predicting resin flow and pressure dynamics. While SEM provides detailed morphological insight into void content and fiber-matrix adhesion, it lacks integration with models that predict resin flow and pressure dynamics. This disconnect limits the optimizing of impregnation parameters necessary for producing high-performance thermoplastic composites.

To bridge this gap, this research integrates advanced mathematical modeling with MATLAB-based simulation to analyze resin flow dynamics, pressure distribution and impregnation efficiency. MATLAB serves as critical tool for simulating the effects of mold temperature, fiber velocity and pressure gradient allowing for precise validation of the theoretical models against experimental results. This approach aims to develop a robust framework that accurately predicts and optimizes processing parameters, enhancing mechanical and thermal properties of thermoplastic composites.

1.9.2 Scope of Study

The aim of this study was to understanding the influence of mold temperature and fiber velocity on resin impregnation, fiber-matrix bonding and resin flow behavior in thermoplastic composites. The research combined theoretical modeling, MATLAB simulations and experimental validation to optimize composite manufacturing processes and ensure high-performances materials.

The study focused on basalt fiber-polypropylene (BFPP) composites, where an integrated mathematical model was developed to predict resin impregnation quality and fiber fracture behavior. The model incorporated Darcy law, Reynolds equation, Weibull distribution, cohesive zone model and Griffith's criterion, MATLAB simulations were performed to analyze pressure distribution and resin flow dynamics in the wedge area of melt impregnation. BFPP samples were prepared using a twin-screw extruder at controlled mold temperature and fiber velocity with a fixed polypropylene-basalt fiber weight ratio 70:30. Basalt fiber (BF813-1200) and polypropylene (PBM100-GH) were selected for their thermal stability and compatibility.

Glass fiber-polypropylene (GFPP) composites were also studied to examined the impact of mold temperature and pressure on resin viscosity, fiber tension and impregnation quality. A temperature-dependent viscosity model and a comprehensive mathematical model incorporating Reynolds equation, Darcy law and the continuity equation were developed to predict resin flow, pressure distribution and fiber tension during melt impregnation. GFPP samples were prepared using a twin-screw extruder with glass fiber (T838P5) and polypropylene stabilized with antioxidants (Irgafos 168 and 1010). Experimental validation and MATLAB simulation confirmed that increasing mold temperature reduced resin viscosity, improved impregnation quality, minimized void content and optimized fiber tension, enhancing overall composites performance.

Further, the impact of fiber velocity on pressure distribution and resin impregnation in GFPP composites was analyzed using a nonlinear Darcy law model. The model captured the interactions between velocity, resin flow and pressure in the wedge area of melt impregnation process. To validate the model, experiments were conducted at varying fiber velocities and scanning electron microscopy (SEM) was used to measure the void content and impregnation quality, Additionally, MATLAB was employed to calculate the pressure distribution in the wedge area of melt impregnation under different fiber velocity.

Furthermore, carbon fiber-polyamide 66 (CFPA66) composites were used for experiments to validate the results through a multi-physics model that integrated Darcy law and the Carreau-Yasuda viscosity model. This model addressed the non-Newtonian behavior of thermoplastic composite melts such as shear thinning and viscoelasticity to predict resin flow dynamics and impregnation efficiency.

1.9.3 Research Objectives

1. To investigate the combine effect of mold temperature and fiber velocity on resin impregnation and fiber fracture in basalt fiber-polypropylene composites using an integrated mathematical model.

This study investigates the combined effect of mold temperature and fiber velocity on resin impregnation and fiber fracture in basalt fiber-polypropylene (BFPP) composites using an integrated mathematical model. The novelty lies in the comprehensive approach, combining Darcy's law, Reynolds equation, Weibull distribution, cohesive zone model and Griffith's criterion to predict impregnation quality and fracture probability. Unlike previous studies, which focused on isolated aspects, this research integrated multiple modeling approaches to provide a holistic understanding of the melt impregnation process.

2. To investigate the effects of mold temperature on resin viscosity, pressure distribution and film thickness on fiber tension in glass fiber-polypropylene composite through mathematical modeling.

This study introduces a novel mathematical modeling approach to analyze the combine effects of mold temperature, resin viscosity, pressure and film thickness on fiber tension in glass fiber-reinforced polypropylene composites during the melt impregnation process by integrating the Arrhenius equation, Oldroyed-B model, Reynolds equation, Darcy law and heat transfer principles, it establishes a quantitative relationship between processing parameter and fiber tension.

3. To analyze the impact of fiber velocity on pressure distribution and resin impregnation in glass fiber-polypropylene composites using a nonlinear Darcy law model.

This study introduces a nonlinear Darcy law model to analyze the effect of fiber velocity on pressure distribution and resin impregnation in GFPP composites, addressing the limitation of traditional linear models. By incorporating nonlinear permeability variation, it provides a more accurate representation of melt flow behavior. The combination of MATLAB simulation and SEM analysis validates the model, offering new insights into optimizing melt impregnation parameters for improved composite quality.

4. To explore the role of mold temperature in resin flow and impregnation efficiency in carbon fiber –polyamide 66 composites using a multi-physics model integrating Darcy law and the Carreau-Yasuda viscosity model.

This study integrates Darcy law with Carreau-Yasuda viscosity model to analyze the impact of mold temperature on resin flow and impregnation efficiency in carbon fiber – polyamide 66 (CFPA66) composites overcoming the limitations of traditional Newtonian-based models. By incorporating non-Newtonian shear-thinning behavior, it provides a more accurate representation of thermoplastic melt dynamics. This combination of experimental validation and mathematical modeling ensures reliability offering a new insights into optimizing processing conditions for enhanced fiber-matrix bonding and reduced void content. This novel approach significantly improves the predictive accuracy of melt impregnation processing, advancing thermoplastic composite manufacturing.

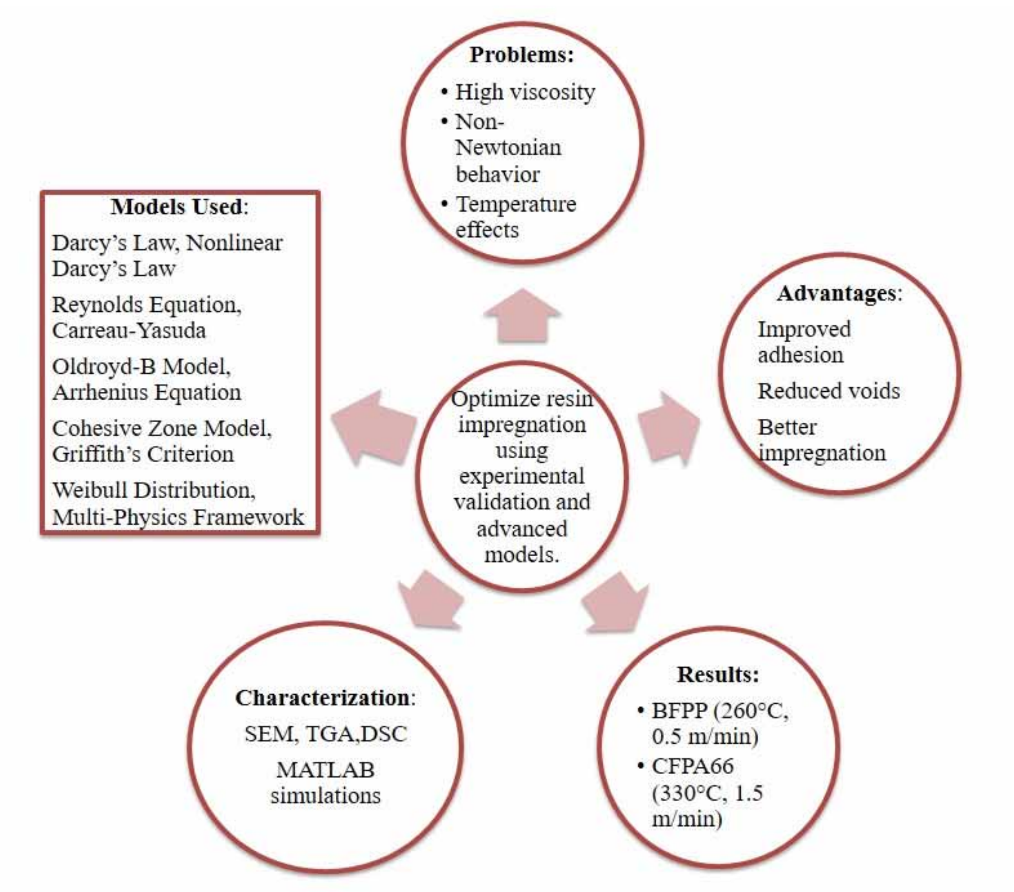

Figure 1-2 Overall structure of thesis

Chapter 2 Effect of Mold Temperature and Fiber Velocity on Resin Impregnation and Fracture Resistance in Basalt Fiber-Polypropylene composites through Integrated Mathematical Modeling

2.1 Introduction

Thermoplastic composites has gained a significant attention in various industries such as aerospace and automotive due to their superior mechanical properties, recyclability and potential for rapid manufacturing processing [23, 158-160]. These composites outperform thermoset material which are often brittle non-recyclable and susceptible to delamination [161, 162]. However, manufacturing high quality basalt fiber reinforced polypropylene (BFPP) composites through melt impregnation is challenging, the issues such as uncompleted impregnation, void formation and fiber fractures significantly degrade composite performance, essentially by weakening fiber matrix adhesion [163]. Previous research studies was focused on improving impregnation quality and interfacial bonding in fiber composites.

Deng et al. [164], investigated the poor fiber matrix compatibility in basalt fiber reinforced polypropylene and found that the compatibilizers such as maleic anhydride-grafted polypropylene (MAPP). Significantly enhance interfacial bonding transforming modes from fiber pull out to fiber fracture Prajapati et al. [165], similarly highlighted the role of compatibilizers in improving mechanical properties in basalt fiber reinforced nylon composites, showed how tailored interfacial bonding reduces voids and increase adhesion. Bashtannik et al. [166], examined the importance of processing temperature that higher extrusion temperatures improved the mechanical performances of basalt fiber reinforced polypropylene composites by enhancing bonding strength and retaining fiber length. Ren et al. [77], used the Weibull distribution to model fiber fracture probability during melt impregnation but its inability to fully account for localized stress at fiber matrix interface. Darcy law has been widely used to model resin flow in porous fiber bundle [167, 168], but if fails to incorporate dynamic stress distribution effect that influence fiber fracture. These limitations expose the need for an integrated modeling framework.

Achieving uniform resin impregnation and minimizing fiber fracture during melt impregnation remains a significant challenge for BFPP composites, parameters such as mold temperature and fiber velocity and fiber tension directly affect resin flow, interfacial bonding and void content. However, exiting models inadequately address the combined effects of these parameters limiting their practical applicability. There is need for robust integrated framework that combines resin flow dynamics, interfacial stress analysis and fractures mechanism to optimize BFPP composite manufacturing.

The aim of this study proposes a novel integrated mathematical modeling to predict resin impregnation quality and fiber fracture behavior in BFPP composites. The framework work combines Darcy law for pressure driven resin flow, Reynolds equation for thin film flow dynamics, Weibull Distribution for fracture probability analysis, cohesive zone model for interfacial stress evaluation and Griffith's criterion for fracture mechanics. However, previous study by Deng et al. [164], and Ren et al. [77], which addressed isolated aspects of composite behavior, this research study integrates multiple approaches to provide a comprehensive understanding of the manufacturing process of BFPP composites.

2.2 Materials and Methods

2.2.1 Materials

In this study, polypropylene (PP) grade (PBM100-GH), purchased from Yangzi Petrochemical Company was used. In order to prevent the oxidative degradation during processing, Irgafos 168 and Irgafos 1010 antioxidants [169-172], were added in the amount of 0.15% each of PP weight. Basalt fiber has high thermal stability and can further improve the structural integrity of the composite under stress; therefore, BF813-1200 was sourced from Sichuan Qanyi composites Co., Ltd direct roving with linear density of 1200 ± 60 tex and monofilament diameters in the range of 16-19 μm was selected.

2.2.2 Fabrication of Basalt Fiber-Polypropylene Composites

The basalt fiber-polypropylene (BFPP) composites samples were fabricated using a twin-screw extruder melt impregnation machine, as shown in Figure 2-1. The impregnation mold temperatures were set at 240 °C, 250 °C, and 260 °C, While fiber velocities were adjusted to be 0.5, 0.8, 1.0, and 1.5 m/min to investigate the effect of processing conditions on resin impregnation and fiber fracture behavior as shown in Figure 2-2 and 2-3. A constant polypropylene resin to basalt fiber weight ratio was kept at 70:30 was maintained, with fiber tension fixed at 10 N throughout experiments.

During preparation, basalt fibers were fed from a creel into a mold where molten polypropylene was uniformly distributed across the basalt fiber bundles using a twin-screw extruder. The impregnated tapes were subsequently cooled by rollers to stabilize the composite structure for further analysis [144, 173-175].

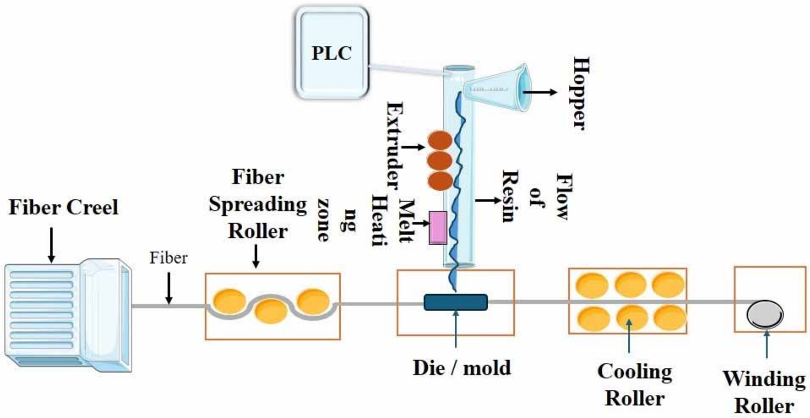

Figure 2-1 Flow diagram of melt impregnation machine

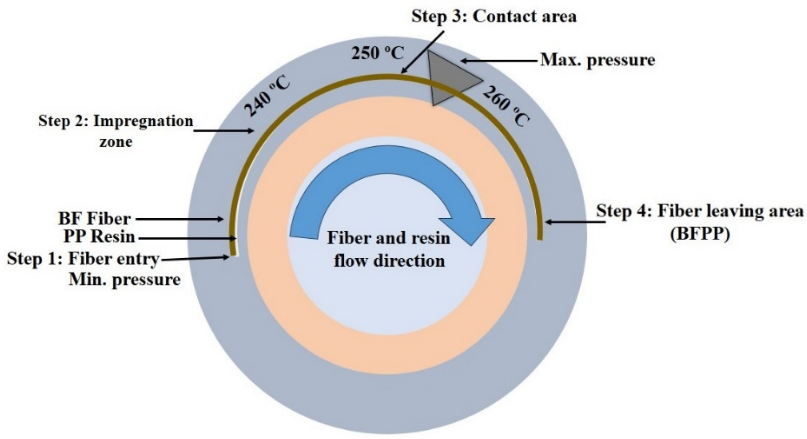

Figure 2-2 Impact of temperature on viscosity in Wedge area of melt impregnation

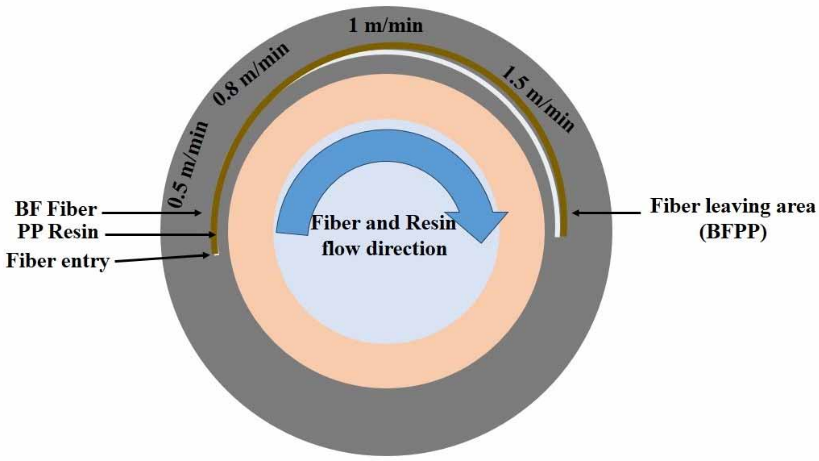

Figure 2-3 Impact of fiber velocity on impregnation in Wedge area of melt impregnation

2.2.3 Model Framework

The study combines multiple theoretical approaches from Darcy's law the Reynolds equation, Weibull distribution, the cohesive zone model (CZM) and Griffith’s criterion, in order to develop a predictive model of fiber fracture and resin impregnation quality.

Darcy's law, defines the flow of polymer melt through the porous structure of the fiber bundle, it calculates the pressure-driven flow within the composite. The Reynolds equation was used to analyze the thin film flow of the resin surrounding individual fibers, predicting the pressure distribution in this thin resin layer. This allows estimation of the contributions of the stress around fibers, which can affect the fracture risk during impregnation. The Weibull distribution describes the statistical possibility of fiber fracture under applied stress, critical in predicting the probability of failure over the composite, with the Weibull distribution, fracture probabilities can be calculated based on localized stress at selected high stress regions using Darcy's law and the Reynolds equation. The CZM considers the fiber matrix interface and, with regard to this boundary between the fibers and the polymer matrix, the distribution of stresses. This helps in establishing debonding risks with modeling of normal and tangential stresses and thus informs optimal impregnation conditions.

2.2.3.1 Model Framework, Assumptions, Boundary Conditions and Mathematical Model

This work uses various models for predicting the resin impregnation quality and the fiber fracture behavior at different processing conditions.

Model assumptions

To establish an accurate predictive model of fiber failure and resin saturation in BFPP composites, key assumptions were adopted.

1) Melt Properties of Polymer: The melt of polypropylene is assumed to be Newtonian fluid with a constant viscosity. In this way fiber bundle can be modelled without complicating it by the flow of fibers.
2) Flow Properties: Steady-state laminar flow of the polymer melt through the fiber bundle is assumed, which appropriately characterizes melt impregnation due to low Reynolds numbers involved in such processes.
3) Porous Medium Representation: The fiber bundle acts as a porous medium where resin flow can be modeled using Darcy's law

4) Pressure Gradient: There is a constant pressure gradient over the entire cross-section of fiber bundle, the flow rate and pressure drop inside resin flow domain can be calculated.
5) Negligible Inertia: The inertial forces in the melt are negligible, so the model can focus on the viscous forces, which dominate the melt flow and impregnation.

Boundary conditions

The resin flow and fiber fracture in the BFPP composite, boundary conditions for each theoretical approach were defined as follows:

1) Pressure Gradient (Darcy's Law): This is the constant pressure gradient over the fiber bundle length, which drives the flow radially inwards through the porous fiber preform.
2) No-Slip Condition: There is a no-slip boundary condition at the fiber-polymer melt interface, meaning the relative velocity of the polymer melt at the surface of the fiber is zero.
3) Radial Flow Velocity Profile (Reynolds Equation): The radial flow velocity decreases uniformly along the fiber direction, as expressed by the pressure gradient and the melt's viscosity through Darcy's Law.
4) Stress Boundary Conditions (Cohesive Zone Model): Both normal and tangential stresses at the fiber-matrix interface are modeled using the cohesive zone parameters to simulate the interfacial bonding forces. Interface failure is predicted when the stress exceeds critical traction limits.
5) Crack Propagation (Griffith's Criterion): The model uses the boundary condition that fracture occurs when the stress in any fiber exceeds the critical value for crack propagation defined by Griffith's Criterion. This boundary condition defines the level of stress necessary to cause fracture of a fiber due to tensile loading.

Mathematical model

Each model component is presented below to predict resin flow behavior and fiber fracture probability within the BFPP composite.

Darcy's law governing, fluid flow through the fiber bundle

A model for the flow of polymer melt in to fiber bundle based on Darcy's Law is presented in this section. It is employed to obtain the pressure changes in the bundle responsible for fiber stress and possible rupture.

The equations of fluid flow begin with the incompressible Navier-Stokes equation:

$$\rho\left(\frac{\partial v}{\partial t}+v.\nabla v\right)= -\nabla v+\mu\nabla^2 v \qquad (2-1)$$

Under the conditions of steady-state (no time dependence), incompressible (constant density) and low Reynolds number (laminar) flow we can further simplify this equation by neglecting the inertial terms:

$$0 = -\nabla v+\mu\nabla^2 v \qquad (2-2)$$

In a porous medium, the velocity “v” of the fluid is proportional to the pressure gradient “∇P”, with the permeability “k” and viscosity “μ” as proportionality factors:

$$\mathrm{v}=-\frac{k}{\mu}\nabla \mathrm{P} \qquad (2-3)$$

So, in order to get the volumetric flow rate “Q”, we need to integrate the velocity over the cross-section “A”:

$$Q=\int_A v.dA=-\frac{k}{\mu}\int_A \nabla P.dA \qquad (2-4)$$

Assuming that the pressure gradient ∇P across this area is uniform simplifies the calculate yield

$$Q= -\frac{k}{\mu}\nabla P.A \qquad (2-5)$$

Final Form of Darcy’s Law:

$$\mathrm{Q}=-\frac{k}{\mu}\nabla \mathrm{P} \qquad (2-6)$$

This equation (2-6) is critical and represents the process of polymer melt flowing in the fiber bundle. This leads to pressure which can create stress in the fibers that exceed the capacity of the fibers, leading to fiber breakage

Pressure drop along the fiber bundle

To calculate the pressure, drop "ΔP" across the length "L" of the fiber bundle, Using Darcy's Law, the pressure drop "ΔP" over a length "L" of the fiber bundle can be found by integrating the pressure gradient:

$$\Delta \mathrm{P} = -\int_0^L \nabla \mathrm{P}\ \mathrm{d}x \qquad (2-7)$$

Substitute Darcy's law into this equation:

$$\Delta \mathrm{P} = \frac{\mu \mathrm{QL}}{KA} \qquad (2-8)$$

Whereas; "L" is the length of the fiber bundle, "A" is the cross-sectional area of the fiber bundle.

The pressure drop gives insight into how the pressure varies along the fiber bundle. Areas with higher pressure drops are more likely to experience higher stress, which can lead to fiber fracture.

Flow around individual fibers using the Reynolds equation

The Reynolds equation to determine the pressure distribution around individual fibers within the polymer melt, which flows as a thin film. This pressure contributes to the stress experienced by the fibers.

Starting with the modified Navier-Stokes equation in the x-direction for thin film flow:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u}{\partial y^2} \qquad (2-9)$$

For thin film flow, assume that the Reynolds number is small, allowing us to ignore inertial terms:

$$0 = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u}{\partial y^2} \qquad (2-10)$$

Velocity: Integrating twice with respect to "y", obtain the velocity profile "u(y)":

$$\mathrm{u(y)} = \frac{1}{2\mu}\frac{\partial p}{\partial x}y^2 + C_1\mathrm{y} + C_2 \qquad (2-11)$$

Applying the no-slip boundary condition at y = 0, u (0) =0, Continuity of stress at y =h(x): u (h) =U (whereas "U" is the velocity of the fiber).

Solving for Constants: Using the boundary conditions, we solve for C_1 and C_2

$$u(y) = \frac{1}{2\mu}\frac{\partial p}{\partial x}(\, y^2 - h(x)^2\,) \qquad (2-12)$$

Flow Rate “Q” Per Unit Width: integrating “u(y)” over the film thickness:

$$Q = \int_0^{h(x)} u(y)dy \; = \; -\frac{h(x)^3}{12\mu}\frac{\partial p}{\partial x} \qquad (2-13)$$

Differentiating with respect to “x” yields, the Reynolds equation:

$$\frac{\partial}{\partial x}\left(h(x)^3\frac{\partial p}{\partial x}\right) = 6\mu U\;\; \frac{\partial h(x)}{\partial x} \qquad (2-14)$$

The Reynolds equation provides a model for the pressure distribution around fibers due to thin-film flow, which influences the overall stress on the fibers a crucial factor for predicting fiber fracture.

Radial and hoop stresses - Lame’s equations

The radial and hoop stresses within cylindrical fibers to understand the internal forces acting on the fibers. These stresses are fundamental in assessing the risk of crack initiation and propagation.

Assumptions: The fibers are cylindrical, the stress distribution is axisymmetric (same in all directions around the fiber’s axis) and the fibers are in equilibrium (no acceleration).

The equilibrium equation in cylindrical coordinates without body forces is:

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \qquad (2-15)$$

Using Hooke’s law: For an isotropic, linear elastic material (material properties are the same in all directions):

$$\sigma_\theta - \;\sigma_r = \frac{r}{E}\frac{d\sigma_r}{dr} \qquad (2-16)$$

Stress equations: Integrating equation (2-16) provides expressions for radial “σr (r)” and hoop “σθ (r)” stresses:

$$\sigma_r(r) = A - \frac{B}{y^2} \qquad (2-17)$$

$$\sigma_\theta(r) = A + \frac{B}{y^2} \qquad (2-18)$$

Applying Boundary Conditions:

At the inner radius “r = ri : σr (ri) = − pi.”

At the outer radius “r=ro: σr (ro) = − po”

Solving for constants “A” and “B”:

$$\sigma_r(r) \ = p_i - \frac{p_i {r_i}^2 - p_0 {r_0}^2 + \dfrac{(p_i - \ p_0){r_i}^2 {r_0}^2}{\mathrm{r}^2}}{{r_0}^2 - {r_i}^2} \qquad (2-19)$$

$$\sigma_\theta(r) \ = \frac{p_i {r_i}^2 - p_0 {r_0}^2 - \dfrac{(p_i - \ p_0){r_i}^2 {r_0}^2}{\mathrm{r}^2}}{{r_0}^2 - {r_i}^2} \qquad (2-20)$$

The equations (2-19) and (2-20) describe the radial stress “σr(r)” and hoop stress “σθ(r)” within the fiber. Both stresses are crucial for understanding the internal forces acting on the fiber. If these stresses exceed the fiber's strength, they can cause fractures, compromising the fiber's integrity.

Integration of Weibull distribution for fiber failure probability analysis

To evaluate fracture risk in high-stress areas, particularly the wedge region, the Weibull distribution is applied to model the probability of fiber fracture:

$$P_f \ = 1 - \ exp\left(-(\frac{\sigma}{\sigma_0})^m \ \right) \qquad (2-21)$$

Axial Stress along the Fiber:

Assuming the fiber is under tension, creating axial stress: The axial stress “σz” is given by;

$$\sigma_z \ = \frac{F_z}{\mathrm{A}} \qquad (2-22)$$

Whereas: “F_z” is the axial force applied along the length of the fiber. “A” is the cross-sectional area of the fiber.

Axial stress adds to the radial and hoop stresses to give the total stress within the fiber. It’s important to include axial stress in the analysis because it contributes to the overall stress that the fiber experiences, which can lead to fracture.

To calculate the total stress “σ_{total}(r)” within the fiber by combining radial, hoop, and axial stresses.

The total stress at any point inside the fiber is:

$$\sigma_{total}\,(r) = \sigma_r\,(r) + \sigma_\theta\,(r) + \sigma_z \tag{2-23}$$

The total stress is crucial for determining whether the fiber will fracture under the given conditions. By calculating the total stress, can assess the possibility of fiber failure during the melt impregnation process.

Fiber-Matrix interface analysis using the cohesive zone model (CZM)

This model assesses stress at the fiber-matrix interface to predict failure. The normal and tangential stresses are modeled as:

Assumptions: The interface between the fiber and matrix behaves according to a nonlinear traction-separation law and failure occurs when the displacement at the interface reaches a critical value.

Traction-separation law:

$$\sigma_n = \sigma_{max}\left(\frac{\delta_n}{\delta_0}\right) exp\left(1 - \frac{\delta_n}{\delta_0}\right) \tag{2-24}$$

Whereas; "σ_n" is the normal stress across the interface, "σ_{max}"is the maximum stress before failure, "δ_n" is the normal displacement across the interface, and "δ0" is the critical displacement at which interface failure begins.

The normal stress across the interface helps us understand the forces trying to separate the fiber from the matrix. If this stress exceeds the bond strength, the fiber start to delaminate, leading to potential fracture.

Tangential stress across the interface:

The tangential (shear) stress "τ_n" at the interface is similarly modeled:

$$\tau_n = \tau_{max}\frac{\delta_t}{\delta_{t0}}\; exp\left(1 - \frac{\delta_t}{\delta_{t0}}\right) \tag{2-25}$$

Whereas: "τ_n" is the shear stress across the interface, "τ_{max}" is the maximum shear stress, "δ_t" is the tangential displacement across the interface, "δ_{t0}" is the critical tangential displacement

Tangential stress contributes to the potential for debonding or delamination at the fiber-matrix interface. It's important to consider both normal and tangential stresses to fully understand the interface behavior.

Energy dissipation at the interface

To calculate the energy required to cause debonding or failure at the fiber-matrix interface.

Energy Release Rate:

$$G_c = \int_0^{\delta(x)} \sigma_n \, d\delta_n + \int_0^{\delta(x)} \tau_n \delta_t \qquad (2-26)$$

Whereas; "G_c" is the critical energy release rate for interface failure. This equation quantifies the energy dissipated during debonding.

Understanding the energy required for interface failure helps predict whether the fiber-matrix bond will hold under stress. If the energy release rate "G" exceeds the critical value "G_c", the interface is likely to fail, leading to potential fiber fracture.

Fracture mechanics using Griffith's criterion

To predict whether and where a crack in the fiber will propagate, leading to fracture.

Griffith's criterion for crack propagation

Assumptions: The fiber behaves as a brittle material and fracture occurs when the energy available for crack propagation exceeds the critical energy required.

Griffith's criterion:

$$\sigma_c = \sqrt{\frac{2E\gamma}{\pi a}} \qquad (2-27)$$

Whereas: "σ_c" is the critical stress required to propagate a crack, "E" is the Young's modulus of the fiber, "γ" is the surface energy, "a" is the length of the crack.

Griffith's criterion gives us the critical stress needed for a crack in the fiber to start growing. This is essential for predicting when and where the fiber will fracture.

Energy release rate for Crack propagation

$$G = \frac{\sigma^2 \pi a}{E} \qquad (2-28)$$

Whereas "G" is the energy release rate during crack growth. If "G" exceeds the critical energy release rate "G_c", the crack will propagate.

The energy release rate helps determine whether the crack were grow, leading to fracture.

2.2.4 Experimental Validation

The predictions of the models have been validated through experimental analysis using scanning electron microscopy (SEM) and thermogravimetric analysis (TGA). Under different parameters of velocity and temperature, fiber-matrix adhesion and thermal stability of the BFPP composites.

SEM analysis was employed to characterize the basalt fiber polypropylene matrix interface, impregnation uniformity, and void contents at various temperatures (240 °C, 250 °C and 260 °C) and fiber velocities (0.5, 0.8, 1.0 and 1.5 m/min). Incomplete impregnation areas with high void content were marked compared to model predictions as a measure of visual validation for areas more likely respond to higher stresses and ultimately fracture surface level fibers.

TGA was used to investigate the thermal stability of the polypropylene (PP) matrix and its decomposition characteristics upon basalt fiber (BF) reinforcement. BFPP composites Samples were analyzed at temperatures of 240 °C, 250 °C and 260 °C, to determine the thermal degradation range. This analysis provided insight into the resin's thermal stability, corroborating model assumptions that higher temperatures support in resin flow and improve fiber impregnation, though excessive heat can impact matrix stability.

2.3 Results and Discussion

2.3.1 Impact of Temperature and Fiber velocity on Pressure in Wedge area of Melt impregnation

MATLAB was used to analyze the effect of basalt fiber polypropylene (BFPP) composite during melt impregnation at varying temperatures (240 °C, 250 °C and 260 °C) and velocity (0.5 m/min, 0.8 m/min, 1 m/min and 1.5 m/min). The results are shown in Figures 2-2, 2-3 and 2-4 to 2-7, while the MATLAB key parameters are summarized in Table 2-1.

As shown in Figure 2-4, the Reynolds equation 2-12, was used to model thin film flow dynamics, it shows that at 0.5 m/min resin film remain stable, and ensuring uniform impregnation, in comparison at 1.5 m/min pressure unbalanced and at arises due to unstable resin flow leading to increased void content. The MATLAB results aligns with experimental results, both results confirming that the lower velocities allow polypropylene resin to flow uniformly around basalt fiber.

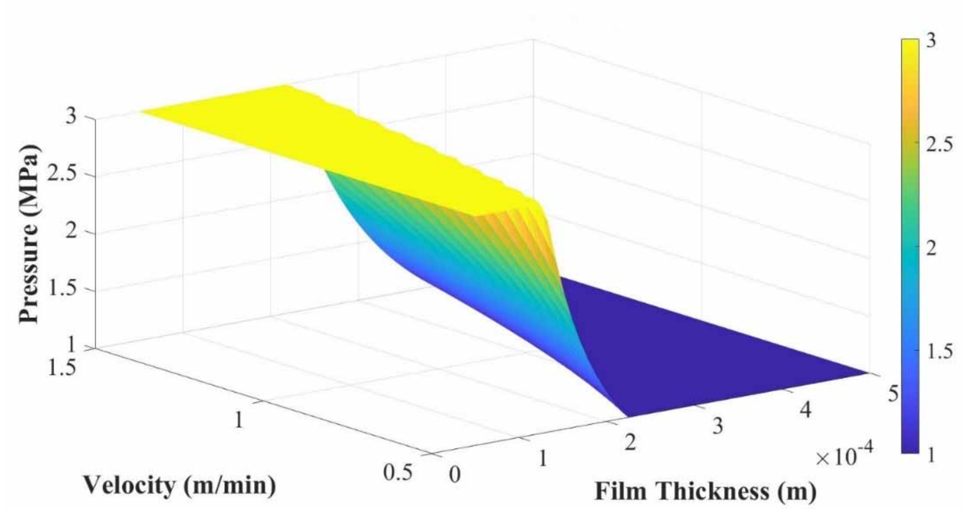

Figure 2-4 Effect of fiber velocity and film thickness on pressure distribution in the wedge area of melt impregnation

With the help of using Weibull distribution equation 2-21, to analyze fiber fractures as shown in Figure 2-5, it explained that the fracture probability is minimal at 260 °C and 0.5 m/min due to reduced stress, as compared to higher fiber velocity the fracture probability increases, significantly as stress exceeds the basalt fiber critical limits. The experimental results validate this prediction with lower velocity fracture rates observed at lower velocities, this validates the Weibull distribution ability to predict fracture behavior based on stress conditions.

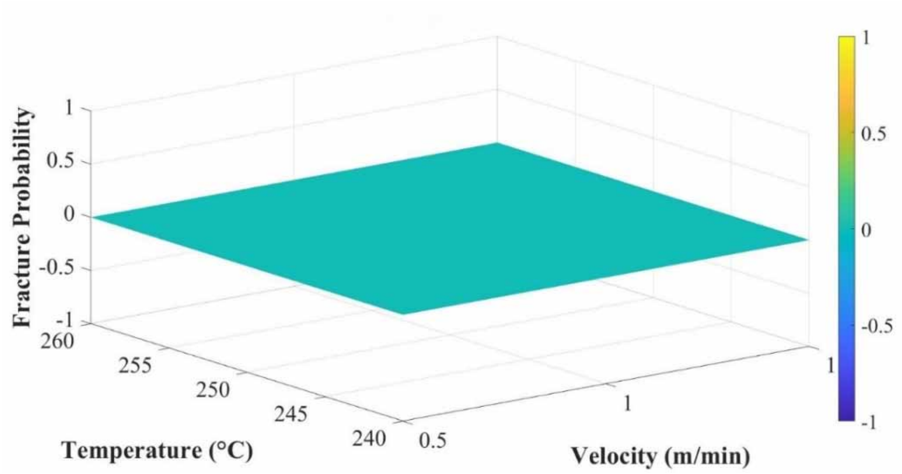

Figure 2-5 Influence of temperature and fiber velocity on fiber fracture

The cohesive zone model equation 2-26, was used to analyze the interface failure as shown in Figure 2-6, it explained that at 260 °C and 0.5 m/min, interfacial stresses remain below the critical traction limit, ensuring strong basalt fiber polypropylene matrix bonding, at higher velocities stress exceed the critical limit leading to reduced impregnation quality, These predictions were validated by observations that better impregnation occurs under low velocity and demonstrating the CZM accuracy in predicting interfacial stress behavior.

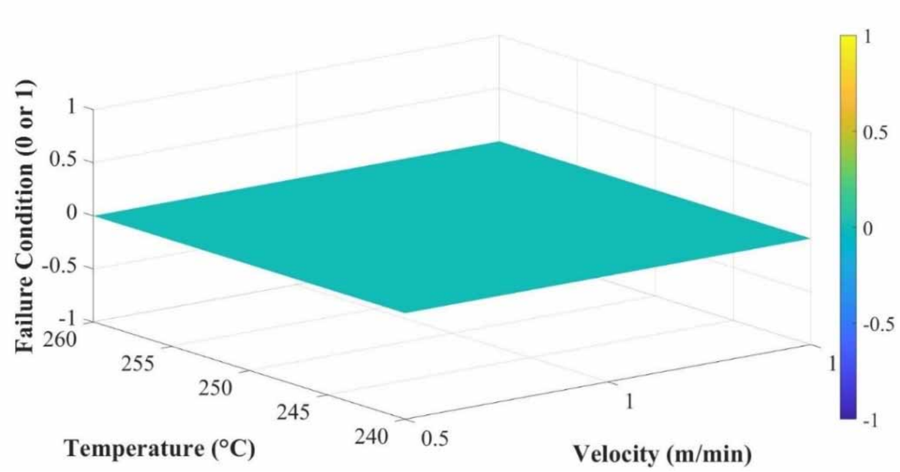

Figure 2-6 Influence of temperature and fiber velocity on interfacial failure

The Griffith criterion equation 2-27, was used to analyze the crack propagation as shown in shown in Figure 2-7, The critical stress threshold is not exceeded at 260 °C and 0.5 m/min, preventing crack initiation, at higher velocities, the stress exceeds the threshold leading to crack growth. The MATLAB predictions validate with experimental results where no cracks were observed at lower velocities and validating Griffith's criterion for crack propagation.

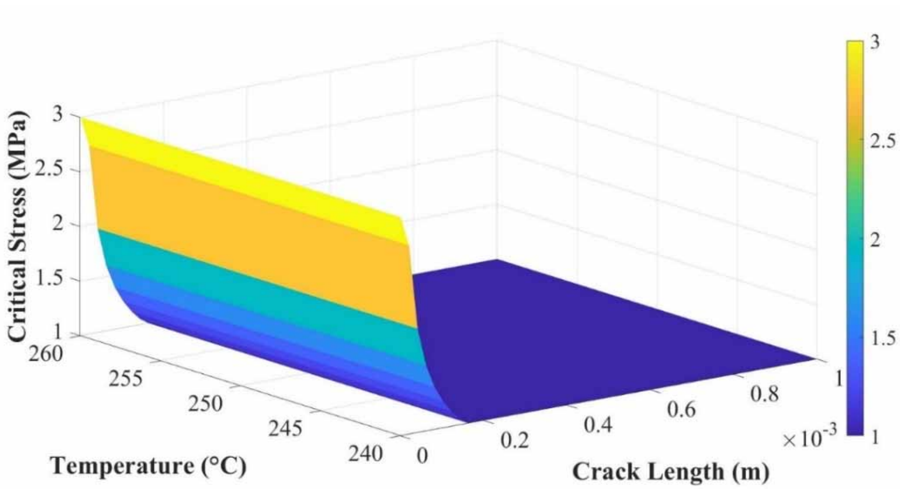

Figure 2-7 Influence of temperature and fiber on crack propagation

Table 2-1 MATLAB parameters

Parameters	**Symbol**	**Values**	**Unit**
Fiber diameter	d_f	16 ~19 μm	m
Linear density		1200	g/km
Porosity	ε_0	0.7	-
PP viscosity	η	80 ~ 174	Pa.s
Temperature	T	240, 250, 260	°C
Velocity	U	0.5, 0.8, 1, 1.5	m/min
Melt film thickness	δ	50 ~ 500 μm	m
Carman kozeny co efficient	k_0	4 ~10	
Length of wedge area	L	30 ~ 100	mm

2.3.2 Comparison of Experimental work vs. Mathematical model

Basalt fiber known for their excellent thermal resistance facilitate uniform resin flow within the BFPP composites, the Darcy law equation 2-6, predict the viscosity behavior of PP resin, it decreases with increasing temperature [176].

As shown in Figure 2-8, the error between experimental result and mathematical model around 1.67 to 2.50% confirm the model capability to define resin flow through the basalt fiber even at higher temperature whereas viscosity significantly reduces. The compatibility of PP resin with basalt fibers ensures consistent impregnation under the model conditions validating the Darcy law as reliable model equation for simulating the resin viscosity and flow in BFPP composites.

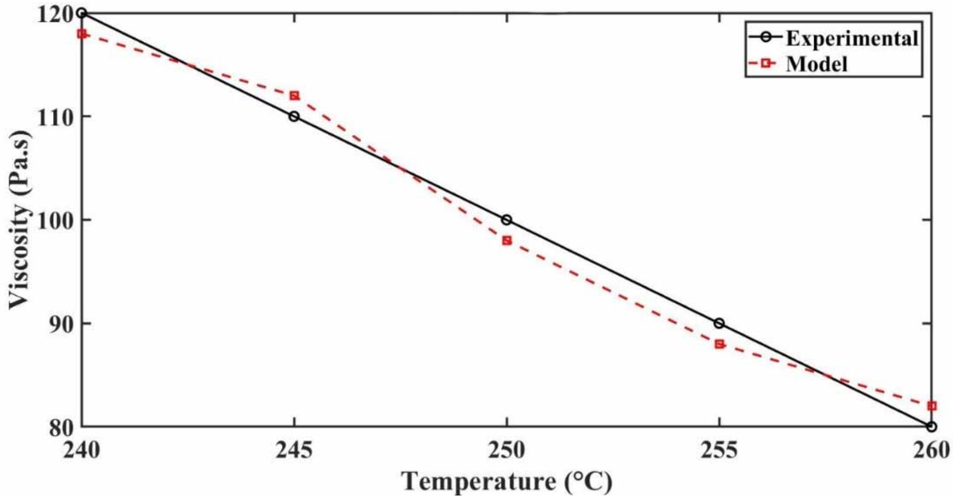

Figure 2-8 Comparison of experimental viscosity of PP and mathematical model at various temperature

Figure 2-9 shows the loss modulus the energy dissipated as heat within the polypropylene (PP) resin during mechanical deformation, for basalt fiber reinforced composites the thin film flow of polypropylene (PP) resin around the basalt fiber play an important role in energy dissipation, Reynolds equation 2-12, which models thin film flow align with experimental results with error around 0.83 to 1.82%, this validated equation for predicting energy dissipation and thin film resin behavior in BFPP composites, the high thermal stability of basalt fiber enhances the heat transfer properties of the BFPP composites ensuring uniform dissipation as modeled.

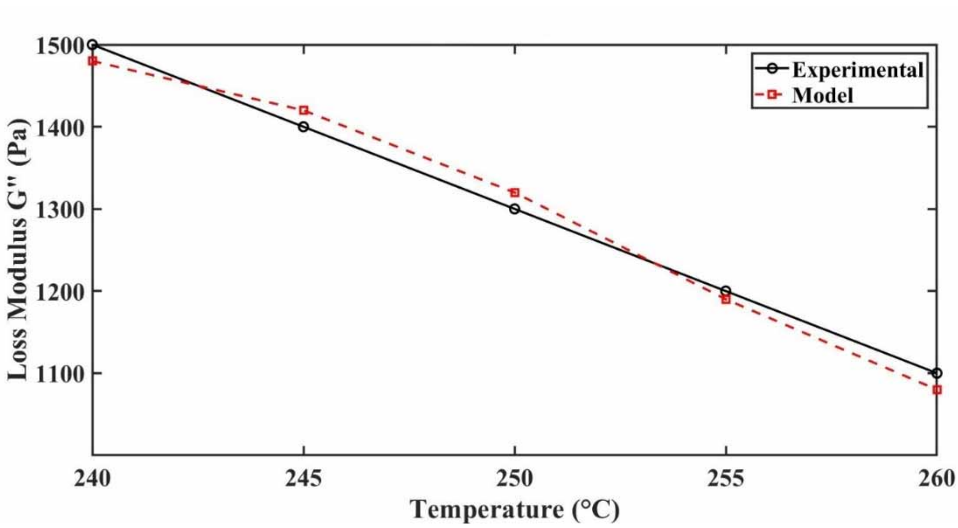

Figure 2-9 Comparison of experimental loss modulus of PP and mathematical model at various temperature

The stiffness of BFPP composites is strongly influenced by the interfacial bonding between basalt fiber and polypropylene resin, as shown in Figure 2-10, the Cohesive zone model equation 2-26, captures this relationship predicting temperature-dependent reduction in storage modulus with error round about 1.8 to 2.86%. The compatibility of basalt fiber with thermoplastic polypropylene resin, mainly at higher temperature where resin flow improves, these results validate equation 2-26, for accurately modeling stiffness and interfacial adhesion BFPP composites.

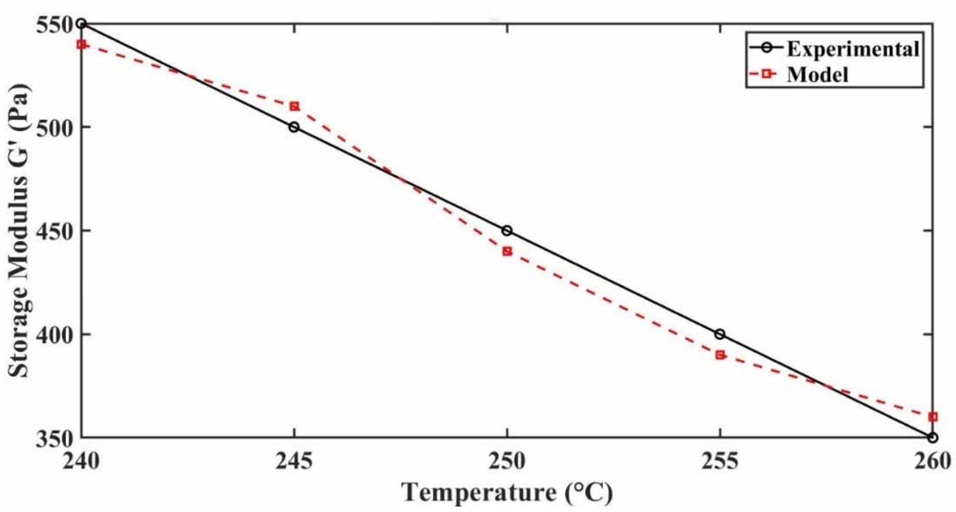

Figure 2-10 Comparison of experimental storage modulus of PP and mathematical model at various temperature

The shear stress at basalt fiber polypropylene matrix interface is critical for understanding the load transfer and fracture behavior of BFPP composites, Griffith's criterion equation 2-27, predict the critical stress for crack propagation as shown in Figure 2-11 , with error range about 0.71 to 4.55%, the higher error at 260 °C reflects the complexity of modeling dynamics resin behavior and shear thinning effects, particularly for polypropylene (PP), despite this general alignment between model and experimental results validate equation 2-27, for assessing fracture risk and stress distribution in basalt fiber composites. The high thermal resistance and strength of basalt fiber reduces the probability of crack beginning, supporting the model assumptions.

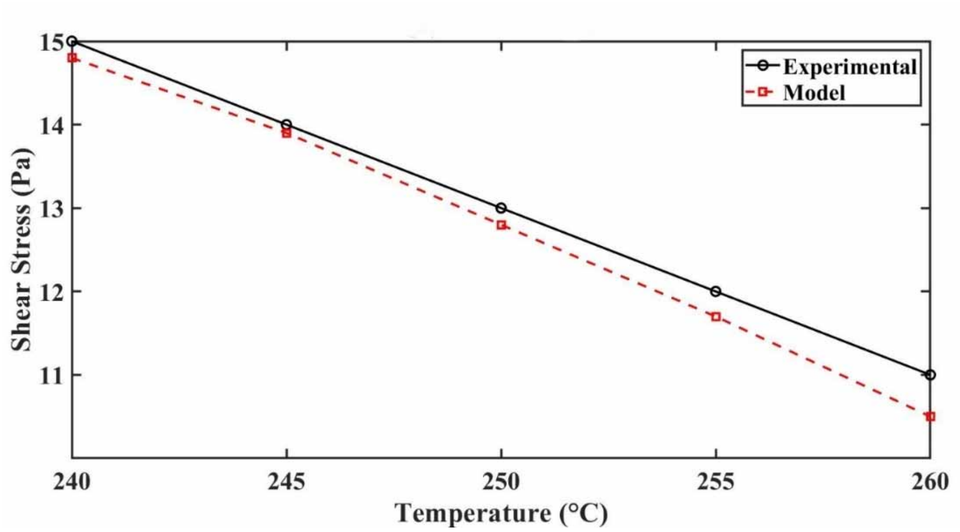

Figure 2-11 Comparison of experimental shear stress of PP and mathematical model at various temperature

2.3.3 Morphological Structure of BFPP composites at various Parameters

In this study, we examined the influence of temperature (240 °C, 250 °C and 260 °C) and fiber velocity (0.5 m/min, 0.8 m/min, 1 m/min and 1.5 m/min) on fiber fracture and melt impregnation quality in terms of void content as well as basalt fiber polypropylene matrix adhesion Figure 2-12 and 2-14 (a to d). The best impregnation quality, highest bonding between the PP and BF, and low void content were achieved under 260 °C and 0.5 m/min conditions as identified by scanning electron microscopy (SEM) analysis as shown in Figure 2-14 (a).

As shown in Figure 2-12 (a), The BFPP sample prepared at the fiber velocity of 0.5 m/min exhibited the best impregnation quality with only small amounts of voids and excellent basalt fiber polypropylene matrix bonding at 240 °C. As shown in Figures 2-12 (b-d), BFPP composites samples prepared at higher fiber velocities (0.8 to 1.5 m/min), exposed an increased void content, suggesting that the quality of impregnation decreased with increase the fiber velocity due to insufficient contact time between PP and BF.

This finding aligns with previous studies [76, 164], that concluded slower fiber velocities improve impregnation due to increased resin-fiber contact time, which is essential for strong bonding in thermoplastic composites.

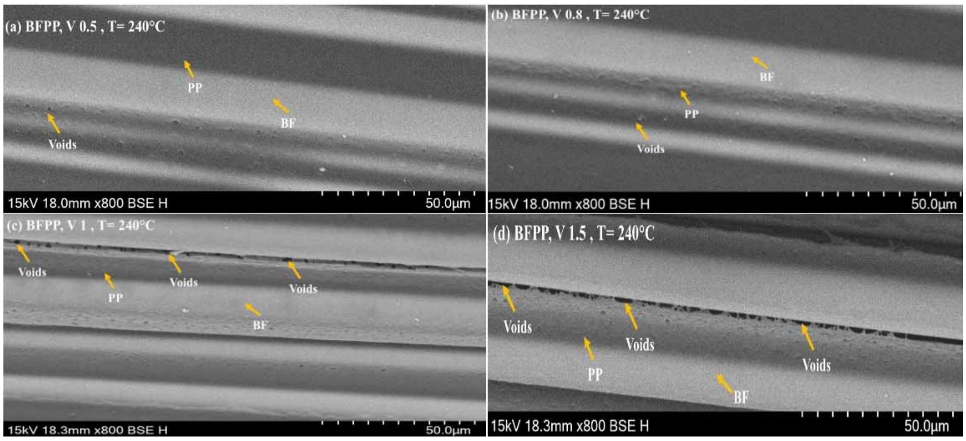

Figure 2-12 Morphological structure of BFPP at 240 °C with different velocity: (a) 0.5 m/min (b) 0.8 m/min (c) 1 m/min and (d) 1.5 m/min

Increasing the temperature to 250 °C improved the flow of the PP matrix, which enhanced basalt fiber wetting and adhesion at 0.5 m/min as shown in Figure 2-13 (a). However, higher fiber velocities again resulted in increasing in void content and less effective impregnation, reinforcing the importance of slower fiber velocity for adequate contact time, as shown in Figures 2-13 (b to c), this observation supports findings by [177, 178], who found that high temperatures improve the flow of thermoplastic resins, supporting better fiber wetting, however, increased fiber velocity limit the effective impregnation time and reduce the benefit of improved impregnation quality. Therefore, the combination of moderate temperatures and slower speeds is essential for achieving optimal composite quality.

At 260 °C, the viscosity of the PP matrix was further reduced, facilitating polypropylene resin flow and improving adhesion to basalt fibers. At 0.5 m/min, as shown in Figure 2-14 (a), indicated uniform impregnation with minimal voids and strong basalt fiber polypropylene matrix adhesion. BFPP composites samples prepared at 0.8 m/min displayed acceptable but slightly compromised impregnation, while those at 1 m/min and 1.5 m/min exhibited significant void content as shown in Figures 2-14 (b to d). These findings confirm the model's predictions regarding pressure distribution from Darcy's Law equation 2-8, and stress concentration, especially in wedge areas where uneven pressure may lead to void formation.

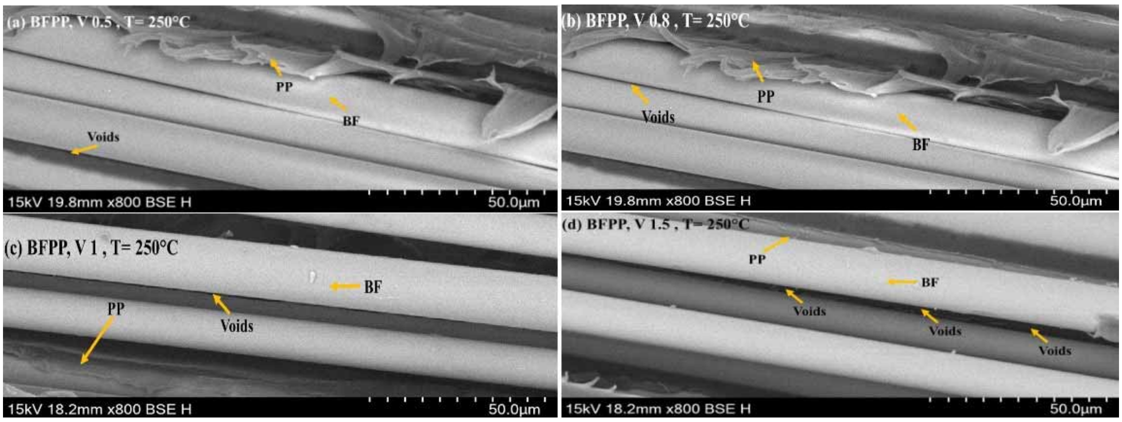

Figure 2-13 Morphological structure of BFPP at 250 °C with different velocity: (a) 0.5 m/min (b) 0.8 m/min (c) 1 m/min and (d) 1.5 m/min

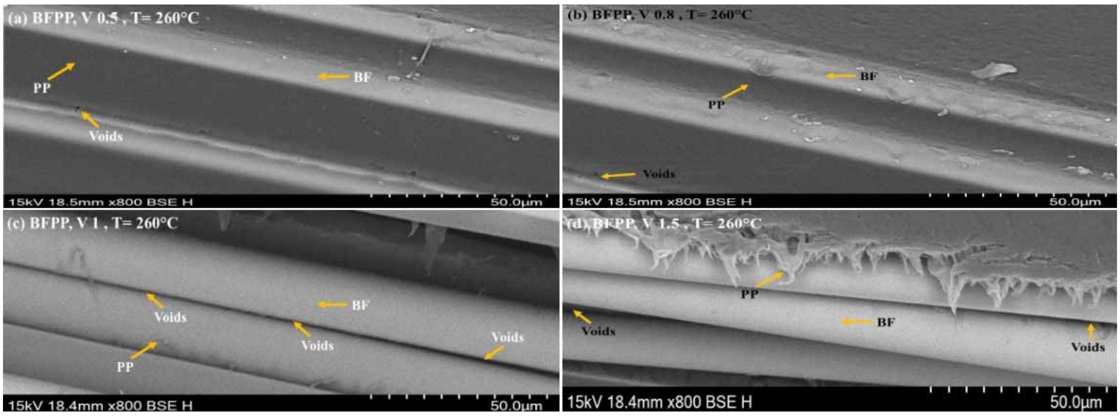

Figure 2-14 Morphological structure of BFPP at 260 °C with different velocity: (a) 0.5 m/min (b) 0.8 m/min (c) 1 m/min and (d) 1.5 m/min

The theoretical mathematical model predictions align closely with SEM Figures 2-12 to 2-14 observations, analysis at each temperature and velocity level revealed details about impregnation quality, void content, and basalt fiber polypropylene matrix adhesion. The Weibull distribution equation 2-21, used to calculate fiber fracture probability, indicated a higher possibility of fracture in high-stress zones, which, Figures 2-12 to 2-14, confirmed by identifying fracture areas. Additionally, the Cohesive Zone Model (CZM) highlighted high stresses at the fiber-matrix interface, likely causing debonding. This was observed in Figures 2-12 to 2-14, where fiber pull-out occurred in high stress areas, emphasizing the need for optimized basalt fiber polypropylene matrix bonding for composite integrity. Finally, fracture mechanics using Griffith's criterion equation 2-27, and total fracture calculations equation 2-29, provided insight into critical stress thresholds for crack propagation within fibers. SEM analysis supported these predictions, revealing fractures in high-stress zones. The optimal conditions of 260 °C and 0.5 m/min as shown in Figure 2-

14 (a), allowed for the best balance between impregnation quality and minimal fracture risk, reinforcing the model's predictions for effective composite production.

2.3.4 Impact of Temperature and Fiber velocity on Fiber Fracture rate in (BFPP) Composites

To analyze the fiber breakage ratio in basalt fiber reinforced polypropylene (BFPP) composites at various mold temperature and fiber velocity, a controlled approach was used to burn off the PP resin matrix from basalt fiber, resin in each of the samples was removed by heating to 550 °C for 4 hours in a muffle furnace, leaving only basalt fibers. After burning the PP from BF, than calculate the fiber fracture of each sample to follow the equation 2-29 [179], obtained result as shown in bellow Table 2-2.

$$P_f = \frac{T_{bef} - T_{aft}}{T_{bef}} \times 100\% \qquad (2-29)$$

Whereas "T_{bef}" represents the initial theoretical linear density, "T_{aft}" is the density measured post-burn, and "P_f" fracture rate

The experimental results show that the BFPP composite sample at temperature 260 °C and 0.5 m/min velocity has better result due to lower fiber fracture rate and porosity, these result align with mathematical model, Darcy law equation 2-6, align this result by predicting flow of PP resin through basalt fiber, where lower velocity promote even pressure distribution and reduced the stress on basalt fiber. Reynolds equation 2-12, also validate this results because the thin film flow of polypropylene resin around basalt fiber, that show that lower velocity allow for better PP resin basalt fiber interface and improved the impregnation quality. The Weibull distribution equation 2-22, further validates by providing a statistical measure fracture possibility in high stress areas, it show that the temperature 260 °C and 0.5 m/min velocity reduce intensities and minimized the fracture rate. Moreover, the cohesive zone model (CZM) equation 2-26, reports the interfacial stress at basalt fiber and PP resin boundary showed that the lower velocity 0.5 m/min, temperature 260 °C support to maintain the stress within critical limit for bonding, thus lower the risk of debonding and enhance the basalt fiber and polypropylene resin adhesion, Finally , Griffith criterion equation 2-27, shows that the critical stress level for crack initiation, confirm that at these conditions stress levels do not exceed the fracture rate, so avoiding the crack propagation in the basalt fiber. The processing temperature and velocity in melt impregnation machine 260 °C and 0.5

m/min suitable for basalt fiber and polypropylene resin better impregnation and reduced the fiber fracture rate. Validating the mathematical model ability the optimize basalt fiber and polypropylene resin composite in thermoplastic melt impregnation applications.

Table 2-2 Fiber fracture of (BFPP) at different temperature and velocity

Temperature (°C)	Velocity (m/min)	Porosity (%)	Fiber Fracture rate (%)
240	0.5	3.20	3.72
240	0.8	4.41	3.77
240	1	4.85	4.16
240	1.5	5.67	4.91
250	0.5	2.59	2.23
250	0.8	3.08	2.79
250	1	4.79	4.12
250	1.5	5.02	4.42
260	0.5	2.94	1.69
260	0.8	3.47	2.98
260	1	4.93	4.12
260	1.5	5.25	4.76

2.3.5 Thermal Stability in Basalt Fiber reinforced Polypropylene (BFPP) Composites

As shown in Figure 2-15 and Table 2-3, the TGA results shown significant thermal stability improvements for polypropylene (PP) composites reinforced with basalt fiber, compared to pure polypropylene. The thermogram for pure PP shows a rapid onset of degradation around 400 °C [180], leading to near-total weight loss by approximately 450 °C, which underscores its limited thermal resistance, the basalt fiber polypropylene composites (BFPP) processed at different temperatures (240 °C, 250 °C and 260 °C) exhibit enhanced thermal stability, with the composite processed at 260 °C showing the best performance. This BFPP composites sample exhibits minimal weight loss within the critical 400–500 °C degradation range [181], suggesting stronger basalt fiber polypropylene matrix adhesion and effective impregnation, These support the model's predictions regarding optimal processing conditions. According to Darcy’s Law equation 2-6, a uniform pressure distribution within the fiber bundle minimizes stress concentrations on individual fibers. The thermal stability observed in the TGA results for the 260 °C BFPP composites sample implies that uniform impregnation was achieved, reducing internal stress concentrations. The TGA results also align with the model’s predictions related to fracture, as defined by the Weibull Distribution for Fiber Failure Probability equation 2-21, and Griffith’s criterion for crack propagation equation 2-28. These above equations predict a higher probability of fracture in high stress areas mostly in wedge area where pressure variations concentrate stress. Though, the enhanced stability observed in the 260 °C BFPP composites sample suggests that stress concentrations were effectively moderated, thus reducing the risk of fracture. According to Griffith’s criterion equation 2-28, areas with reduced thermal degradation experience lower fracture probabilities due to enhanced fiber strength and improved fiber-matrix bonding at elevated temperatures. The TGA result thus support the model predictions by indicating that optimal temperature processing at 260 °C decreases fracture risk by minimizing stress induced damage within the BFPP composite.

The TGA BFPP results substantiate the model predictions, representative that optimized impregnation and processing temperature contribute to thermal stability and reduced fracture potential. Basalt fiber polypropylene precise temperature control at 260 °C enhance the composites durability, suggesting that this temperature is ideal for achieving both thermal stability and reduced fracture risk within BFPP composites.

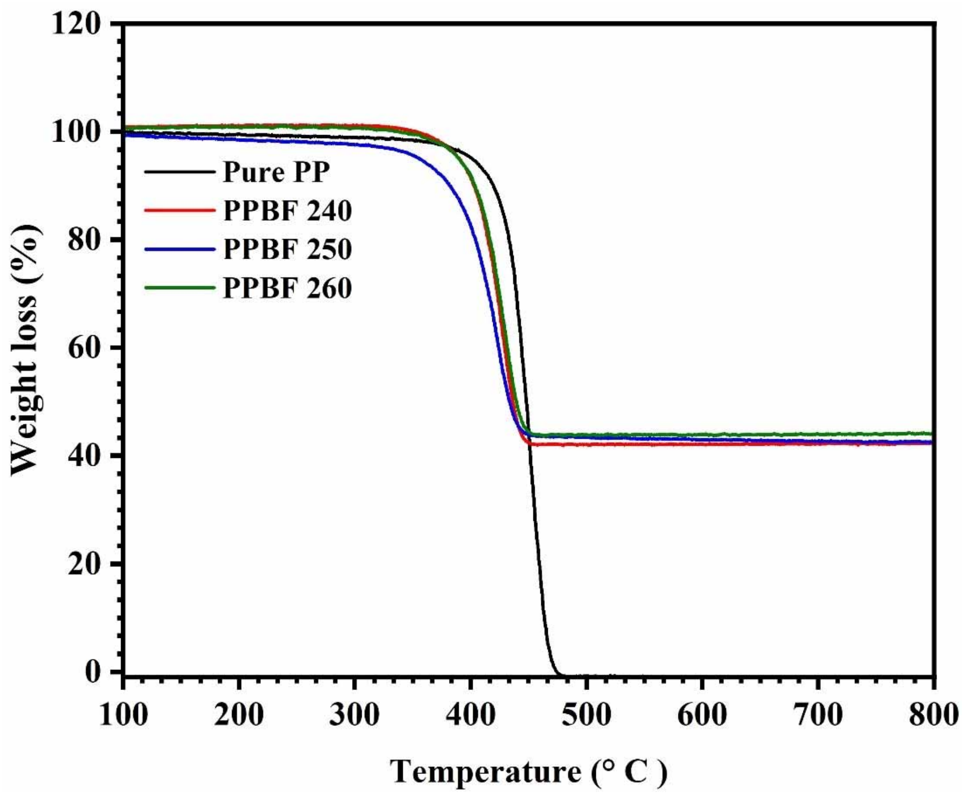

Figure 2-15 TGA curve of PPBF composites at different temperature

Table 2-3 Weight retention and loss % of PP and BFPP composite from TGA

SN	Sample	Weight Retention (%)	Weight Loss (%)
1	Pure PP	0	100
2	BFPP 240 °C	42	58
3	BFPP 250 °C	40	60
4	BFPP 260 °C	41	59

2.4 Summary

The aim of this study to optimize resin impregnation and minimize fiber fracture in basalt fiber reinforced polypropylene (BFPP) composites.

- The complete impregnation and lowest fractures rate were achieved at 260 °C and fiber velocity of 0.5 m/min, these parameters results in minimal void content, strong basalt fiber-polypropylene (BFPP) composites bonding and enhanced durability.

- The integrated mathematical model combining Darcy law, Reynolds equation, Weibull distribution, cohesive zone model and Griffith's criterion successfully predicted resin impregnation quality and fractures probabilities, validation through experimental results showed strong validate with mathematical model predictions.
- Higher temperature reduced polypropylene resin viscosity, improving flow and adhesion to basalt fibers, lower velocities allowed adequate basalt fiber-polypropylene (BFPP) contact time, reducing stress and fracture probability.

Chapter 3 Mathematical Modeling of Temperature and Pressure Effects on Resin Viscosity, Fiber Tension and Impregnation Quality in Glass Fiber-Reinforced Polypropylene Composites

3.1 Introduction

Thermoplastic composites have been gained a significant attention in various industries due to their superior mechanical properties, recyclability and potential for rapid manufacturing processing [23, 158-160]. The fiber reinforced thermoplastic composites are mainly produced through powder impregnation, solvent impregnation and melt impregnation process technologies [71, 174, 182]. Among these the melt impregnation techniques stands out as it contains the pulling fiber through guide pins submerged in molten thermoplastic [1, 81, 183, 184]. This technique significantly affects the mechanical properties of the final composite product, mainly fiber tension and molten polymer pressure which are critical parameters required for optimization [76, 183-185]. However, essentially high viscosity of molten thermoplastic ranges effective impregnation, leading to void contents and weak fiber matrix adhesion [81, 182, 186]. The melt impregnation process is considered by complex interactions involving heat transfer, viscoelastic behavior and resin fiber interaction. Advances in analytical modeling have attempted to address these challenges, yet experimental validation remains insufficient. The melt impregnation process an essential technique in thermoplastic composite manufacturing includes the infiltration of resin into fiber bundles under controlled conditions, key parameters such as temperature, pressure and film thickness play major role [144, 182, 187, 188]. Temperature governs the viscosity and flow behavior of the resin [189, 190], whereas pressure simplifies resin penetration into the fiber bundles [191-193]. Film thickness affects impregnation uniformity and void content both of which are critical to the structural reliability of the thermoplastic composite [145]. Improper control of these parameters can lead to imperfections such as void content and weak fiber-resin bonding is most significant remain underexplored and require further validation.

Höftberger et al. [194], found that incorporating glass fiber with a β-nucleating agent significantly improved the mechanical properties of polypropylene composites, particularly under varying temperatures, which enhanced fiber-matrix adhesion. Rahman et al. [195], found that impact properties were influenced by fiber loading and specimen geometry, with temperature also playing a significant role in determining the overall performance of glass fiber-reinforced polypropylene composites. Kabiri et al. [196], assessed the mechanical properties and biocompatibility of glass fiber/polypropylene composites, finding that optimal processing conditions could enhance both strength and compatibility for biomedical applications. Asoodeh et al. [197], indicated that processing parameters such as mold temperature and pressure significantly affect mechanical properties, with higher temperatures generally leading to improved tensile strength due to better fiber dispersion and matrix adhesion. Lauke et al. [198], showed that both fiber length and content had a pronounced effect on the mechanical properties of glass fiber reinforced polypropylene composites with optimal processing conditions enhancing these effects. Lyngdoh et al. [199], studied the interfacial bonding behavior of Carbon Fiber Reinforced Laminated Polyaryletherketone (CFR-LMPAEK) and Short Fiber Reinforced Polyetheretherketone (SFR-PEEK) composites. They used experiments and simulations to evaluate their elastic properties and bonding performance, finding that CFR-LMPAEK had a higher elastic modulus and strength compared to SFR-PEEK, making it more mechanically robust. Song et al. [200], used a mathematical model to analyze the impregnation process in long glass fiber-reinforced polypropylene composites. The model assessed how processing conditions, melt viscosity, and fiber structure influence the time required for complete fiber impregnation, validated through experiments and interlaminar shear strength (ILSS) measurements. Liu et al. [201], Studied ultrasonic-assisted melt impregnation in basalt fiber-reinforced polypropylene composites, optimizing parameters to achieve 2.99% porosity, 3.36% fracture rate, and improved mechanical properties. Their model highlighted that higher power and reduced frequency and action distance enhanced impregnation quality.

Moghadamazad et al. [202], developed a two-dimensional transient heat transfer model for AFP thermoplastic structures heated by a hot gas torch, highlighting the influence of temperature gradient on impregnation. Viscoelastic behavior play an important role as study by Mukherjee et al. [156], who explored the solute dispersion in viscoelastic flows under temperature gradient , highlighting the importance of accounting for viscoelastic effects during impregnation. Similarly Steggall-Murphy et al. [147], modeld thermoplastic melt impregnation which capturing the interplay between heat transfer, multi transfer, multiphase flow and viscoelastic dynamics. Processing temperature is a critical factor on resin viscosity reduction and resin penetration, as temperature increases viscosity decreases allowing better resin movement and fiber wetting. However higher temperature may cause resin over softening leading to compromised fiber matrix bonding. Tipboonsri et al. [157] highlighted the balance between temperature and pulling speed in optimizing mechanical properties of thermoplastic composites. Zhao et al. [154], Abouhamzeh et al. [203], further explored how temperature and viscoelastic response's influence resin infiltration and stress distribution highlighting the necessity for precise thermal cool.

Despite progress in the thermoplastic composite field several gaps persist in current research. There is insufficient integration of experimental validation with theoretical models limited focus on the dynamics of the wedge area during melt impregnation and a lack of studies addressing multi-parameter effects on fiber tension and resin flow [53, 119, 200, 201].

The production of high performance glass fiber reinforced polypropylene (GFPP) composites faces significant challenges due to high viscosity molten during impregnation. This limitation often leads to void formation poor interfacial adhesion and reduced mechanical properties in final composite of GFPP. Effective optimization processing temperature is essential to overcoming these challenges though ensuring resin flow and fiber wetting. Previous studies shows that the temperature play a critical role in reducing resin viscosity, improving impregnation efficeincy and enhancing the fiber matrix interaction, however excessive temperature may comporomise matrix stability leading to void contents [76, 81, 147, 156, 157, 203]. Additionally, the viscoelastic behavior of thermoplastic under shear stress add another layer of complexity influencing resin flow dynamics and stress distribution during impregnation [147, 202]. However, the aim this research work to develop an integrated mathematical model that incorporates temperature dependent resin viscosity, viscoelastic behavior, pressure distribution and film thickness impact on fiber tension to predict and optimize impregnation quality of GFPP composites. The model was experimentally validated to recognize the optimal processing conditions that minimize the void content and maximize glass fiber polypropylene matrix bonding.

3.2 Materials and Methods

3.2.1 Materials

In this research, the matrix material used was polypropylene (PP), specifically the (PBM100-GH) grade purchased from Sinopec Yangzi Petrochemical Co. LTD, Nanjing, China, it was chosen due to its thermal stability and compatibility with melt impregnation processing. To improve the stability of the matrix during high-temperature processing, antioxidants Irgafos 168 (0.15%) and Irgafos 1010 (0.15%) [169-172], were added to prevent oxidative degradation. The reinforcement material comprised continuous glass fiber roving (T838P5), procured from CTG Company, each with a linear density of 1200 Tex and a fiber diameter of 17 μm, optimized for fiber-matrix interaction and ease of impregnation.

3.2.2 Preparation of GFPP composites Sample using Twin-screw Melt impregnation

The glass fiber polypropylene (GFPP) composites samples were prepared using a custom-designed twin-screw melt impregnation machine as shown in Figure 3-1, configured to achieve uniform impregnation and consistent matrix flow throughout the process. The impregnation mold temperatures were set to 220 °C, 230 °C, 240 °C, and 250 °C to examine mold temperature effects and pressure on resin viscosity, tension, fiber impregnation quality and impact of varying temperature in wedge area of melt impregnation as shown in Figures 3-2 and 3-3.

A controlled fiber velocity of 0.7 m/min was maintained, and the polymer feed rate was set at 0.956 kg/h to ensure stable resin flow and matrix distribution. Throughout the process, the fiber-matrix ratio was controlled to 70% PP and 30% glass fiber by weight. Fiber tension was held constant at 10 N to promote consistent fiber wetting and prevent any fiber misalignment within the matrix, ensuring uniform impregnation.

The glass fiber roving’s were first supplied from a creel, then passed through a mold where the twin-screw extruder was used to the molten polypropylene by extrusion and impregnate it into the glass fibers. After impregnation the fiber tapes were passed through series of rollers for cooling and stabilize the matrix around the fibers and minimize the void formation [144, 173, 174].

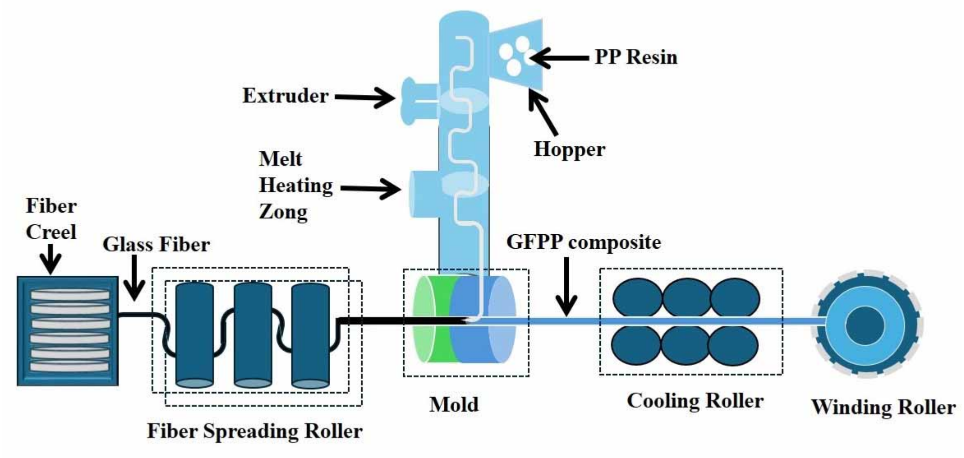

Figure 3-1 Flow diagram of melt impregnation machine

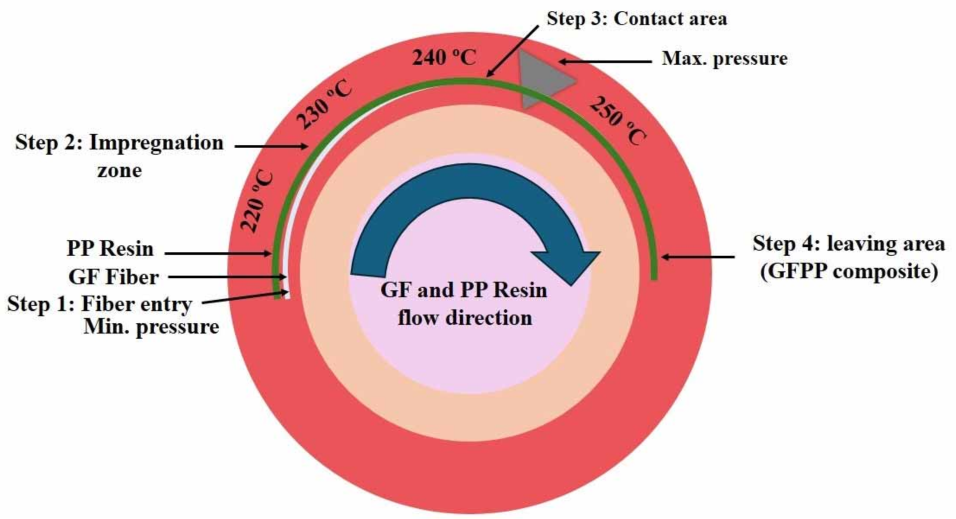

Figure 3-2 Impact of mold temperature in wedge area of melt impregnation

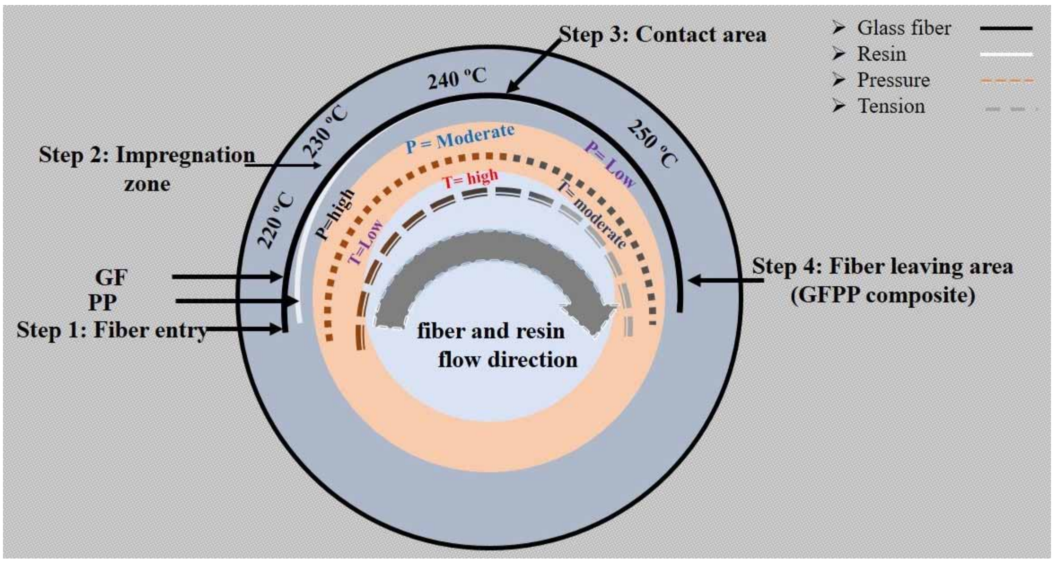

Figure 3-3 Effect of mold temperature on Pressure and Tension GFPP composites samples

3.2.4 Temperature-Dependent Resin viscosity Mathematical model

To quantify the competition among temperature, viscosity and viscoelasticity during melt impregnation as means to reliably predict fiber impregnation quality in thermoplastic composites, a mathematical model was developed. Our model includes both temperature-dependent viscosity dynamics and the viscoelastic characteristics of the polypropylene matrix to simulate complex resin flow through glass fibers. The assumptions and governing equations that define the model are as follows.

3.2.4.1 Assumptions

1) Viscosity Dependent on Temperature: The viscosity of the matrix tends to follow an Arrhenius-like relationship with respect to temperature so that if the temperature is increased there will be a reduction in viscosity and therefore, better resin penetration through the fiber tow.
2) No-slip condition: A no-slip condition is applied at the fiber-matrix interface to ensure adequate wet-out of the fibers by maximizing bond surface area for the resin.
3) Steady-State Flow: The fluid is assumed steady (which is also true) with no noticeable variations over time ensuring complete and even impregnation through the length of the composite.
4) Laminar Flow Regime: The resin flow is laminar due to the confined nature of the flow provided between the narrow inter-fiber spaces, which makes using Reynolds equation a simplified approach.
5) Incompressible Matrix: The resin is treated as incompressible, meaning that its density does not change after impregnation; this greatly simplifies continuity and momentum equations.
6) Fiber Volume Fraction (FVF) Constancy: In this method the fiber volume fraction is kept constant along a composite, which leads to predictable resin distribution in-between the fiber bundles.
7) Negligible Capillary Forces: Capillary effects within the fibers are considered negligible due to the dominance of external pressure-driven flow.
8) Viscoelastic Behavior: The viscoelastic response of the PP matrix is represented by the Oldroyd-B model, accounting for the material’s dual viscous and elastic behavior under shear stress.

3.2.4.2 Boundary Conditions and Constants

Boundary conditions are set to define temperature, film thickness, and pressure distribution along the resin flow path:

At x = 0

- Temperature: $T(0) = T_0$
- Film thickness: $h(0) = h_0$
- Pressure gradient: $\frac{\partial P}{\partial x}\Big|_{x=0} = \acute{P}$

3.2.4.3 Governing Equations

The model is governed by a set of equations that capture the temperature-dependent resin flow and viscoelastic stress distribution within the fiber bundles:

Temperature-dependent viscosity modeled using the Arrhenius equation:

$$\eta(T) = \eta_0 \exp\left(\frac{E_a}{RT}\right) \qquad (3-1)$$

Whereas; "η(T)" is Resin viscosity at temperature T (Pa·s), "η_0" : Reference viscosity (Pa·s) "Ea" is Activation energy (J/mol) , "R": Universal gas constant (8.314 J/(mol·K)), "T" is Absolute temperature (^{0}C)

According to equation (3-1) state that how resin viscosity decreases with increasing temperature. Higher temperatures lower the viscosity, making the resin flow more easily through the fiber bundle. This leads to improved impregnation of fibers. However, maintaining a controlled temperature is important to prevent excessive flow, which may lead to poor bonding or voids in the final composite material.

Void content calculation

$$\emptyset = \frac{T_d - M_d}{T_d} \times 100 \qquad (3-2)$$

Whereas; "ϕ" is void content (%), "T_d"is theoretical density of the composite, "M_d" is measured density of the composite.

The Equation (3-2) calculates the degree of impregnation by measuring void content, optimizing resin flow through fibers to enhance impregnation quality.

Oldroyd-B model for viscoelastic behavior

$$\tau + \lambda_1 \left(\frac{\partial \tau}{\partial t} + v . \nabla\tau - \tau . \nabla v - \nabla v^T . \tau \right)$$
$$= 2\eta \left(\mathrm{D} + \lambda_2 \left(\frac{\partial \mathrm{D}}{\partial t} + v . \nabla \mathrm{D} \right)\right) \qquad (3-3)$$

Whereas; “τ” is the Stress tensor (represents internal forces in the fluid due to deformation), “λ_1” is the relaxation time (a measure of how quickly the fluid relaxes stress after deformation) and “λ_2”is the retardation time (a measure of the fluid's response to deformation), and “D” is the rate of deformation tensor or strain rate tensor (describes how the fluid is being deformed). "v" is the Velocity vector (describes the fluid's flow), "∇v" is the velocity gradient tensor (rate of change of velocity), "∂t" is the time derivative (rate of change over time), "∇" is the gradient operator (describes spatial variation).

The model equation (3-3) describes the viscoelastic behavior of the resin during the impregnation process. It captures both the elastic (recoverable) and viscous (permanent) responses of the resin. In high-stress conditions, such as those in the impregnation process, the viscoelastic properties are crucial for understanding how the resin flows and redistributes stress within the fiber bundle, ensuring uniform impregnation and reducing the chances of fiber damage.

Reynolds equation for pressure distribution

$$\frac{\partial}{\partial x}\left(h^3 \frac{\partial P}{\partial x}\right) + \frac{\partial}{\partial z}\left(h^3 \frac{\partial P}{\partial z}\right) = 6\eta U h \frac{\partial h}{\partial x} + 6\eta h U + 12\eta V \qquad (3-4)$$

Whereas “h” is the film thickness, “P” is the pressure distribution, “η” is the dynamic viscosity of resin, “U” is the relative velocity of fiber, and “V” is the volume flow rate

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta(\mathrm{T})} \frac{\partial P}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{h^3}{\eta(\mathrm{T})} \frac{\partial P}{\partial z}\right)$$
$$= 6\eta(T) U h \frac{\partial h}{\partial x} + 6\eta(T) h U + 12\eta(T) V \qquad (3-5)$$

Whereas “η(T)” is the temperature dependent viscosity derived from equation (3-1).

Equation (3-5), integrates temperature dependent viscosity from equation (3-1), into the Reynolds equation (3-4), represents how variations in temperature control flow and consequently pressure distribution within the resin. The viscosity decreases with temperature so the resin can easily flow which leads to better fiber impregnation. The resin film flow behavior and its correlation with temperature and pressure is presented in this equation, which helps control the impregnation as a function of time.

Energy conservation (heat transfer)

$$\rho C_\rho \ (v . \nabla T) = k\nabla^2 T + Q \qquad (3-6)$$

Whereas; "ρ" is the density, "Cp"is the specific heat capacity, "k" is the thermal conductivity, and "Q" is the heat source term.

Equation (3-6) models shows how temperature changes propagate through the resin thus affecting its flow and impregnation ability.

Temperature-dependent shear stress (Oldroyd-B model)

$$\tau(T) + \lambda_1 \frac{D\tau}{Dt} = 2\eta(T)\gamma^{\cdot} + \lambda_2 \frac{D\gamma^{\cdot}}{Dt} \qquad (3-7)$$

Whereas; $\frac{D\tau}{Dt}$ is Material derivative, $\gamma^{\cdot}$ is shear rate

Effective viscosity adjusted fiber content:

$$\eta_{eff=} \eta(T) \left(1 - \emptyset_f\right) \qquad (3-8)$$

Whereas; $\eta_{eff=}$ is Effective viscosity considering fiber content (Pa·s), "$\emptyset_f$" is Fiber volume fraction (dimensionless)

In the equation (3-8) it modifies resin viscosity due to fibers in volume in composites. That is why the higher amount of fiber volume fraction makes the effective viscosity more and that will effect on the flow behavior of resin. More fibers mean more resistance to flow, thus temperature and pressure has to be carefully controlled in order to enable enough impregnation. However, this equation has to be taken into account to properly model resin flow in the presence of fibers.

Substitute "η_{eff}" into equation (3-4)

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta_{eff}}\frac{\partial P}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{h^3}{\eta_{eff}}\frac{\partial P}{\partial z}\right) = 6\eta_{eff} Uh \frac{\partial h}{\partial x} + 6\eta_{eff} Uh + 12\eta_{eff} V \qquad (3-9)$$

Note: Since "$\emptyset_f$" constant (fiber content is fixed), η_{eff} varies only with temperature.

Film Thickness Variation along Flow Direction

$$h = h_0 \left(1 - \frac{x}{R}\right) \qquad (3-10)$$

Where "h_0"is the initial film thickness at the entry point, "x" is the distance along the flow direction, and "R" is the pin radius.

The equation (3-10) describes how the film thickness changes along the flow direction, depending on the fixed pin radius. As the distance increases, the film thickness decreases, affecting how the resin impregnates the fibers. Thicker films near the start may slow down impregnation, while thinner films further down may improve flow. Controlling film thickness is key to achieving optimal impregnation quality.

Substitute k(x) in the place of "h"

$$k(x) = k_0 \left(1 - \frac{\mathrm{x}}{R}\right) \qquad (3-11)$$

Whereas "k_0"is the initial thermal conductivity

The equation (3-11) like film thickness, thermal conductivity also varies with distance along the flow direction. As the pin radius affects film thickness, it also influences how heat is conducted through the resin. Proper heat transfer is essential for maintaining the correct temperature, ensuring that the resin stays at an optimal viscosity for impregnation.

Substitute "k(x)" into Heat Transfer Equation (3-6)

$$\mathrm{\rho C_\rho} v_x \frac{\partial P}{\partial x} = k_0 \left(1 - \frac{\mathrm{x}}{R}\right) \frac{\partial^2 T}{\partial x^2} \qquad (3-12)$$

Whereas"v_x" velocity in the "x"-direction (m/s)

$$\theta = \frac{\mathrm{T}}{T_0} \qquad (3-13)$$

Whereas "T_0" is a reference temperature (°C).

$$X = \frac{\mathrm{x}}{\mathrm{L}} \qquad (3-14)$$

Whereas; "L" is the length of the impregnation zone (m).

$$\mathrm{\rho C_\rho} v_x \frac{\partial \theta}{\partial X} = \frac{k_0}{L} \left(1 - \frac{\mathrm{XL}}{R}\right) \frac{\partial^2 \theta}{\partial X^2} \qquad (3-15)$$

Simplify:

Let, apply "β" in equation (3-15) left and right side

$$\beta = \frac{\mathrm{\rho C_\rho} v_x}{k_0} \qquad (3-16)$$

$$\beta \frac{\partial \theta}{\partial X} = = \left(1 - \frac{\mathrm{XL}}{R}\right) \frac{\partial^2 \theta}{\partial X^2} \qquad (3-17)$$

Solve for Temperature gradient

Assuming: Linear temperature profile, so $\frac{\partial^2 \theta}{\partial X^2} = 0$

$$\frac{\partial \theta}{\partial X} = constant \qquad (3-18)$$

Integrate equation (3-18) for temperature;

$$\theta(X) = \theta_0 + \left(\frac{\partial \theta}{\partial X}\right) X \qquad (3-19)$$

Whereas; "θ_0" is the dimensionless temperature at X=0.

Rewriting equation (3-19) in terms of "T" and "x":

$$T(x) = T_0 + \left(\frac{\partial T}{\partial x}\right) x \qquad (3-20)$$

Whereas; $\frac{\partial T}{\partial x}$ is the temperature gradient

Substitute "T(x)" from Equation (3-20) into the Arrhenius equation (3-1):

Final temperature gradient model and viscosity relation

$$\eta(\mathrm{x}) = \eta_0 \exp\left(\frac{\mathrm{E}_a}{R\left(T_0 + \left(\frac{\partial T}{\partial x}\right) x\right)}\right) \qquad (3-21)$$

The equation (3-21) models state that how viscosity changes as a function of position along the flow direction due to the temperature gradient. As temperature increases along the resin flow, viscosity decreases, making the resin easier to flow and penetrate the fiber gaps. This is a critical factor in ensuring uniform impregnation across the length of the composite material.

Stress distribution using Oldroyd-B model

$$\gamma^{\cdot}(x) = \frac{\partial u}{\partial y} \qquad (3-22)$$

Whereas; "u" is the velocity component in the "x"- direction (m/min).

$$\tau(\mathrm{x}) = \eta(\mathrm{x}) \gamma^{\cdot}(x) \qquad (3-23)$$

The equation (3-23) relates the shear stress to viscosity and the shear rate. As the resin flows through the fibers, the temperature gradient changes the viscosity, which directly impacts the stress distribution. A well-distributed stress ensures better impregnation, reducing the risk of fiber breakage or poor bonding.

Substitute η(x) from equation (3-21)

$$\tau(\mathrm{x}) = \eta_0 \exp\left(\frac{\mathrm{E}_a}{R\left(T_0 + \left(\frac{\partial T}{\partial x}\right) x\right)}\right)\gamma^{\cdot}(x) \qquad (3-24)$$

Oldroyd-B model Simplified (Assuming Steady-State and Neglecting Convective Terms):

$$\tau(\mathrm{x}) + \lambda_1 \frac{d\tau}{dt} = 2\eta(\mathrm{x})\gamma^{\cdot}(x) \qquad (3-25)$$

Assuming steady-state $\frac{d\tau}{dt} = 0$

$$\tau(\mathrm{x}) = 2\eta(\mathrm{x})\gamma^{\cdot}(x) \qquad (3-26)$$

Simplifies the model under steady-state conditions and emphasizes the direct relationship between stress, viscosity, and shear rate.

Combine equation (3-24 and 3-25) get final expression for stress as a function of position:

$$\tau(\mathrm{x}) = 2\eta_0 \exp\left(\frac{\mathrm{E}_a}{R\left(T_0 + \left(\frac{\partial T}{\partial x}\right) x\right)}\right)\gamma^{\cdot}(x) \qquad (3-27)$$

The equation (3-27) final expression captures the combined effect of temperature, viscosity, and stress distribution along the flow direction. It highlights how stress evolves with the changing viscosity due to the temperature gradient, directly influencing the impregnation quality. Lower viscosity (due to higher temperatures) reduces stress, leading to smoother resin flow and better fiber impregnation.

Comprehensive model for stress and pressure distribution

Substitute “τ(x)” into Reynolds equation (3-9)

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta_{eff}}\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{h^3}{\eta_{eff}(x)}\frac{\partial P}{\partial z}\right)$$
$$=6\eta_{eff}Uh\frac{\partial h}{\partial x}+6\eta_{eff}(x)Uh+12\eta_{eff}(x)V \qquad (3-28)$$

Whereas; $\eta_{eff}(x) = \eta(x)(1-\emptyset_f)$

The comprehensive equation models (3-28) how temperature gradients affect viscosity and stress distribution, influencing the flow and impregnation quality.

3.2.5 Mathematical Model of Temperature and Pressure effect on Tension

3.2.5.1 Assumptions:

1. Negligible volume forces: Volume forces are negligible compared to viscous forces within resin flow.
2. No slip condition: The molten PP resin exhibits no slip relative to the surfaces of the fiber bundle.
3. Incompressible fluid behavior: The PP resin behaves as an incompressible fluid maintaining a constant density throughout the process.
4. Laminar flow regime: The flow is laminar, with no turbulence formation.

   The laminar flow is assumed because the processing condition (low Reynolds number and controlled resin flow) favor smooth, non-turbulent motion. This assumptions simplifies the equation while remaining representative of real-world impregnation process.
5. Newtonian fluid characteristics: The resin exhibits Newtonian behavior, meaning viscosity is independent of the shear rate.
6. Radial flow dominance: The resin flow occurs only in the radial direction of the fiber bundle, simplifying the impregnation dynamics.
7. Steady-state conditions: The system operates under steady-state conditions with no time-dependent changes in flow
8. Negligible Transverse pressure gradient: The pressure gradient in the transverse direction (Perpendicular to the flow) is negligible compared to the longitudinal direction.

   The assumption of negligible transverse pressure gradient is inherently built into the mathematical model to simplify the governing equation. This is justified because the resin flow primarily occurs in the longitudinal direction driven by the geometry of the wedge area. The transverse gradient are significantly smaller and have a negligible effect on overall impregnation process, by neglecting these gradient the model becomes computationally efficient without compromising its accuracy as validated by experimental results and MATLAB simulations.

9. Uniform fiber bundle cross-section: The cross-sectional area of the fiber bundle remains constant along its length.
10. Linear pressure distribution: The pressure decreases linearly along the length of the fiber bundle starting at the inlet pressure and reaching zero at the outlet.
11. Film thickness variation: The film thickness fluctuates linearly along the fiber length due to pressure gradient.
12. Temperature-dependent viscosity: The resin viscosity depends on temperature modeled using an Arrhenius equation
13. Simplified wedge geometry: The wedge area is idealized with varying film thicknesses (h_1 and h_2) at entrance and exit.
14. Adjustable fiber velocity and mold temperature: fiber velocity and mold temperature are adjustable experimental parameters influencing the resin flow dynamics and impregnation quality.

3.2.5.2 Model Equations:

1. Reynolds equation:

The Reynolds equation describes pressure distribution within a thin fluid film critical for understanding fluid dynamics in various applications. In the context of melt impregnation of thermoplastic composites this equation models pressure distribution in the wedge area formed between fiber bundles and pins. It governs how molten resin flows into fiber bundles. The equation considers factors like fluid density, film thickness and velocity gradients. By solving it, predict impregnation efficiency and optimize process parameters. Understanding pressure dynamics aids in designing efficient impregnation processes for composite manufacturing.

$$\frac{\partial}{\partial x}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial z}\right)$$

$$=6(U_1-U_2)\rho h\frac{\partial h}{\partial x}+6\rho h(U_1+U_2)+12\rho(V_1-V_2)\ (3-29)$$

In our specific setup, with assumptions such as $U_2 = 0, V_2 = 0, U_1 = U,\ V_1 = V$, and the pin being static, this simplifies to:

$$\frac{\partial}{\partial x}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial z}\right)$$

$$= 6(U-0)\rho h\frac{\partial h}{\partial x}+6\rho h(U+0)+12\rho(V-0) \qquad (3-30)$$

$$\frac{\partial}{\partial x}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{\rho h^3}{\eta}\frac{\partial P}{\partial z}\right)=6U\rho h\frac{\partial h}{\partial x}+6\rho hU+12\rho V \qquad (3-31)$$

This is the constant factor "ρ"

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta}\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{h^3}{\eta}\frac{\partial P}{\partial z}\right)=6Uh\frac{\partial h}{\partial x}+6hU+12V \qquad (3-32)$$

$$\frac{\partial}{\partial x}\left(h^3\frac{\partial P}{\partial x}\right)+\frac{\partial}{\partial z}\left(h^3\frac{\partial P}{\partial z}\right)=6\eta Uh\frac{\partial h}{\partial x}+6\eta hU+12\eta V \qquad (3-33)$$

Equation 3-33, which describe the pressure distribution and fluid flow in the context of varying thickness and velocities.

2. Darcy's Law:

Darcy law is a fundamental principle in fluid dynamics describing fluid flow through porous media. It quantifies the flow rate of a fluid through a porous material, such as a fiber bundle in thermoplastic composite manufacturing. The law states that the flow rate is directly proportional to the pressure gradient and inversely proportional to the material's permeability. In the context of melt impregnation Darcy's law helps predict how resin flows through the fiber bundles under different pressure conditions aiding in process optimization and impregnation quality control.

$$q = -K\frac{1}{\eta}\nabla P \qquad (3-34)$$

Making the simplification step on this:

$$q_x = -K\frac{1}{\eta}\frac{\partial P}{\partial x} \qquad (3-35)$$

$$q_y = -K\frac{1}{\eta}\frac{\partial P}{\partial y} \qquad (3-36)$$

$$q_z = -K\frac{1}{\eta}\frac{\partial P}{\partial z} \tag{3-37}$$

For our purposes considering a fiber bundle where flow is primarily in the radial direction:

$$q_y = -K\frac{1}{\eta}\frac{\partial P}{\partial y} \tag{3-38}$$

3. Continuity Equation:

The continuity equation is a fundamental principle in fluid dynamics that expresses the conservation of mass within a fluid flow. It states that the rate of change of mass within a control volume is equal to the net flow rate of mass into or out of the control volume. In mathematical terms, the continuity equation is often expressed as the divergence of the fluid velocity field which describes how the fluid velocity changes throughout the flow domain. In the context of melt impregnation in thermoplastic composite manufacturing the continuity equation helps ensure that mass is conserved as the resin flows through the fiber bundle providing insights into how the impregnation process affects resin distribution and composite quality.

$$\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z} = 0 \tag{3-39}$$

For our purposes considering a fiber bundle where flow is primarily in the radial direction:

$$q_y = -K\frac{1}{\eta}\frac{\partial P}{\partial y} \tag{3-40}$$

$$V = \frac{\partial q_y}{\partial y} \tag{3-41}$$

$$V = -K\frac{1}{\eta}\frac{\partial^2 P}{\partial y^2} \tag{3-42}$$

Combing the equations we get:

$$\frac{\partial}{\partial x}\left(h^3\frac{\partial P}{\partial x}\right) + \frac{\partial}{\partial z}\left(h^3\frac{\partial P}{\partial z}\right) = 6\eta U h\frac{\partial h}{\partial x} + 6\eta h U - 12K\frac{\partial^2 P}{\partial y^2} \tag{3-43}$$

For a simplified 2D flow (ignoring$\partial/\partial z$):

$$\frac{\partial}{\partial x}\left(h^3\frac{\partial P}{\partial x}\right)=6\eta Uh\frac{\partial h}{\partial x}+6\eta hU-12K\frac{\partial^2 P}{\partial y^2} \quad (3-44)$$

$$\frac{\partial}{\partial x}\left(h^3\frac{\partial P}{\partial x}\right)=6\eta Uh\left(\frac{\partial h}{\partial x}+1\right)-12K\frac{\partial^2 P}{\partial y^2} \quad (3-45)$$

Whereas: "h" is the film thickness. "η" is the fluid viscosity. "P"is the pressure. "U" is the relative velocity of the moving surface.

To simplify the given partial differential equation (PDE):

Simplifying Assumptions:

Assume steady-state conditions where all time derivatives are zero.

The pressure variation in the y-direction is significant. Hence, we focus on the x-direction for simplification.

Assume $\frac{\partial^2 P}{\partial y^2}=0$ because it represents the pressure gradient in the transverse direction, which may be negligible compared to the longitudinal direction.

Neglecting the Transverse Pressure Gradient:

$$\frac{\partial}{\partial x}\left(h^3\frac{\partial P}{\partial x}\right)=6\eta Uh\left(\frac{\partial h}{\partial x}+1\right) \quad (3-46)$$

Expanding the Derivatives:

Expand the left-hand side:

$$h^3\frac{\partial^2 P}{\partial x^2}+3h^2\frac{\partial h}{\partial x}\frac{\partial P}{\partial x}=6\eta Uh\left(\frac{\partial h}{\partial x}+1\right) \quad (3-47)$$

Combining the equation 3-46, we get:

$$h^3\frac{\partial^2 P}{\partial x^2}+3h^2\frac{\partial h}{\partial x}\frac{\partial P}{\partial x}=\left(6\eta Uh\frac{\partial h}{\partial x}+6\eta Uh\right) \quad (3-48)$$

Introduce Non-Dimensional Variable:

To further simplify, introduce a non-dimensional variable $\xi=\frac{x}{L}$ where L is a characteristic length.

First-Order Approximation:

Assume that $h(x)$ is slowly varying, i.e;

$\frac{\partial h}{\partial x}$ and higher-order terms are small.

Integrating Both Sides:

Integrate equation 3-48, both sides with respect to x:

$$\int \left(h^3 \frac{\partial^2 P}{\partial x^2} + 3h^2 \frac{\partial h}{\partial x}\frac{\partial P}{\partial x} \right) dx = \int \left(6\eta U h \frac{\partial h}{\partial x} + 6\eta U h \right) dx \quad (3-49)$$

Assuming $\frac{\partial^2 P}{\partial x^2}$ is small, focus on dominant terms:

It is very easy to integrate from equation 3-49, to this step

$$\int \frac{\partial}{\partial x}\left(h^3 \frac{\partial P}{\partial x} \right) dx = \int \left(6\eta U h \frac{\partial h}{\partial x} + 6\eta U h \right) dx \quad (3-50)$$

As we can cancel out the integral and derivative directly from the left side and simplifying the right side we get:

$$h^3 \frac{\partial P}{\partial x} + c_1 = \int \left(6\eta U h \frac{\partial h}{\partial x} + 6\eta U h \right) dx \quad (3-51)$$

Now making the simplifications on the right hand side.

$$h^3 \frac{\partial P}{\partial x} = 6\eta U h + C \quad (3-52)$$

Where, C is an integration constant.

Divide through byh^3:

$$\frac{\partial P}{\partial x} = \frac{6\eta U}{h^2} + \frac{C}{h^3} \quad (3-53)$$

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta} \frac{\partial P}{\partial x} \right) = 6\eta U \frac{\partial h}{\partial x} + C \quad (3-54)$$

This equation 3-54, provides a relation between the pressure gradient and the film thickness in the x-direction under the given assumptions and approximations. This form is useful for further analysis and solution depending on specific boundary conditions and known parameters of the system.

Simplification of the Pressure Distribution

Follow Equation 3-46,

$$\frac{\partial}{\partial x}\left(h^3 \frac{\partial P}{\partial x} \right) + \frac{\partial}{\partial z}\left(h^3 \frac{\partial P}{\partial z} \right) = 6\eta U \frac{\partial h}{\partial x} + 12\eta V \quad (3-55)$$

Given that the flow is primarily in the x-direction, and to simplify, we neglect variations in the z-direction:

$$\frac{\partial}{\partial x}\left(h^3 \frac{\partial P}{\partial x}\right) = 6\eta U \frac{\partial h}{\partial x} \qquad (3-56)$$

Assumptions:

Steady and Laminar Flow: The flow is steady, meaning there are no time-dependent changes. The flow is also laminar, implying smooth fluid motion.

Newtonian Fluid Behavior: The resin behaves as a Newtonian fluid, meaning viscosity "η" does not depend on shear rate but may depend on temperature.

Considering these assumptions, simplify the equation 3-55, to focus on how changes in film thickness influence relative pressure differences:

Integration of the simplified equation 3-55,

To analyze the pressure distribution along the fiber length, integrate equation 3-55, with respect to "x" over the length "L":

$$\int_0^L \frac{\partial}{\partial x}\left(h^3 \frac{\partial P}{\partial x}\right) dx = \int_0^L 6\eta U \frac{\partial h}{\partial x} dx \qquad (3-57)$$

This integration gives:

$$h^3 \frac{\partial P}{\partial x}\bigg|_{x=0}^{x=l} = 6\eta U \, [\, h\,(L) - h(0)] \qquad (3-58)$$

The boundary condition assumes that the pressure at the inlet is equal to the applied pressure while the pressure at the outlet is zero due to open system. These conditions provides a realistic basis for calculating pressure distribution along the fiber bundle.

Apply Boundary Conditions on equation 3-57;

At x = 0: P = P_0 (inlet pressure).

At x=L: Since there is no defined outlet pressure, leave P (L) unspecified and instead focus on the pressure gradient across the length.

Pressure Gradient:

At x = 0

$$h_0^3\left(\frac{\partial P}{\partial x}\right)_{x=0} = 6\eta U h_0 \qquad (3-59)$$

Whereas "h_0"is the film thickness at the inlet.

At x=L: The pressure gradient is unknown at x=L, so consider how the pressure changes relative to the inlet.

Thus, the equation 3-57, simplifies to:

$$h_L^3\left(\frac{\partial P}{\partial x}\right)_{x=L} = -h_0^3\left(\frac{\partial P}{\partial x}\right)_{x=0} = 6\eta U[\,h\,(L) - h(0)] \qquad (3-60)$$

Temperature Dependence

Follow equation 3-57,

$$\frac{\partial}{\partial x}\left(h^3\,\frac{\partial P}{\partial x}\right) = 6\eta U\frac{\partial h}{\partial x} \qquad (3-61)$$

The viscosity "η" of the resin is temperature-dependent, especially in a process involving heating like melt impregnation. To account for this, introduce the Arrhenius-type temperature dependence for viscosity:

Temperature-dependent viscosity was incorporated to reflect the behavior of polypropylene under varying thermal conditions, using the Arrhenius equation (3-62) allows the model to accurately capture the exponential decrease in viscosity with increasing temperature which directly impacts resin flow dynamics and impregnation quality.

$$\eta(T) = \eta_0\exp\left(\frac{-Ea}{RT(x)}\right) \qquad (3-62)$$

Assumptions:

Relative Temperature Dependence: focus on how changes in temperature influence viscosity, recognizing that temperature might vary non-linearly across the system.

Local Temperature Effects: consider the effects of local temperature variations rather than attempting to model the entire temperature distribution precisely.

By focusing on relative changes in viscosity due to temperature, capture the essential effects of temperature on the resin's behavior, avoiding the complexities of dealing with exact temperature measurements across the entire system.

Substituting Temperature-Dependent Viscosity

Substitute the temperature-dependent viscosity from Equation 3-62, into Equation 3-55;

$$\frac{\partial}{\partial x}\left(h^3\,\frac{\partial P}{\partial x}\right) = 6\eta_0 \exp\left(\frac{-Ea}{RT(x)}\right) U\frac{\partial h}{\partial x} \qquad (3-63)$$

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta\,(T)}\frac{\partial P}{\partial x}\right) = 6\eta(T)U\frac{\partial h}{\partial x} \qquad (3-64)$$

Pressure-Temperature-Film Thickness Relationship

Follow equation (3-64)

The relationship between pressure, temperature, and film thickness can now be explored by integrating the modified Reynolds equation with respect to “x”over the length “L” of the fiber bundle:

$$\int_0^L \frac{\partial}{\partial x}\left(h^3\,\frac{\partial P}{\partial x}\right)dx\ = \int_0^L 6\eta_0 \exp\left(\frac{-Ea}{RT(x)}\right) U\frac{\partial h}{\partial x}\ \ dx \qquad (3-65)$$

Integration equation 3-63, of the Left Side:

The left side is an exact derivative with respect to “x”, so the integration simplifies directly:

$$\int_0^L \frac{\partial}{\partial x}\left(h^3\,\frac{\partial P}{\partial x}\right)dx = \ \left[h^3\frac{\partial P}{\partial x}\right]_{x=0}^{x=L} \qquad (3-66)$$

This results in

$$h^3\frac{\partial P}{\partial x}\bigg|_{x=L} - h^3\frac{\partial P}{\partial x}\bigg|_{x=L} \qquad (3-67)$$

Integration equation 3-67, of the Right Side:

$$\int_0^L 6\eta_0 \exp\left(\frac{-Ea}{RT(x)}\right) U\frac{\partial h}{\partial x}\ \ dx \qquad (3-68)$$

Since “η_0” and “R” are constants, the integration focuses on the product of the exponential function of temperature and the derivative of “h(x)”:

$$6\eta_0 U\int_0^L \exp\left(\frac{-Ea}{RT(x)}\right)\frac{\partial h}{\partial x}\ \ dx \qquad (3-69)$$

This integral can be complex depending on how “T(x)” and “h(x)” vary with “x”. In general, without specifying a particular form for “T(x)”, the integral remains:

However, if "T(x)" is assumed constant over small sections or if "h(x)" changes slowly,

$$6\eta_0 U \ \exp\left(\frac{-Ea}{RT(x)}\right)[\,h\,(L)-h(0)] \tag{3-70}$$

Combining Both Sides equation 3-67 and 3-70;

After integrating both sides, the final result is:

$$h^3\frac{\partial P}{\partial x}\bigg|_{x=L} - h^3\frac{\partial P}{\partial x}\bigg|_{x=L} \quad = 6\eta_0 U \ \exp\left(\frac{-Ea}{RT(x)}\right)[\,h\,(L)-h(0)] \tag{3-71}$$

$$\frac{\partial}{\partial x}\left(\frac{h^3}{\eta_{0\exp\left(\frac{-Ea}{RT(x)}\right)}}\frac{\partial P}{\partial x}\right) = 6\eta_0 \ \exp\left(\frac{-Ea}{RT(x)}\right) U\frac{\partial h}{\partial x} \tag{3-72}$$

Relating Pressure to Fiber Tension

The fiber tension "T" is influenced by the pressure exerted by the resin on the fiber bundle.

The tension in the fiber can be approximated as being proportional to the integrated pressure over the length of the fiber bundle. Therefore, express the fiber tension "T" as:

$$T \propto \frac{1}{A}\int_0^L P(x)dx \tag{3-73}$$

To move from a proportional relationship to an equality, introduce a proportionality constant "C":

$$T \propto \frac{C}{A}\int_0^L P(x)dx \tag{3-74}$$

Whereas: "T" is the fiber tension. "C" is the proportionality constant that depends on the material properties and specific setup. "A" is the cross-sectional area of the fiber bundle.

"P(x)" is the pressure distribution along the fiber length "L".

Apply the Assumptions

Assumption 1: Uniform Cross-sectional Area "A"

Since the cross-sectional area "A" is assumed to be uniform along the fiber bundle, it simplifies the integral. This assumption allows us to factor "A" out of the integral, leaving the focus on the pressure distribution:

$$T = \frac{C}{A}\int_0^L P(x)dx \qquad (3-75)$$

Assumption 2: Pressure Distribution

Let's consider that the pressure gradually drops along the length "L" of the fiber bundle, but we do not know the exact distribution. We can express the pressure distribution as:

$$P\,(x) = P_0 + \left(1 - \frac{x}{L}\right) \qquad (3-76)$$

This equation assumes a linear drop from the inlet pressure "P_0" to zero at the outlet due to the open system.

Whereas: "P_0" is the pressure at the inlet (x=0), "x" is the position along the length of the fiber.

Integrate the Pressure Distribution

Substitute the pressure distribution into the integral:

$$\int_0^L P(x)dx = \int_0^L P_0 + \left(1 - \frac{x}{L}\right)\,dx \qquad (3-77)$$

Perform the integration:

$$\int_0^L P(x)dx = \int_0^L \left(P_0 - \frac{P_0 x}{L}\right)\,dx \qquad (3-78)$$

This results in:

$$\int_0^L P(x)dx = \left[P_0 x - \frac{P_0 x^2}{2L}\right]_0^L \qquad (3-79)$$

Evaluate the integral:

$$\int_0^L P(x)dx = P_0 L - \frac{P_0 L^2}{2L} \qquad (3-80)$$

Simplify the expression:

$$\int_0^L P(x)dx = P_0 L - \frac{P_0 L}{2} \qquad (3-81)$$

$$\frac{P_0 L}{2} \qquad (3-82)$$

The model establishes that fiber tension is directly proportional to the pressure gradient along the fiber bundle. This relationship is derived through the integration of pressure equation, particularly using the linear pressure distribution expressed as Reynolds equation. It emphasizes that resin flow resistance governed by viscosity and processing conditions directly influences the axial forces acting on fiber during the impregnation process. Higher pressure gradient correspond to greater fiber tension reflecting the increased resistance the resin exerts on the fibers. Conversely reduced viscosity at higher temperature lower pressure gradient thereby decreasing fiber tension and promoting better impregnation quality.

Substitute Back equation 3-82, into the Tension Equation 3-74;

Substitute the result into the equation for "T":

$$T = \frac{C}{A} . \frac{P_0 L}{2} \qquad (3-83)$$

Simplifying:

$$T = \frac{C P_0 L}{2A} \qquad (3-84)$$

The equation 3-83; result connects the fiber tension "T" directly to the inlet pressure "P_0" the length of the fiber bundle "L", and the cross-sectional area "A". It shows that the tension in the fibers increases with the inlet pressure and the length of the fiber.

Relating Pressure to Film Thickness:

Now, explore how the pressure affects the film thickness "h(x)" along the fiber. The film thickness is assumed to vary linearly with the pressure.

In equation 3-76, given the pressure distribution "P(x)" along the fiber length, where $(x) = P_0 + \left(1 - \frac{x}{L}\right)$, express the film thickness as;

$$h(x) = h_0 + \left(1 + \alpha P(x)\right) \qquad (3-85)$$

Substituting, equation 3-75, into equation 3-84;

$$h(x) = h_0 + \left(1 + \alpha P_0 \left(1 - \frac{x}{L}\right)\right) \qquad (3-86)$$

Expanding equation 3-86;

$$h(x) = h_0 + \left(1 + \alpha P_0 - \alpha P_0 \frac{x}{L}\right) \qquad (3-87)$$

Simplifying equation 3-87, further:

$$h(x) = h_0 + \left(1 + \alpha P_0\right) - h_0 \frac{\alpha P_0 x}{L} \qquad (3-88)$$

The equation 3-88; shows that the film thickness decreases linearly along the fiber length due to the pressure drop, starting from an initial thickness influenced by the inlet pressure "P_0" and the material constant "α".

3.3 Results and Discussion

3.3.1 Impact of Temperature on Pressure Distribution in the Melt impregnation Wedge area

The pressure distribution at different mold temperature (220 °C to 250 °C) in the wedge area of melt impregnation as show in MATLAB Figure 3-4. It shows that how temperature influence the resin flow during the impregnation process, at higher temperature reduced the PP resin viscosity and enable smoother PP resin flow and better glass fiber impregnation, while the pressure decrease along the length of wedge area due to the thinning of resin film. At higher temperature 250 °C the pressure is constantly lower through the wedge area as compared to low temperature 220 °C. This explained by Arrhenius equation 3-1, which predicts an exponential decrease in PP resin viscosity with increasing temperature. The lower viscosity at higher temperature reduces the resistance to resin flow and allowing the resin to more penetrate onto the glass fiber. This reduction in viscosity at higher temperature 250 °C directly affect the pressure gradient as explained in Reynolds equation 3-5, the pressure gradient is inversely proportional to the viscosity and film thickness, as the viscosity decrease with increasing the temperature, the pressure required to drive the resin flow also decreases as shown in Figure 3-4, the pressure and stress distribution equation 3-28, also validate. This equation integrates the combined effect of shear stress, film thickness and viscosity to predict the pressure distribution along the wedge area of melt impregnation. This result confirms that at 250 °C temperature the lower viscosity reduced the shear stress and ensures a more uniform pressure distribution and better impregnation of PP resin into the glass fiber.

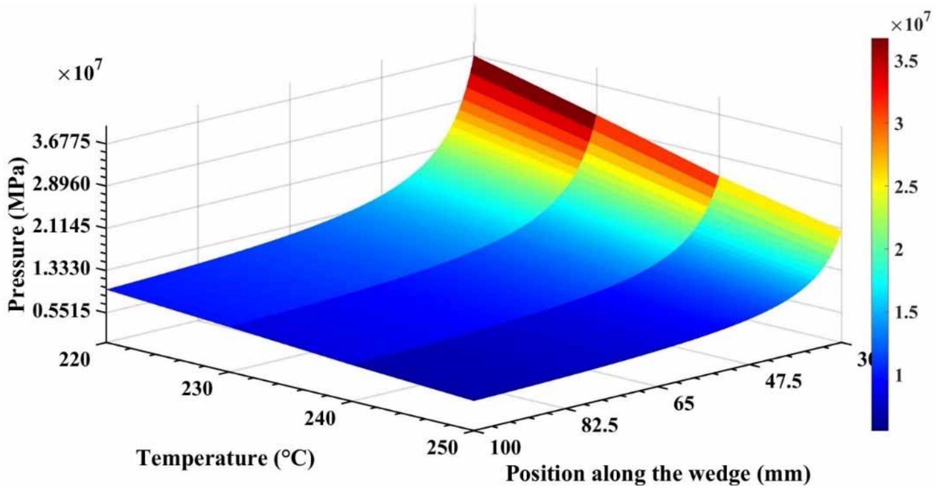

Figure 3-4 Pressure distribution along the melt impregnation wedge area at different temperature (220 °C to 250 °C)

The MATLAB Figure 3-3 and Figure 3-5 show the effect of mold temperature on the pressure distribution in the wedge area during the melt impregnation process. The pressure decreases gradually along the wedge area, consistent with the prediction of the Reynolds equation 3-29. This equation governs the pressure distribution in thin fluid films capturing the influence of resin viscosity, film thickness and flow resistance. As the mold temperature increases from 220 °C to 250 °C, the pressure significantly decreases due to decreasing in resin viscosity, which is modeled by the Arrhenius equation 3-62. This equation explained how resin viscosity decrease exponentially with increasing temperature, simplifying improved resin flow and reducing resistance.

The pressure distribution trends aligns with the theoretical relationship between pressure and film thickness equation 3-40, which states that pressure is inversely proportional to film thickness (h(x)). At higher temperature, the lower viscosity leads to a thinner melt film, further reducing the pressure across the wedge. The pressure reduction observed in the MATLAB Figure 3-5, also validates the integration of the pressure gradient in the fiber tension equation 3-43, which links the pressure drop along the wedge to the fiber tension, higher temperature results in lower tension due to decreased resistance within the resin. The agreement between the experimental results and model confirm the validity of these equations, specifically, equation 3-29, predicts the pressure gradient along the wedge, equation 3-37, explains the viscosity temperature dependent and equation 3-40, highlights the relationship between pressure and film thickness. These equations mutually validate the observed pressure reduction at higher mold temperatures. The results highlight the importance of controlling mold temperature to optimize pressure distribution, minimize flow resistance and enhance resin impregnation quality in thermoplastic composites.

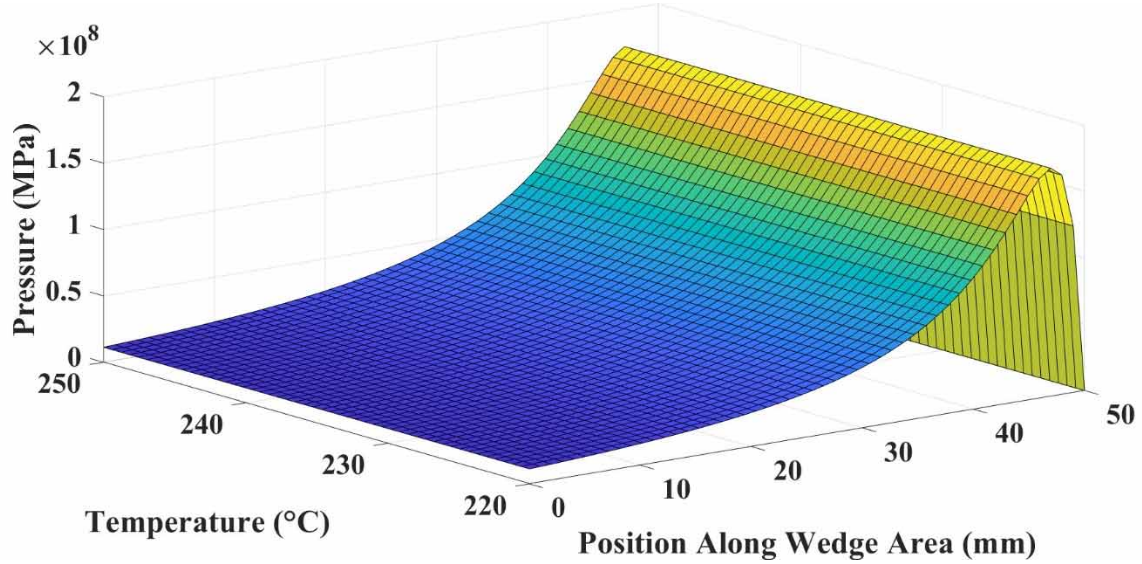

Figure 3-5 Effect of mold temperature on pressure in wedge area of melt impregnation

The MATLAB Figure 3-6 and Figure 3-3 show the effect of mold temperature and pressure on fiber tension of glass fiber-polypropylene (GFPP) composites during the melt impregnation process. As the mold temperature increases from 220 °C to 250 °C, a significant reduction in fiber tension is observed due to the exponential decrease in resin viscosity, governed by Arrhenius equation 3-62. The lower viscosity at higher temperature reduces the resistance to PP resin flow, allowing smoother impregnation into the glass fiber bundle and decreasing the tension exerted on glass fibers. On the other hand, increasing pressure results in higher fiber tension, consistent with the pressure distribution modeled by the Reynolds equation 3-46. The fiber tension is directly proportional to the integrated pressure along the fiber length as explain in equation 3-83. This relationship is further simplified by the linear pressure distribution in equation 3-82, which assumes a gradual pressure drop.

The viscosity-temperature dependent relationship also influences the PP resin film thickness (h(x)), which decreases linearly along the fiber length due to the pressure drop as expressed by equation 3-88. This relationship validates how variation in pressure and temperature simultaneously affect resin flow and fiber tension. At higher temperature, reduced viscosity leads to thinner resin film, lowering the pressure gradient along the fiber length and reducing fiber tension. Substituting temperature-dependent viscosity into the Reynolds equation simplifies the analysis, revealing that pressure and film thickness are inversely proportional under steady-state flow conditions. At lower temperature, higher viscosity increases the pressures influence on tension, while at higher temperature reduced viscosity.

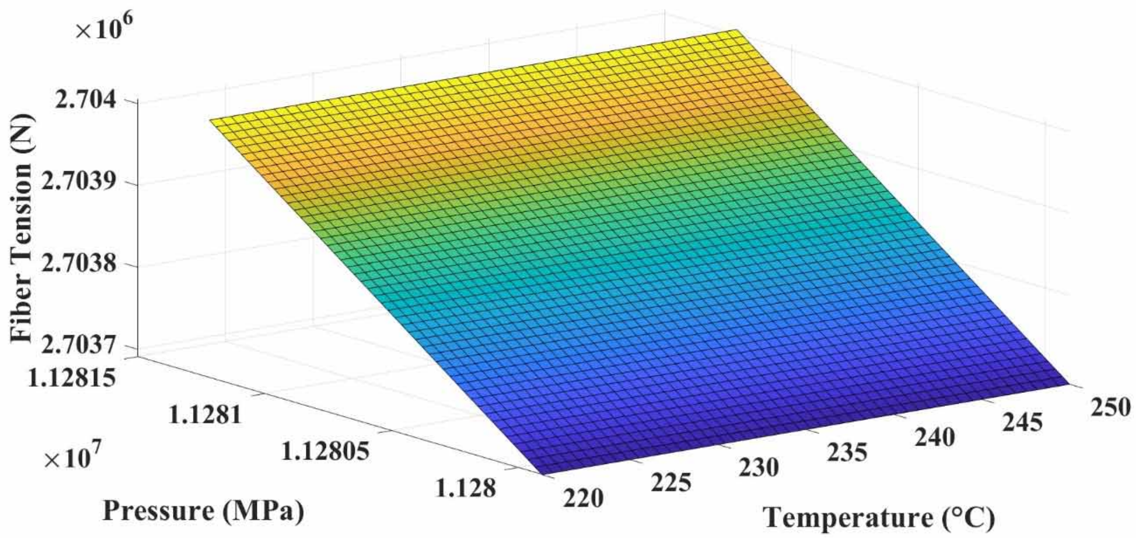

Figure 3-6 Effect of mold temperature and pressure on GFPP composites Tension

The MATLAB Figure 3-7 show the relationship between fiber tension, film thickness and mold temperature during the melt impregnation process, showing that fiber tension increase as temperature rises from 220 °C to 250 °C and as film thickness decreases slightly. This trend aligns with the equation 3-88, and equation 3-83, in the model. The reduction in void content at higher mold temperature improves the interfacial bonding between glass fibers and polypropylene matrix. This enhancement in bonding contributes to increased tensile strength and structural integrity of the GFPP composite, these improvements are particularly critical for high performance application in automotive and aerospace industries, where mechanical reliability and durability are important. Equation 3-87, explain the linear variation of film thickness (h(x)) along the fiber length due to the pressure gradient, stating that film thickness decreases linearly. Where film thickness decreases as temperature increases, influenced by the interplay of resin viscosity and pressure distribution. The temperature dependence of viscosity is captured by the Arrhenius equation 3-62, in the model. Simultaneously, equation 3-83, links fiber tension to the integrated pressure along the fiber length showing that tension increases with the fiber pressure gradient. The reduced resin viscosity at higher temperature increases the pressure gradient due to thinner films, resulting in greater fiber tension. This behavior observed in the MATLAB Figure 3-7, validates the model's assumptions about the inverse relationship between film thickness and pressure and the direct proportionality of fiber tension to the pressure distribution. Together, these equations highlight the critical role of temperature and film thickness in determining fiber tension, reinforcing the models ability to predict and explain the MATLAB simulation trends shown in Figure 3-7. This highlights the importance of precise control over processing parameters to optimize resin impregnation and fiber tension during GFPP composite manufacturing.

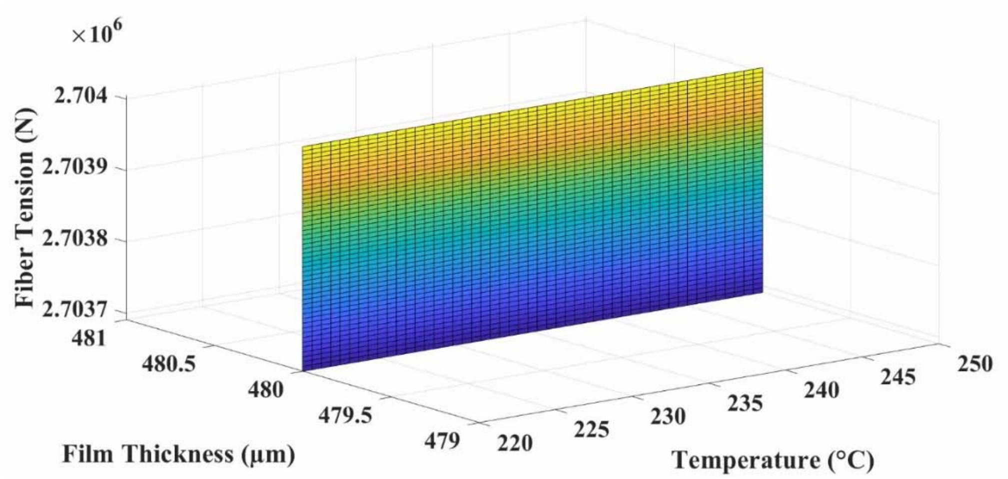

Figure 3-7 Effect of mold temperature and film thickness on GFPP composites Tension

The key parameters used to model and simulate the impregnation process are shown in Table 3-1.

Table 3-1 MATLAB parameters

Parameters	Symbol	Values	Unit
Fiber diameter	d_f	17 μm	m
Fiber monofilament	n_f	4400	-
Porosity	ε_0	0.7	-
PP viscosity	η	98 ~ 174	Pa.s
Temperature	T	220 ~ 250	°C
Velocity	U	0.7	m/min
Melt film thickness	δ	40 ~ 500 μm	m
Carman kozeny co efficient	k_0	5	-
Length of wedge area	L	30 ~ 100	mm

3.3.2 Comparison of Experimental Result with Mathematical model

The relationship between temperature and viscosity explains the model Arrhenius-based assumption equation 3-1, that an increase in temperature should decrease viscosity and increase the movement of resin through fiber bundles, as showed in Figure 3-8. This is confirmed by the results at 250 °C wherein the viscosity was decreases enough to allow effective resin penetration onto the fiber. This confirms the model's prediction that 250 °C gives optimum impregnation by reaching an ideal viscosity.

The experimental and predicted viscosity values across the temperature range of 220 °C to 250 °C showed excellent validation. The error percentage range from 2.8% to 4.5%, with an average of 3.6%. This validates the Arrhenius based viscosity equation 3-1, which accurately captures the exponential reduction in viscosity with increasing temperature, facilitating optimal resin flow and fiber impregnation.

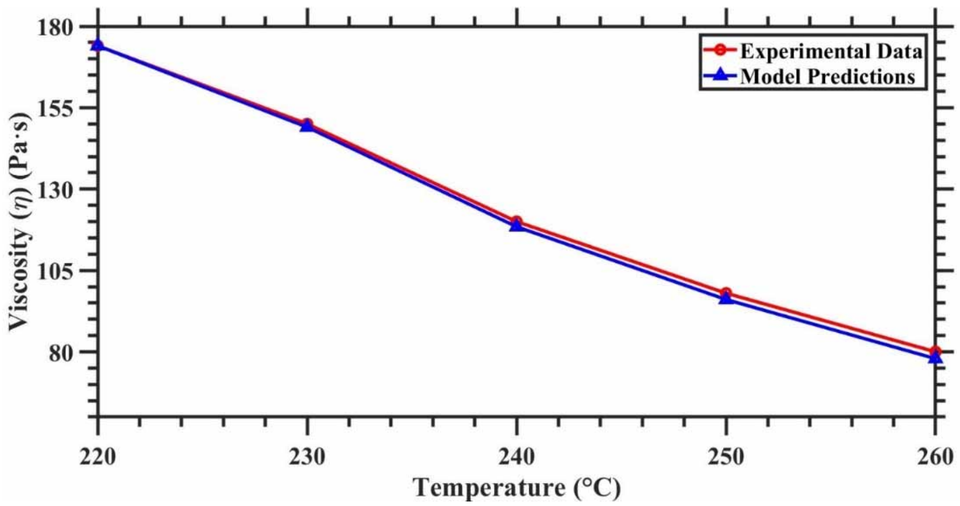

Figure 3-8 Comparison experimental viscosity of PP results vs. mathematical model predictable

The effect of temperature on shear stress shown in Figure 3-9 was consistent with the model viscoelastic behavior Oldroyed–B model equations 3-3 and 3-23. According to the model, decreasing viscosity with increasing temperature should enable stress to distribute evenly across the fiber resin interface. This balance is confirmed in the experimental results at 250 °C based on reduced shear stress and decreasing fiber fracture. This supports the model conclusion that a uniform stress distribution at 250 °C is important for effective impregnation.

Using Oldroyed-B model for viscoelastic behavior equation 3-3 and 3-23, the predicted shear stress closely validated the experimental results. The error percentage range from 3.1%

to 5.2% with an average of 4.1%, minor deviations were observed due to slight variation in fiber alignment and experimental temperature fluctuations, especially at lower temperatures.

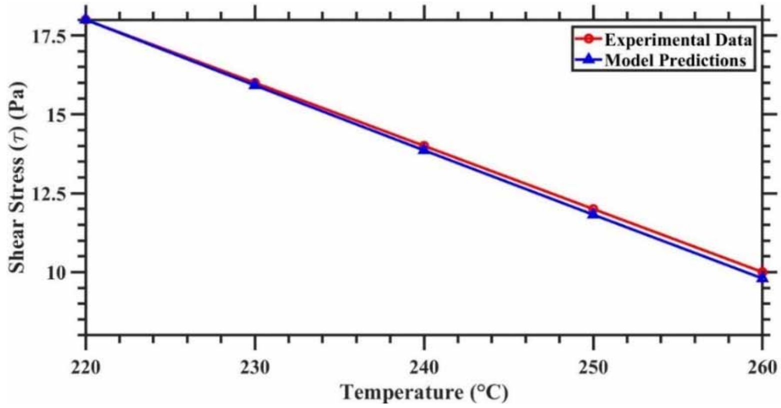

Figure 3-9 Comparison experimental shear stress of PP results vs. mathematical model predictions

Storage modulus, reflecting the elastic property of resin by Oldroyed–B model equation 3-3, as function of temperature is shown in Figure 3-10. While the model states that increased temperature should lead to reduced elasticity (facilitating a more fluid resin) and thus allow for better fiber impregnation. At 250 °C the results supports again, with a lower storage modulus reinforcing the model's declaration of increased resin flow and fiber wetting due to decreased elasticity at 250 °C temperature.

The storage modulus, representing the elastic properties of resin, exhibited and error percentage between 2.5% and 4.2% with an average of 3.4%. The model accurately predicted the decrease in elasticity with increasing temperature, particularly at 250 °C, where reduced storage modulus enhanced resin flow and impregnation.

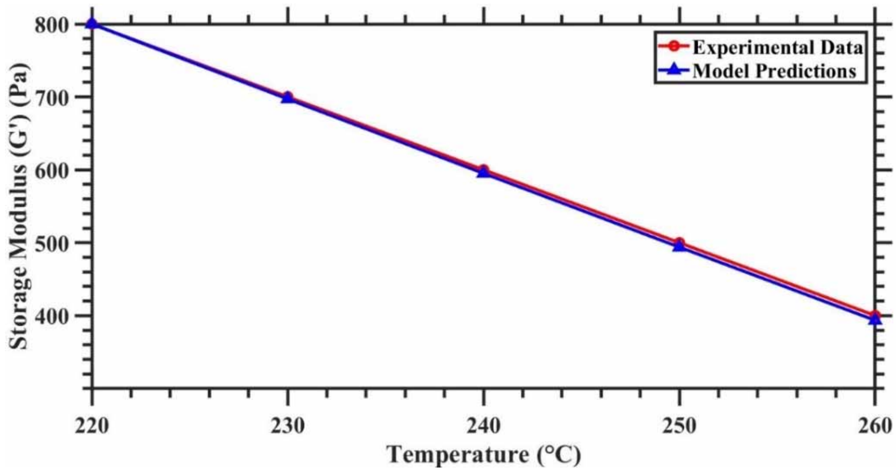

Figure 3-10 Comparison experimental storage modulus of PP results vs. mathematical model predictions

Theoretical similarly in Figure 3-11, the temperature influences loss modulus (the viscous resistance of the resin, Model equation 3-3. The resistivity should theoretically decrease with increased temperatures and help resin flow more freely. The decrease in loss modulus observed at 250 °C also validates the model's prediction that this temperature induces the highest level of fluidity for sufficient impregnation and strong fiber-matrix bonding.

The loss modulus comparison indicated and error percentage range of 3.2% to 4.7%, with an average 3.9%. The results validated the model’s prediction that higher temperature reduced viscous resistance allowing improved resin infiltration into fiber bundles.

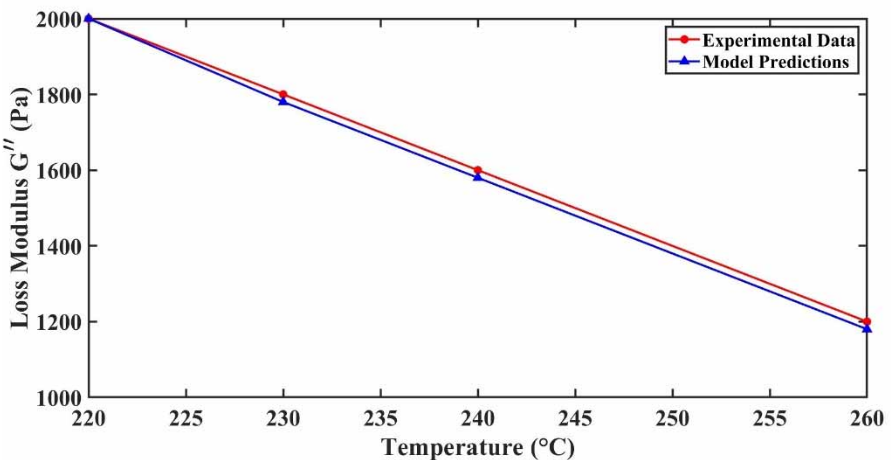

Figure 3-11 Comparison experimental loss modulus of PP results vs. mathematical model predictions

From Figures 3-8 to 3-11, confirm and indicate that 250 °C is the suitable temperature for melt impregnation for PP. Experimental data support the model’s predictions of lower viscosity, homogeneous stress distribution and balanced viscoelastic properties at this temperature so that 250 °C is validated as the optimal condition for better fiber impregnation.

The overall error percentage for all parameters remained within the range of 2-5%, confirming the reliability of the mathematical model. The observed discrepancies are attributed to small experiment variations such as temperature fluctuations, resin flow irregularities and slight fiber alignment.

The Figure 3-12 show the relationship between pressure and temperature in thermoplastic composite impregnation process, comparing experimental data with model prediction. The model effectively captures the trend of decreasing pressure with increasing temperature with error between experimental and predicted data as follows; 2.5% at 220 °C, 1.1. % at 230 °C, 1.4% at 240 °C and 5.0% at 250 °C.

The 5.0% error observed at 250 °C undergoes the exceptional performance of the model in capturing pressure dynamics, even under challenging high mold temperature conditions. At higher mold temperature factors such as; rapid reduction in resin viscosity can introduce a minor variation in resin flow dynamics due to mold temperature fluctuations, fiber misalignment, additionally model operates on idealized assumptions such as; uniform viscosity and steady state flow, it continuous to deliver results closely aligned with experimental results.

The Figure 3-13 illustrate the relationship between fiber tension and temperature comparing experimental results with model prediction. The tension decrease as the temperature increases, reflecting the impact of reduced PP resin viscosity on the pressure gradient and fiber tension. The small error percentage between experimental and model data 2.0% at 220 °C, 1.4. % at 230 °C, 2.0% at 240 °C and 2.7% at 250 °C.

The slightly higher error at 250 °C (2.7%) maybe attributed to experimental uncertainties such as minor temperature fluctuations, fiber misalignment or non-ideal resin flow dynamics which are not fully captured by the model assumptions. Despite these minor deviations the strong alignment between experimental and predicted values confirms the reliability of the model in capturing the interplay between temperature, viscosity, pressure and tension.

The model equation validate this relationship such as Arrhenius equation 3-62, for viscosity reduction, the Reynolds equation 3-46, for pressure distribution, the pressure film thickness equation 3-40, for film thinning and fiber tension equation 3-83, linking pressure to fiber tension.

These findings hold significant implication for aerospace and automotive industries, where precise control processing parameters is essential for manufacturing lightweight, and high-strength thermoplastic composites. The model ensures uniform fiber wetting, minimal void content and improved glass fiber-polypropylene adhesion resulting in superior mechanical properties and thermal stability with error percentage validating its reliability for practical applications.

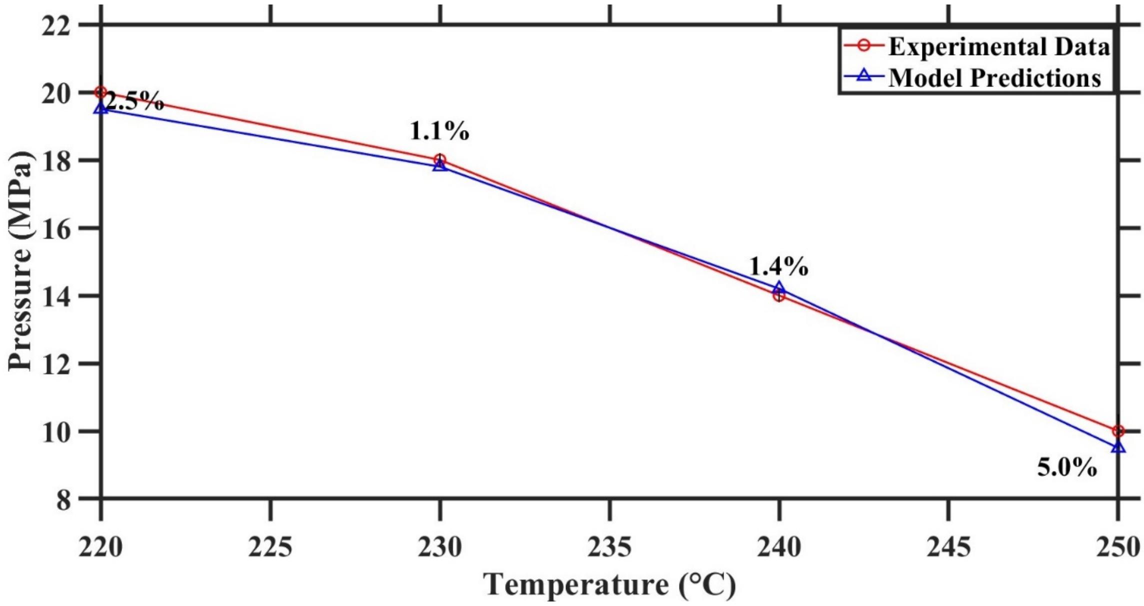

Figure 3-12 Experimental results Pressure and Temperature VS mathematical model

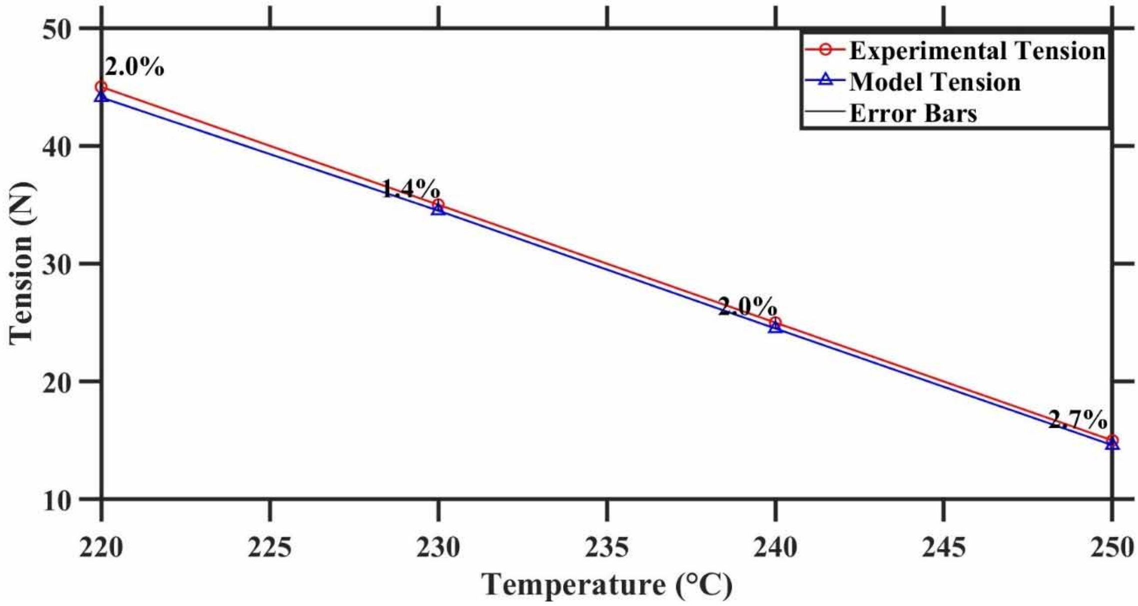

Figure 3-13 Experimental results Pressure and Temperature VS mathematical model

3.3.3 Morphological Structure of GFPP composites at various Mold temperature

The morphological assessment of the fiber-matrix interface across a temperature range of 220 °C to 250 °C highlights the critical role of processing temperature on impregnation quality in glass fiber-reinforced polypropylene (GFPP) composites, as shown in Figure 3-14 to 3-17

At 220 °C, the high viscosity of the polypropylene matrix contains resin infiltration, resulting in limited fiber impregnated and significant interfacial void content as shown in Figure 3-14. This observation is consistent with findings by [81], who reported that prominent viscosity at lower temperatures delays matrix flow into the fibers, leading to insufficient impregnation and visible voids that weaken the composite.

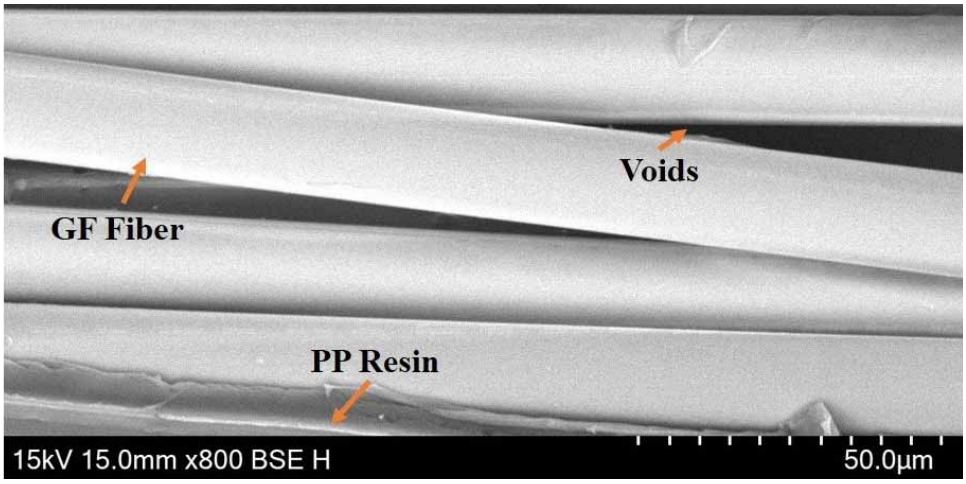

Figure 3-14 Morphological structures of PPGF at 220 °C

As temperature increases to 230 °C, viscosity decreases slightly, promoting improved resin penetration into the fiber tow and reducing void content, although resin penetration remains incomplete as shown in Figure 3-15. This finding aligns with [157], who noted that partial impregnation and incomplete matrix infiltration at this temperature result in similar limitations in impregnation quality.

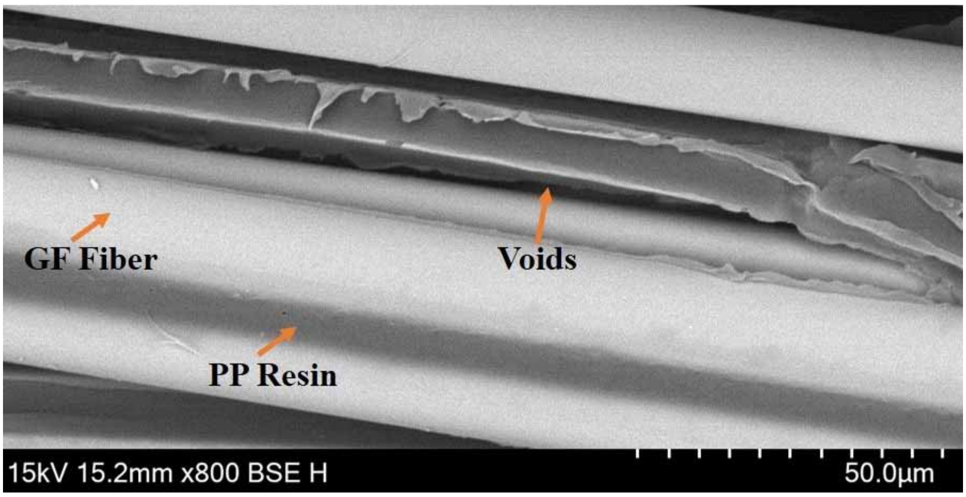

Figure 3-15 Morphological structures of PPGF at 230 °C

An obvious improvement in matrix distribution appears at 240 °C, where Figure 3-16, analysis shows uniform matrix coverage around the fiber bundles and minimized void content. This temperature supports a viscosity reduction that enables more effective fiber impregnated and improved fiber-matrix interfacial bonding. These results are consistent with [204], who proved that moderate temperatures achieve a viscosity level conducive to thorough impregnation without excessive flow variation. This temperature appears to slowdown an optimal balance for efficient impregnation [204], which highlights the advantages of moderate temperatures in enhancing composite structural quality.

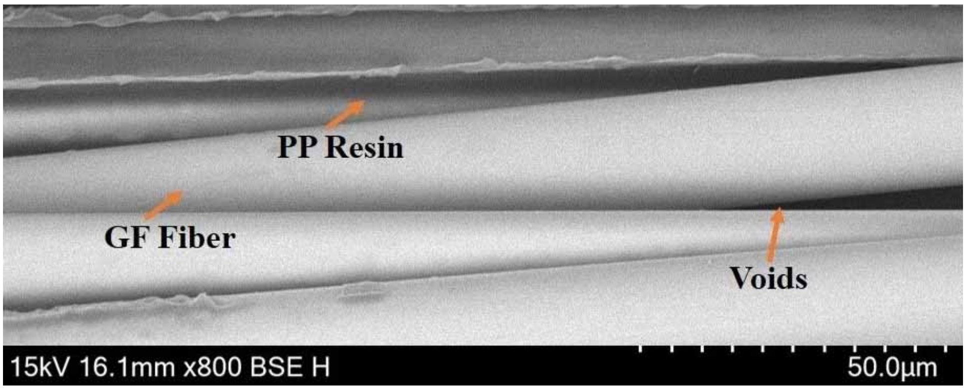

Figure 3-16 Morphological structures of PPGF at 240 °C

At 250 °C, Figure 3-17 expose extensive fiber coverage with minimal voids, indicating high impregnation quality and strong fiber-matrix adhesion [153]. However, cautions that excessively high temperatures can lead to an overly molten resin state, potentially compromising bond stability if the resin's solidification behavior is affected. This observation suggests that, while 250 °C achieves superior impregnation, maintaining controlled processing conditions is essential to preserve bond integrity. The SEM Figure 3-17, shows the optimal impregnation at 250 °C mold temperature as compared to 220 °C, 230 °C and 240 °C mold temperature [205].

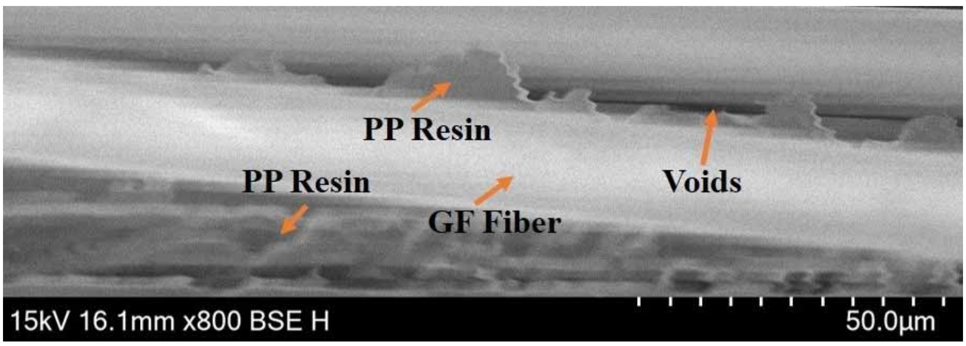

Figure 3-17 Morphological structures of PPGF at 250 °C

The optimal temperature range for GFPP composites lies between 240 °C and 250 °C, achieving a balance between matrix infiltration and interfacial bond stability. Observations at 240 °C, with uniform matrix coverage and minimal voids, and at 250 °C, with maximized fiber impregnation, validate findings from previous studies and provide additional insights into temperature's impact on resin impregnation quality in thermoplastic composites.

These SEM Figures 3-14 to 3-17, observations validate the mathematical model developed for the melt impregnation process. The Arrhenius equation for temperature-dependent viscosity equation 3-1 and 3-62, accurately predicts that increased temperatures lower viscosity, facilitating matrix infiltration into the fiber bundles. This relationship is observed in the Figures 3-14 to 3-17, where progressive improvements in matrix distribution and fiber coverage are evident from 220 °C to 250 °C. The model's temperature viscosity relationship is essential for enabling effective resin flow into the fibers, resulting in more complete impregnation, as demonstrated by the reduced void content at higher temperatures.

The Oldroyd-B model equation 3-3, which captures the viscoelastic behavior of the matrix, also finds validation in these Figures 3-14 to 3-17, as temperature rises, the matrix viscosity decreases, leading to adjusted stress distribution that allows the matrix to flow more uniformly around the fibers without excessive stress concentration. This behavior is evident in the Figures 3-16 to 3-17 at 240 °C and 250 °C, where enhanced flow properties enable seamless fiber-matrix bonding, supporting the model's prediction and confirming that the dual viscous elastic nature of the matrix is optimally balanced at these temperatures.

Equation 3-2 measures void content as a measurable indicator of impregnation quality, and the Figure 3-17, show a significant decrease in void content with increased temperature, especially at 250 °C. This supports the model's assertion that void formation diminishes as viscosity is optimized for impregnation, resulting in stronger composite mechanical properties. The heat transfer equation 3-6, further describes temperature propagation through the matrix, influencing viscosity and flow behavior. Strong glass fiber-polypropylene matrix bonding observed at higher temperatures suggests effective heat distribution, keeping the resin within an optimal viscosity range for impregnation. At 250 °C, efficient heat transfer ensures that temperature dependent viscosity is low enough to achieve thorough fiber impregnated and minimize voids, thus validating the heat transfer model. Equation 3-23, relating shear stress to viscosity and shear rate, reinforces the model's predictions. The Figure 3-17 show that, at 250 °C, lower viscosity allows for more uniform shear stress distribution across the fiber-matrix interface, reducing void content and enabling smoother matrix flow. This alignment with the model's stress distribution equation confirms that reduced viscosity directly enhances impregnation by enabling uniform shear stress across the fibers. Overall, the Figures 3-14 to 3-17, provide robust empirical validation for the mathematical model, showing that increased temperature enhances polypropylene matrix infiltration, glass fiber impregnation, and void reduction. Each model equation from the Arrhenius temperature viscosity relationship to the heat transfer and stress distribution equations finds support in the morphological changes observed across temperatures. These findings highlight the model's accuracy in capturing the relationship between temperature, viscosity, and impregnation quality in GFPP composites, highlighting the importance of controlled temperature elevation in optimizing the melt impregnation process to achieve high-performance thermoplastic composites.

The Arrhenius equation 3-62, accurately predicts the exponential decrease in resin viscosity with increasing temperature [206]. At lower temperature 220 °C and 230 °C, the high viscosity limits resin flow causing voids and incomplete impregnation, at higher temperatures 240 °C and 250 °C, the significant viscosity reduction facilitates better resin flow, improving impregnation quality and reducing voids.

The Reynolds equation 3-46, states that the pressure distribution in wedge area, at lower temperature, higher viscosity increases the pressure gradient, leading to incomplete resin penetration and poor impregnation. At higher temperature reduces viscosity decrease the pressure gradient resulting in smoother resin flow and better fiber wetting as observed from Figure 3-14 to 3-17 SEM images.

The pressure-film thickness relationship equation 3-40, align with the SEM results. At higher temperature 240 °C and 250 °C, the resin forms a thinner melts film due to reduced viscosity. This allows for more efficient pressure distribution leading to improved resin penetration and minimal void contents [207, 208].

The fiber tension equation 3-83, explains the reduced tension observed at higher temperature. Lower viscosity and pressure reduce fiber tension enabling better resin wetting and adhesion. The SEM images at 250 °C reflect the minimal void content and optimal fiber-resin bonding predicted with equation 3-83.

3.3.4 Thermal Stability and Degradation behavior in GFPP composites

The thermogravimetric analysis (TGA) of glass fiber-reinforced polypropylene (GFPP) composites processed at different mold temperatures shows the relationship between processing conditions and thermal stability as shown in Figure 3-18 and Table 3-2, aligning well with predictions from the mathematical model. The TGA data indicates that pure polypropylene (PP) begins degrading around 450 °C, showing rapid weight loss rapidly after, consistent with findings from [209], However, GFPP composites demonstrate improved thermal stability compared to pure PP, with delayed beginning of degradation and reduced weight loss, particularly at higher processing temperatures. The composite processed at 250 °C shows the highest thermal stability, with approximately 48% weight retention at 480 °C, validating the model's prediction that higher processing mold temperatures enhance fiber impregnation and overall composite performance.

The Arrhenius law for temperature dependent viscosity equation 3-1, describes how viscosity decreases as mold temperature increases. This reduced viscosity allows the resin to infiltrate fiber bundles more effectively, achieving better fiber impregnated and stronger interfacial bonding with the fibers. Consistent with the TGA results, composites processed at 250 °C exhibit superior thermal stability, where the lower viscosity enables thorough impregnation, providing greater resistance to degradation under thermal stress. These observations align with previous work by [145], which reported that elevated processing temperatures enhance fiber matrix bonding and thermal performance.

The Oldroyd-B model for viscoelastic equation 3-3, is essential for understanding stress distribution within the fiber matrix interface. As mold temperature rises, the resin's viscosity decreases, allowing it to respond to stress changes without exerting excessive strain on the fibers. TGA findings show that composites processed at 250 °C retain greater weight under thermal stress, indicating that lower viscosity optimizes stress distribution and helps prevent untimely separation at the fiber-matrix interface. Equation 3-23, which relates shear stress to viscosity and shear rate, further supports this by showing that lower viscosity under thermal stress enhances shear stress distribution, which reinforces the stability of fiber-matrix bonding

In areas with higher fiber content, effective viscosity equation 3-8, considers fiber volume fraction to account for the increased resistance to polypropylene resin flow. Higher fiber content requires an optimized viscosity to achieve uniform impregnated and to limit thermal expansion stresses. The TGA results align with this prediction, showing that composites processed at 250 °C, where effective viscosity is balanced for fiber content, validate enhanced thermal stability.

Temperature gradients and heat transfer consistency equations 3-16 and 3-17, within the resin illustrate how steady heat distribution maintains the resin in an ideal viscosity state for effective impregnated. The TGA data confirms that GFPP composites processed at 250 °C exhibit improved thermal stability due to consistent resin flow, facilitated by efficient heat transfer.

Controlled heat distribution during processing prevents overheating reducing degradation and preserving fiber matrix bonding in glass fiber reinforced polypropylene composites, According to shear rate and viscosity adjustment along the resin flow path equation 3-27, processing mold temperature affects shear rate, with optional viscosity achieved at 250 °C, This ensures even shear stress distribution enhancing thermal stability. The stress distribution model across the resin matrix equation 3-28, describes how temperature gradient influence bond stability with TGA finding showing high weight retention at 480 °C for GFPP proceed at 250 °C. These results confirm that controlled temperature adjustment as outlined in equations 3-27 and 3-28, enhance the impregnation quality and thermal resistance, making GFPP composite suitable for high temperature applications.

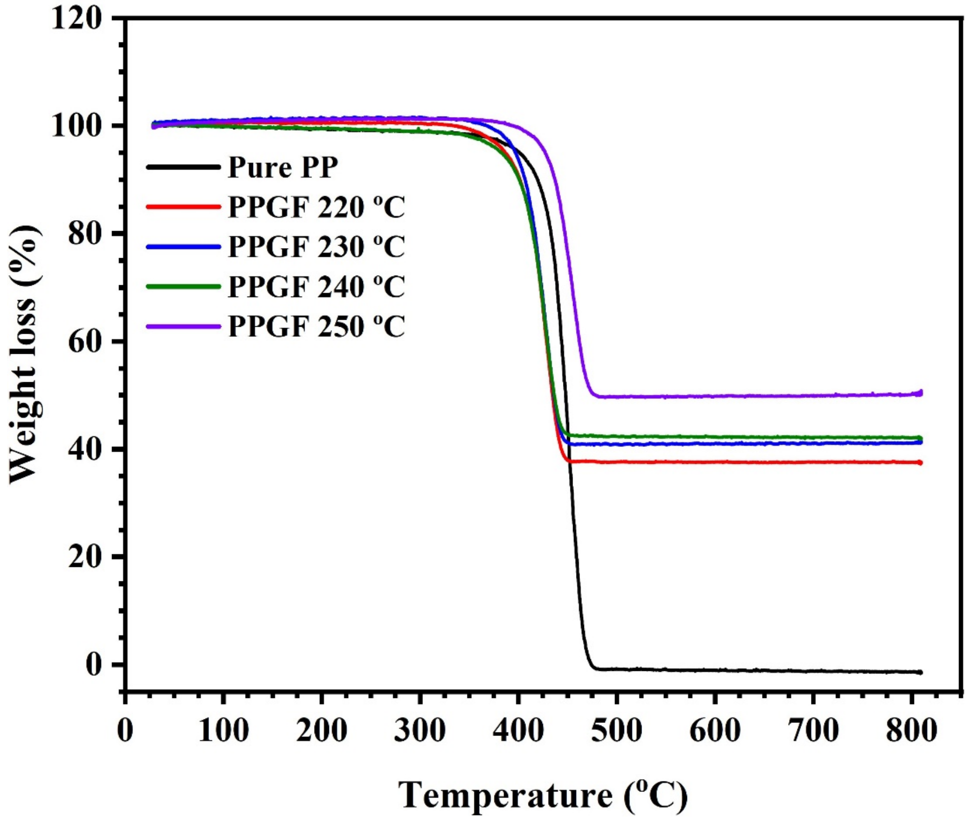

Figure 3-18 TGA curve of PPGF composite at various temperature

Table 3-2 Weight retention and loss % of PP and GFPP composite from TGA

SN	Sample	Weight Retention (%)	Weight Loss (%)
1	Pure PP	2	98
2	GFPP 220 °C	35	65
3	GFPP 230 °C	38	62
4	GFPP 240 °C	39	61
5	GFPP 250 °C	18	52

3.3.5 Impact of Mold temperature on Crystallinity and Melting of GFPP composites

Figures 3-19 (a and b) and Table 3-3 show the results of the differential scanning calorimetry (DSC) analysis conducted to study the crystallization and melting behavior of polypropylene (PP) and glass fiber-reinforced polypropylene (PPGF) at processing temperatures ranging from 220 °C to 250 °C. The analysis was performed using a DSC Q2000 under a nitrogen atmosphere. Samples weighing approximately 7.5 mg were heated from room temperature to 300 °C at a rate of 10 °C/min, held at 300 °C to eliminate any thermal history, cooled to 50 °C at 10 °C/min, and then reheated to 300 °C at the same rate.

The degree of crystallinity (X_c) for each sample was calculated from the second heating cycle using the equation:

$$X_c = \frac{\Delta H_m}{\emptyset \Delta H_m^0} \times 100\% \qquad (3-89)$$

Whereas ΔH_m is the melting enthalpy measured by DSC, ΔH_m^0 is the melting enthalpy of fully crystalline PP (207 J/g) [210], and $\emptyset$ represents the weight fraction of PP in the composite, which is 0.7 for the PPGF samples.

The DSC thermograms showed that the pure PP sample exhibited a sharp melting peak at approximately 165 °C [211], corresponding to a high degree of crystallinity (65.35%) in the polymer. In contrast, the PPGF samples displayed melting peaks around 163 °C, with broader and less intense profiles, reflecting the reduction in crystallinity due to the presence of glass fibers.

Table 3-3 Crystallization and melting of PPGF with different temperature

Sample	T_c (ºC)	ΔH_c (J/g)	T_m (ºC)	ΔH_m (J/g)	X_c (%)
PP	131.36	94.70	163.97	102.8	65.35
PPGF 220 ºC	128.44	51.48	163.29	41.60	35.52
PPGF 230 ºC	127.99	55.47	163.60	51.89	38.28
PPGF 240 ºC	128.20	57.37	163.16	47.77	39.59
PPGF 250 ºC	128.03	58.80	163.55	52.14	40.57

The PPGF samples processed at 240 °C and 250 °C demonstrated more distinct melting behavior compared to those processed at 220 °C and 230 °C. The sample processed at 250 °C exhibited the highest crystallinity (40.57%), and the most defined melting peak as show in table 3-3. This indicates that higher processing temperatures promote improved crystallization and enhance fiber-matrix interactions. The increased crystallinity at higher temperatures suggests improved mechanical performance and reduced void content in the thermoplastic composite, which is critical for applications involving melt impregnation

The Arrhenius equation 3-1 and 3-62, is validate because the exponential decrease in viscosity with increasing temperature explain the improvement in crystallinity (X_c) and melting enthalpy (ΔH_m) observed in the DSC results as show in Table 3-3. At higher temperature reduced viscosity facilitates smoother resin flow and better polymer chain alignment, enhancing crystallinity. The fiber tension and pressure relationship equation 3-83, is align with DSC results due to reduced viscosity at higher temperature lower the pressure gradient reducing fiber tension and enabling improved fiber-matrix adhesion.

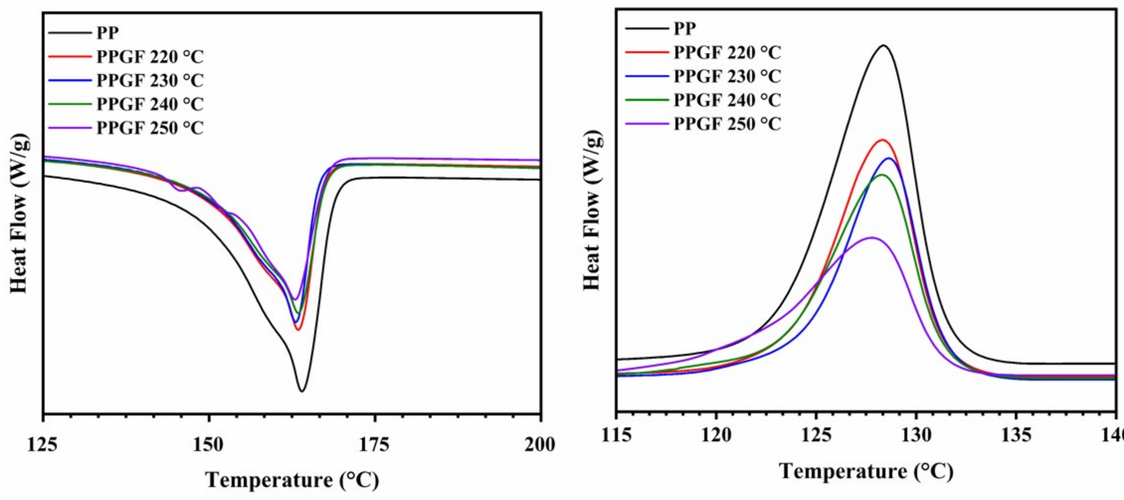

Figure 3-19 DSC of GFPP composites at various mold temperature (a) Melting (b) Crystallinity

3.4 Summary

- This research demonstrated the significant impact of mold temperature on resin viscosity, pressure and film thickness of fiber tension during the melt impregnation of glass fiber-reinforced polypropylene composites, using a mathematical model.
- The model predictions were validate through experimental results as well as MATLAB simulation which provide visual confirmation of the relationship between temperature and resin viscosity, pressure, film thickness and fiber tension.
- The experimental results show that a higher mold temperature 250 °C improves impregnation quality, minimizing void content, enhancing resin flow and improving fiber-matrix adhesion in glass fiber-polypropylene composite bonding
- MATLAB simulations complemented these finding confirming the relationship between temperature, pressure, film thickness and fiber tension reinforcing the reliability of the developed mathematical model.
- Future study will enhance the model by incorporating parameters such as; thermal gradient, shear-dependent viscosity, fiber velocity, mold gap to improve its predictive accuracy , additionally exploring advanced thermoplastic resins and the effect of cooling rates and fiber orientation will deeper insights into optimizing composite performances.

Chapter 4 Impact of Fiber Velocity on Pressure Distribution in the Melt Impregnation Wedge Area of Thermoplastic Composites Using Nonlinear Darcy Law Mathematical model

4.1 Introduction

Thermoplastic composites have become high interest in the aerospace, automotive and infrastructure industries due to their unique properties such as good mechanical properties along with recyclability and compatibility with rapid manufacturing processes [182, 212, 213]. In comparison to the thermoset composites which are often unfriendly for recycling, thermoplastic can be melted and remolded, thus more environmentally friendly and perfect for applications requiring durability and flexibility, due to their impact resistance, moisture resistance, and light weight [71, 158]. However, achieving high-quality thermoplastic composites relies on control of processing parameters to ensure uniform flow and impregnation of resin onto fibers. One of the important methods is melt impregnation, which has been shown to improve fiber resin impregnation properties. The melt impregnation process consists in pulling fibers through molten thermoplastic resin, where the resin will flow through it allowing fiber impregnation producing a composite [91, 214-216]. The process of melt impregnation itself is complex, and many aspects that influence the quality of the final composite. Such Parameters comprise of a pulling speed or velocity of the fiber, pressure temperature and viscosity of the resin. Such properties determine the appropriate resin for achieving full impregnation of fiber bundles, this uniformity is critical to create the maximum mechanical properties for the composite, while incomplete impregnation leads to voids and loss of strength and durability [217]. Though, achieving optimal impregnation remain challengeable, the viscosity of the molten polymer as well as the pulling speed of consecutive fibers and applied pressure behave in complex interactions that often produce compromises, high fiber velocities lead to higher production rates but on the other hand they do not allow enough contact time between fibers and resin which often results in poor impregnation quality, Improper velocity or pressure can cause voids and therefore poor adhesion, either of which reduces composite performance [76, 218-220]. However, the pulling speed of the fibers plays an important role in pressure distribution and impregnation quality. At high velocities, the pressure gradient needed to push resin through the fiber bundle is usually much higher. A higher pressure gradient like this means less time for effective resin penetration, often resulting in incomplete impregnation [221]. Thus, optimal velocity is not just about maximizing the speed to produce but balancing that with enough time for impregnation. It has been shown that the pulling velocities should be moderate allow sufficient time for resin penetration of the fiber bundle and composites with better mechanical properties can be obtained.

Although Darcy´s Law is an effective model for porous media flow, it becomes inadequate for the simulation of highly viscous fluids under conditions of varying pressure gradients, molten thermoplastic polymers display complex rheological behavior that cannot be accurately represented by a linear model relating flow rate and pressure gradient. However, in practical use, the resistance to flow within a thermoplastic composite is not constant and depends on deformation rate (pressure difference), as well as upon temperature and fiber orientation changes. This is because Darcy's Law assumes constant permeability, ignoring the phenomenon that molten polymer may nonlinear permeate into fiber matrix. However, under higher pressure gradients fiber bundles can compact and this will change the permeability and resistance to flow of the material (this intrinsic limitation requires modifications on the original law that accounts for nonlinear flow behavior of high-viscosity thermoplastic resins) [222, 223].

Shi H et al. [224], analyzed void formation in thermoplastic composites during the welding process, their results showed that higher processing temperature and pressure increased void formation, particularly at high pulling speed, adjusting these parameters reduced void contents and improved bonding strength. Wiedmer et al. [155], focused on pultruded CF/ PA12 and conducting an experimental process at different pulling speeds from 100 to 400 mm/min and find out that with increase the speed the voids content also increasing. Ren et al. [76], developed model with the help of Darcy law and Reynolds equation to show the pressure distribution, and optimize the melt impregnation parameters temperature and pin configuration but did not calculate the pressure at different velocity. Heflin et al. [5], investigated the effect of fiber alignment and processing conditions on void formation in thermoplastic composites, their results showed that slower fiber pulling speed significantly reduced void content due to enhanced resin infiltration which lead to improve the impregnation quality. Alsinani et al. [225], found that increasing the pultrusion speed from 50 to 1000 mm/min resulted in higher polymer pressure within the die/mold but did not use mathematical model and MATLAB to calculate it. Zhang et al. [80], developed a slot die and impregnation model for continuous fiber-reinforced thermoplastic composite UDtapes, their study showed that optimizing die design parameters such as; fiber velocity and resin flow rate lead to significant improvement in impregnation quality, the results showed that slower velocities achieved lower void content and uniform pressure distribution across the tape width. Yifei et al. [81], investigated the effect of melt impregnation process parameters on the permeability of fibers bundles, their study highlighted that lower velocities and moderate resin viscosity improved resin flow and reduced void formation within the fiber bundles. Chakraborty et al. [226], explored non-Newtonian fluid flow in porous media relevant for understanding resin flow in thermoplastic composites, their study highlighted how resin viscosity impacts permeability under varying pressures. Steggall et al. [227], developed a model for the consolidation phase during compression molding of continuous fiber thermoplastic matrix composites, their research demonstrated that higher consolidation pressures and longer dwell times reduce void content, enhancing composite quality. Song et al. [228], investigated the impregnation of ε-caprolactam with polyamide 6 oligomer in thermoplastic resin transfer molding, they found that regulating key parameters such as contact angle, surface tension and viscosity is crucial for minimizing void content and achieving uniform impregnation. Baochao et al. [229], analyzed the intra-yarn impregnation in commingled yarn thermoplastic composites, their study highlighted the influence of pressure and temperature on impregnation quality, noting that optimal conditions reduced void content within the composite structures. Huang et al. [230], conducted a multiphysics coupling study on the impregnation process of hot-melt resin in fiber fabrics, their research providing insights into optimizing the flow process to enhance resin impregnation uniformity and adequacy, thereby reducing void contents.

While previous research [5, 76, 80, 227, 231, 232], has explored the relationship between fiber velocity and impregnation quality, these studies mainly used Darcy law model, which did not capture the complex effects of velocity on pressure distribution in wedge area. However, this study introduces Nonlinear Darcy flow mathematical model to analyze how fiber velocity affect pressure distribution in melt impregnation wedge area of glass fiber reinforced polypropylene composites. Furthermore, MATLAB simulation were used to calculate the pressure in wedge area, while SEM analysis was used for void content and impregnation quality at different fiber velocity.

4.2 Materials and Methods

4.2.1 Materials

In this research work, the thermoplastic polymer used was polypropylene (PP), specifically the (PBM100-GH) grade from Yangzi Petrochemical Company along with antioxidant to prevent degradation during processing. The antioxidant included Irgafos 168 (0.15%) and Irgafos 1010 (0.15%) [169, 233], which were added to the PP polymer. The glass fiber employed was a 2400 Tex direct wound fiber product, specifically developed for melt impregnation compounding processes by JUSHI Group CO. LTD. The fiber consisted of 4400 sized glass monofilaments, each with a diameter of 17 μm.

4.2.2 Fabrication of Glass fiber-reinforced Polypropylene tapes

The experimental setup utilized a custom-designed twin-screw melt impregnation machine, which allowed for precise control over the experimental parameters, as illustrated in Figure 4-1. The impregnation mold temperature was maintained at 250 °C to ensure that the PP polymer was fully molten and capable of effectively impregnating the glass fiber bundles. Fiber velocities were varied at 0.5, 0.8, 1 and 1.5 m/min with corresponding polymer flow rates of 0.670, 0.956, 1.340 and 1.912 kg/h, respectively as shown in Figure 4-2, to maintain a consistent polymer supply throughout the experiments. Fiber tension was held constant at 10 N to ensure uniformity across trials, with the fibers released from a creel.

The glass fiber were first supplied from a creel, then passed through a mold where the twin-screw extruder was used to the molten polypropylene by extrusion and impregnate it into the fibers. After impregnation the fiber tapes were passed through series of rollers for cooling and stabilize the matrix around the fibers and minimize the void formation [144, 173, 174].

4.2.3 Characterizations

Scanning electron microscopy (SEM) was used to study the morphological structure of the glass fiber-polypropylene matrix interface at different fiber velocity impregnation. The SEM study concentrated on the quality of fiber wetting, distribution of matrix and interfacial voids which enlightened on the effect of varying velocity on uniform impregnation.

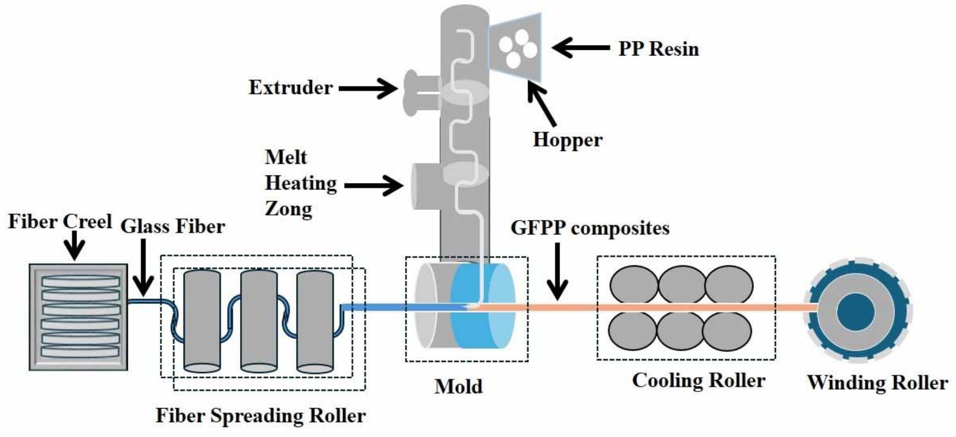

Figure 4-1 Schematic flow diagram of melt impregnation machine

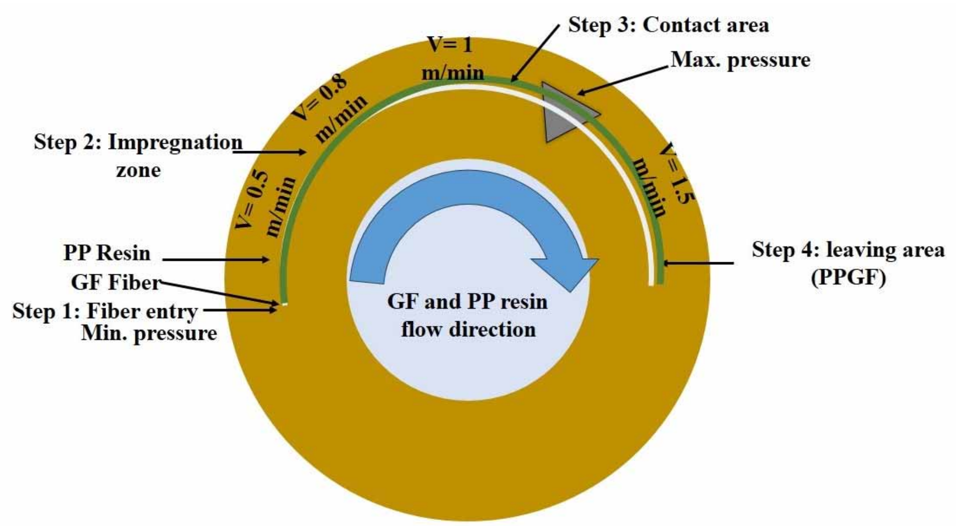

Figure 4-2 Impact of fiber velocity in melt impregnation wedge area

4.2.4 Mathematical Model

4.2.4.1 Assumptions

Nonlinear Permeability: The permeability “k” of the fiber bundle varies in a nonlinear manner with changes in the pressure gradient “∇P”

This assumption allows the model to capture how fiber bundles might respond differently under different pressure levels. In real conditions, higher pressure can affect the fiber structure, which in turn influences permeability. Accounting for nonlinearity in permeability makes the model more accurate in predicting behavior under various loads, especially at higher pressures. Nonlinear permeability accounts for fiber comparison at higher pressure gradient which reduces effective flow paths for molten polymer infiltration, this adjustment improves the model accuracy in predicting real-world impregnation behavior under varying fiber velocities.

Constant Viscosity: The molten polymer's viscosity "μ" is assumed to stay constant.

This keeps the model manageable by assuming that viscosity does not change with shear rate and temperature within the experiment.

While this assumptions simplifies the model, we recognize that viscosity with temperature and shear rate in real world scenarios, if these variation were included the results would predict more dynamic pressure distribution, for instance reduced viscosity at higher shear rate or temperature could lower the pressure gradient required for polymer flow enhancing impregnation at moderate fiber velocity.

Steady-State, Incompressible Flow: The flow remains steady (unchanging over time) and incompressible, meaning density stays constant.

This simplifies the modeling process, assuming that flow conditions are stable and density is uniform throughout the fiber bundle. If there are density variations due to temperature changes, the model may not capture these effects, potentially reducing accuracy in such setups.

Uniform Fiber Distribution: Fibers are uniformly distributed across the bundle.

This ensures that permeability can be treated as a uniform property throughout the fiber bundle. However, in practical applications, even minor inconsistencies in fiber arrangement could impact flow locally. So, this model is more applicable to experiments with tightly controlled fiber distributions.

No Slip Condition: The molten polymer fully adheres to the fiber surfaces, meaning there is no slip at the interface. This maximizes fiber-matrix interaction in the model. However, if there is slip perhaps with specific polymers or fibers treated to reduce adhesion the assumption could lead to overestimated impregnation quality. Thus, this model suits fiber-resin pairs where no slip is realistic.

However, deviations from no-slip condition such as fiber coating or low surfaces can reduce fiber matrix interaction altering pressure perdition and resin infiltration dynamics. Similarly, non-uniform fiber distribution can create localizes pressure variations leading to uneven resin flow and impregnation.

Rigid Fiber Structure: Fibers are considered rigid, without deformation under applied pressure. This implies that any changes in flow are due to polymer behavior rather than fiber deformation. For fiber materials prone to deformation under pressure, this could underestimate void formation or impregnation differences, limiting accuracy in certain cases.

Pressure Gradient: The polymer flow through the fiber bundle is driven entirely by the pressure gradient "∇P",

This keeps the model focused solely on pressure-driven flow, ignoring other potential forces. In applications where gravitational effects or other forces are relevant, the model may require additional terms to account for these factors.

Constant Temperature: The temperature remains constant throughout the experiment.

It is assumed that the temperature does not change, allowing viscosity and other properties that are highly dependent on temperature to remain stable, and simplifying the equations used. In contrast, either temperature fluctuation based on application or viscosity changes influence the quality of impregnation by decreasing. Thus, the model might need to be adjusted in order for results to be accurate under these varying conditions.

Velocity as the Only Variable: The variation is done only in velocity "U", keeping other parameters such as temperature, and viscosity constant.

This facilitates the controlled investigation of the impact on impregnation quality caused only by velocity, which is useful for experiments that isolate its effects. In most applications, when multiple factors change at once, the model would need some changes to be made for forming interaction amongst variables.

Understanding nonlinear Darcy's law

Providing a great theoretical background, Darcy's Law explains the flow rate in terms of driving force (in our case pressure) and resistant faced by fluid that is a function of permeability and viscosity. This is commonly used in some applications related to fluid flow through porous media, and specifically in the case of melt impregnation of thermoplastic composites.

In melt impregnation, a polymer melt have (thick fluid) to perfuse inside the fiber bundle. The demanding variables are: The stress using to push the melt through the fibers.

The basic Darcy's Law equation tells us that the flow rate “q” will increase if: The pressure gradient “∇P” increases (we push harder). The permeability “k” increases (the fibers are less resistant). The viscosity “μ” decreases (the melt is less thick).

Darcy's Law is a fundamental equation that describes how a fluid flows through a porous medium. In its simplest form, it is written as:

$$\mathrm{q} = -\frac{k}{\mu}\nabla \mathrm{P} \qquad (4-1)$$

Whereas; “q” is the Darcy velocity, which is the flow rate per unit area of the fluid through the porous medium (measured in meters per minutes, m/min). “K” is the permeability of the porous medium, which tells us how easily the fluid can flow through it (measured in square meters, m^2). “μ" is the dynamic viscosity of the fluid, which is a measure of the fluid's resistance to flow (measured in Pascal-seconds, Pa·s). “∇P” is the pressure gradient, representing how much the pressure changes over a certain distance within the porous medium (measured in Pascal’s per meter, Pa/m).

Why Nonlinearity?

In many practical situations, especially with thick fluids like molten polymers, the relationship between the pressure gradient and the flow rate isn't straightforward (linear). This non linearity can happen due to various reasons, such as: The complex interaction between the fluid and the fibers. Changes in the fluid properties at different flow rates and to capture this behavior, we modify Darcy's Law to include a nonlinear relationship.

Modifying Darcy's law

We introduce a nonlinearity into the permeability “k” by making it a function of the pressure gradient “∇P”. The modified equation looks like this:

$$\mathrm{q} = -\frac{k(\nabla \mathrm{P})}{\mu}\nabla \mathrm{P} \qquad (4-2)$$

Whereas “k (∇P)” represents the permeability that changes depending on the pressure gradient. This allows us to account for the fact that the resistance offered by the fibers might change as the flow rate or pressure changes.

Incorporate nonlinear permeability

To make the model practical, we need to define how “k(∇P)” changes with “∇P”. Common choice for modeling this nonlinearity is to use a power-law relationship:

$$k(\nabla \mathrm{P}) = k_0(1 + \alpha(\ |\nabla \mathrm{P}|)^{\ n-1}) \qquad (4-3)$$

Whereas; “k0” is the base permeability, which is the permeability when the pressure gradient is small (measured in m²). “α” is a dimensionless coefficient that controls the strength of the nonlinearity. “P0” is a reference pressure, which helps normalize the pressure gradient (measured in Pa). “n” is the flow regime exponent, which determines how strongly the permeability depends on the pressure gradient.

Why we use this model?

This model is flexible and can capture different types of nonlinear behavior.

Let’s suppose:

If n=1, the permeability doesn't change with the pressure gradient (linear case).

If n >1, the permeability increases as the pressure gradient increases (which might happen if the fibers "open up" under high pressure).

If n <1, the permeability decreases as the pressure gradient increases (which might happen if the fluid becomes more resistant under pressure).

Substituting the permeability model

We now substitute the nonlinear permeability model back into Darcy's Law:

$$q = -\frac{k_0(1 + \alpha(\ |\nabla \mathrm{P}|\ )^{\ n-1})}{\mu}\nabla \mathrm{P} \qquad (4-4)$$

This equation tells us how the flow rate depends on the pressure gradient when the permeability is nonlinear. The term inside the parentheses, adjusts the permeability based on the pressure gradient.

Mass conservation: The continuity equation

Why Use the Continuity Equation?

In any fluid flow problem, we need to ensure that mass is conserved that is, what goes in must come out unless it's accumulating somewhere. The continuity equation is the mathematical expression of this principle.

$$\nabla . \mathrm{q} = 0 \qquad (4-5)$$

This equation states that the divergence (or spread) of the flow rate “q” is zero, meaning the amount of fluid entering any region is equal to the amount leaving it.

Substituting the nonlinear flow rate

We substitute our expression for "q" from the modified Darcy's Law into the continuity equation:

$$\nabla.\left(-\frac{k_0(1+\alpha(\mid\nabla\mathrm{P}\mid)^{n-1})}{u}\nabla\mathrm{P}\right)=0 \tag{4-6}$$

Simplifying the continuity equation

$$-\frac{k_0}{u}\nabla.[\ (1+\alpha(\mid\nabla\mathrm{P}\mid)^{n-1})\nabla\mathrm{P}]=0 \tag{4-7}$$

Since "k_0" and "μ" are constants, they can be moved outside the divergence operator:

$$\nabla.[\ (1+\alpha(\mid\nabla\mathrm{P}\mid)^{n-1})\nabla\mathrm{P}]=0 \tag{4-8}$$

This is the continuity equation for our nonlinear flow model. It ensures that the mass conservation principle is respected, even with a nonlinear permeability.

Combine pressure gradient to velocity: The momentum balance

The momentum balance equation relates the pressure gradient to the velocity of the fluid. It's based on Newton's second law (force equals mass times acceleration) but adapted for fluid flow in a porous medium.

For steady-state, incompressible, and laminar flow, the momentum balance in the porous medium is simplified to:

$$\nabla\mathrm{P}=-\frac{\mu}{k(\nabla\mathrm{P})}\,\mathrm{u} \tag{4-9}$$

Whereas; "u" is the velocity of the fluid.

Substituting the Nonlinear Permeability

We substitute our expression for "k (∇P)" into the momentum balance equation:

$$\nabla\mathrm{P}=-\frac{\mu}{k_0(1+\alpha(\mid\nabla\mathrm{P}\mid)^{n-1})}\,\mathrm{u} \tag{4-10}$$

This equation now relates the pressure gradient directly to the velocity of the fluid, taking into account the nonlinear effects of the permeability.

4.3 Results and Discussion

The SEM analysis of the glass fiber polypropylene (GFRPP) composites at various fiber velocities such as, 0.5 m/min, 0.8 m/min, 1 m/min and 1.5 m/min, it shows that glass fiber velocity has a significant impact on impregnation quality.

At the low fiber velocity of 0.5 m/min, show thick polymer infiltration with minimal voids, indicating strong adhesion between the polymer matrix and the glass fibers as shown in Figure 4-3.

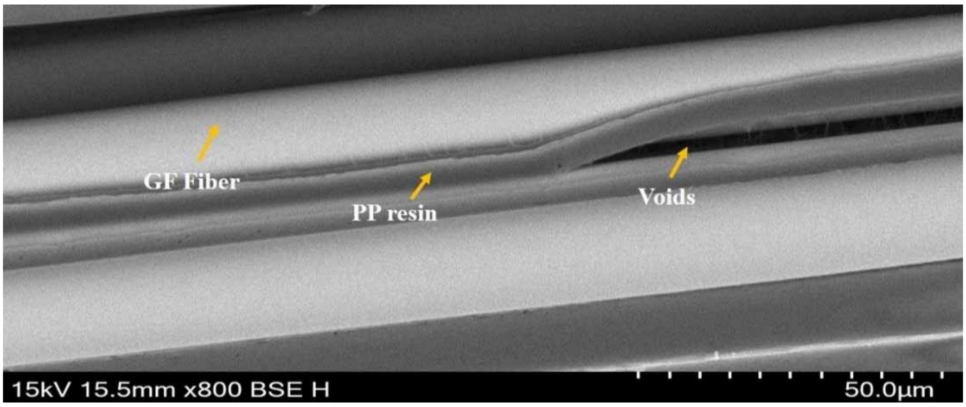

Figure 4-3 Morphological structure of GFPP at 0.5 m/min fiber velocity

If the polymer viscosity were lower, it would further enhance impregnation by reducing the pressure gradient required for polymer flow allowing better penetration even at higher fiber velocities. Conversely larger fiber diameter might hinder impregnation due to a reduced surface area to volume ratio, which could necessitate adjustment in pressure or fiber velocity to ensure uniform wetting and bonding

As shown in Figure 4-4, as the fiber velocity increases to 0.8 m/min, there is a slight increase in void content, though impregnation quality remains relatively effective.

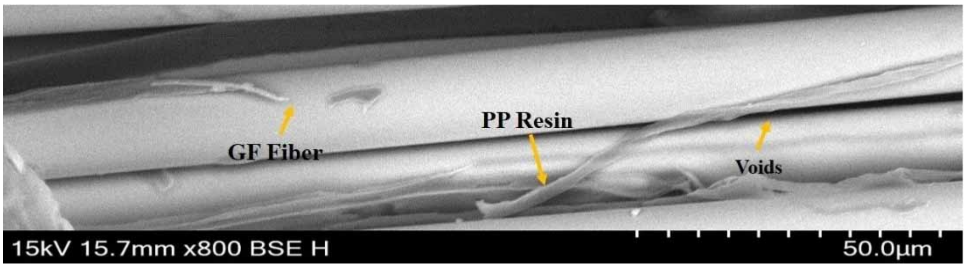

Figure 4-4 Morphological structure of GFPP at 0.8 m/min fiber velocity

However, at the highest fiber velocity 1 m/min and 1.5 m/min, void content becomes more prominent, as shown in Figures 4-5 and 4-6, showing compromised impregnation quality due to limited polymer infiltration time [155, 221, 225], where fiber velocity are inversely related to void content, with lower fiber velocity allowing more complete impregnation and reduced voids.

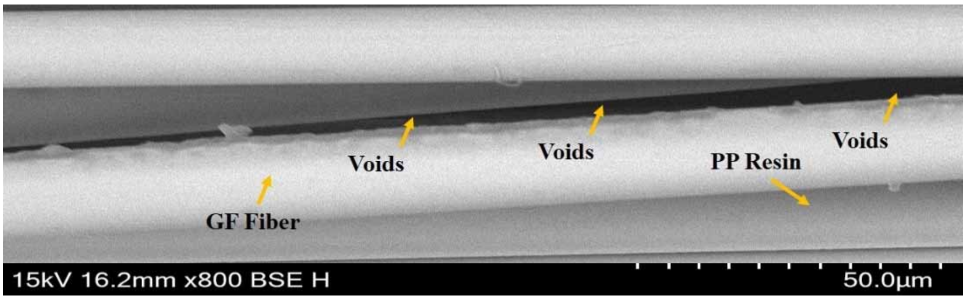

Figure 4-5 Morphological structure of GFPP at 1 m/min fiber velocity

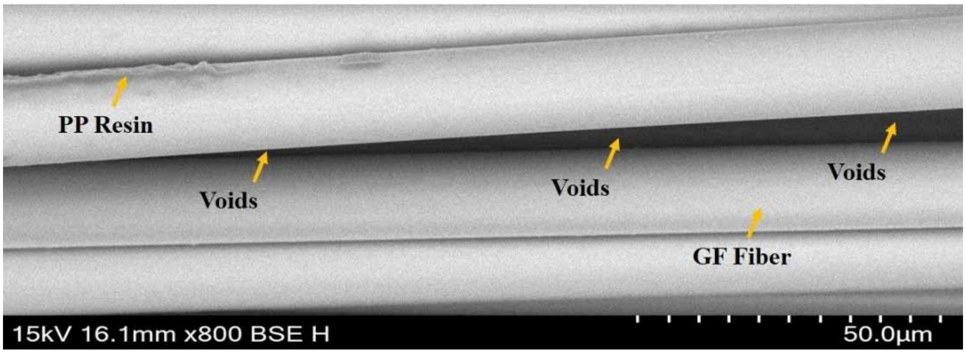

Figure 4-6 Morphological structure of GFPP at 1.5 m/min fiber velocity

According to the Nonlinear Darcy Flow model, the fiber velocity has a direct impact on the pressure gradient required to achieve impregnation. As defined by equation 4-1, a higher fiber velocity requires a greater pressure gradient across the glass fibers to sustain polymer flow through the fibers. When fiber velocity increases, the system compensates by increasing the pressure gradient, but this creates a shorter contact time for the molten PP polymer within the glass fiber. As a result, although the PP polymer moves faster, it lacks sufficient time to fully impregnate each glass fiber tows, compromising the impregnation quality as shown in above Figures 4-5 and 4-6.

Studies by [225], support this model prediction by demonstrating that as fiber velocity increases, pressure also rises, but often at the expense of impregnation quality due to decreased contact time, resulting in lower impregnation quality. The Nonlinear Permeability in equation 4-2, further illustrates how permeability decreases as the pressure gradient rises, particularly when, n <1 at high fiber velocity, this increased pressure gradient reduces effective permeability, making it more challenging for the polymer to infiltrate the fibers.

Lower fiber velocity, however maintain a more intermediate pressure gradient so there is sufficient permeability and full impregnation as shown in Figure 4-3, and have minimum void content, it show that empirical evidence for model predictions that validate the nonlinear Darcy flow approach to capture relationship between velocity and pressure in impregnation quality. As expected, that the model correctly predicts moderate pressure gradients that maintain permeability and allow polypropylene resin to penetrate into the glass fiber bundles for complete impregnation with low void content at lower velocities. The GFPP SEM Figures validate well with the equations 4-2, of the model, which states that smaller pressure gradient maintains higher effective permeability and more promising fiber wetting and bonding. Nonlinear Darcy flow model validate with GFPP SEM Figures provide an actual structure for understanding and controlling the impregnation behavior in thermoplastic composite material.

Using an exponential nonlinearity in the model could enhance its ability to capture more rapid changes in permeability with pressure gradient. This approach may better represents the physical behavior of molten polymer flow under high-pressure conditions, particularly at higher fiber velocities, where permeability and pressure interactions becomes more complex. The results could show sharper sensitivity in pressure distribution highlighting more distinct optimal operating points for impregnation. While the power law model provides smooth transitions, the exponential model might offer better predictive accuracy for extreme velocity ranges supporting more previse optimizing of impregnation quality and production speed.

Comparison of MATLAB simulation results, SEM analysis and mathematical model predictions simplifies a detailed understanding of the impregnation quality in thermoplastic composites for various velocity. Each method presents its own particular perspective and the validation of our theoretical predictions, in addition to focusing on particular conditions that favor melt impregnation.

The MATLAB simulation are highly sensitive to porosity “ε_0” and moderately sensitive to the Carman-Kozeny coefficient “K_0” as both directly influence permeability and pressure gradient. Changes in porosity raised to the 3rd power in permeability calculations, have larger effect especially at higher fiber velocities where nonlinearities amplify.

The MATLAB as shown in Figures 4-7 to 4-10, simulated plots of pressure distribution across the melt impregnation wedge area at different velocities (0.5 m/min, 0.8 m/min, 1.0

m/min and 1.5 m/min), it show that with increasing fiber velocity speed, the change in pressure required for driving polymer flow also increases with increase in fiber velocity with the lowest speed, 0.5 m/min as shown in Figure 4-7, pressure is not very high where around 0.8 MPa and this indicates low pressure gradient to provide longer contact time between polypropylene and glass fiber. Table 4-1, summarize the key parameters of used in MATLAB simulations for pressure distribution across the wedge area, including fiber velocity, temperature and polypropylene viscosity.

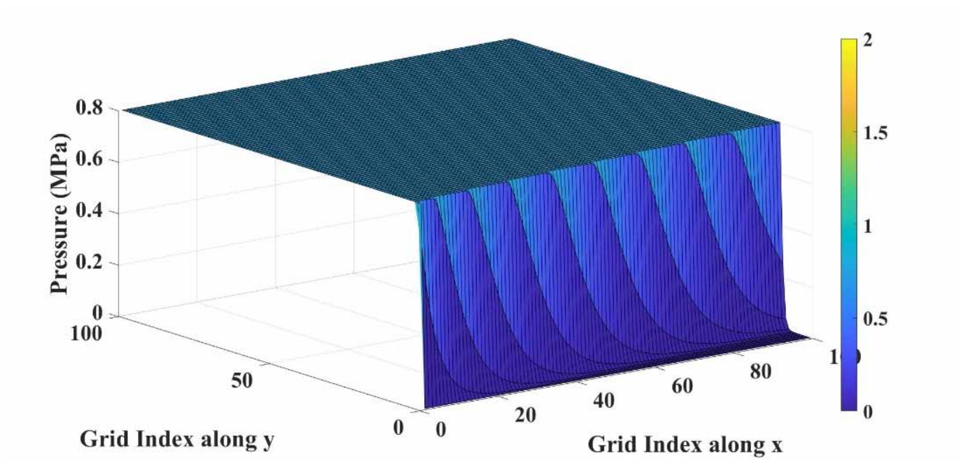

Figure 4-7 Pressure distribution in wedge area at 0.5 m/min fiber velocity

As shown in Figure 4-8, low fiber velocity and pressure makes it very suitable for thorough impregnation. However, increasing pressure of approximately 1.2 MPa when velocity around 0.8 m/min.

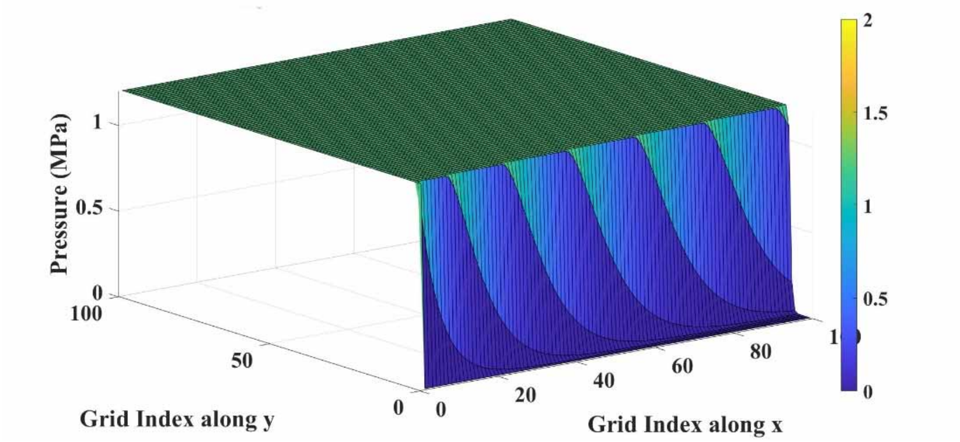

Figure 4-8 Pressure distribution in wedge area at 0.8 m/min fiber velocity

Increase of pressure up to 1.0 m/min and 1.5 m/min again continues by increasing the fiber velocity it reaches pressures of further 1.5 MPa and 2.0 MPa, as shown in Figure 4-9 and 4-10.

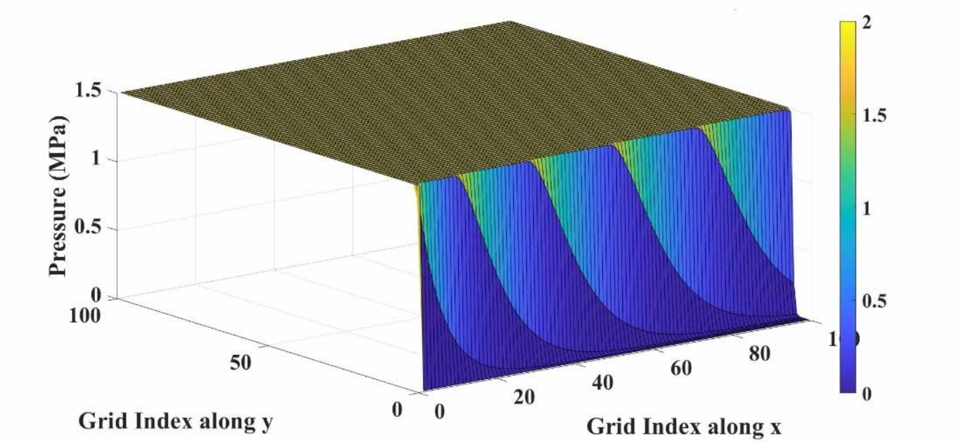

Figure 4-9 Pressure distribution in wedge area at 1 m/min fiber velocity

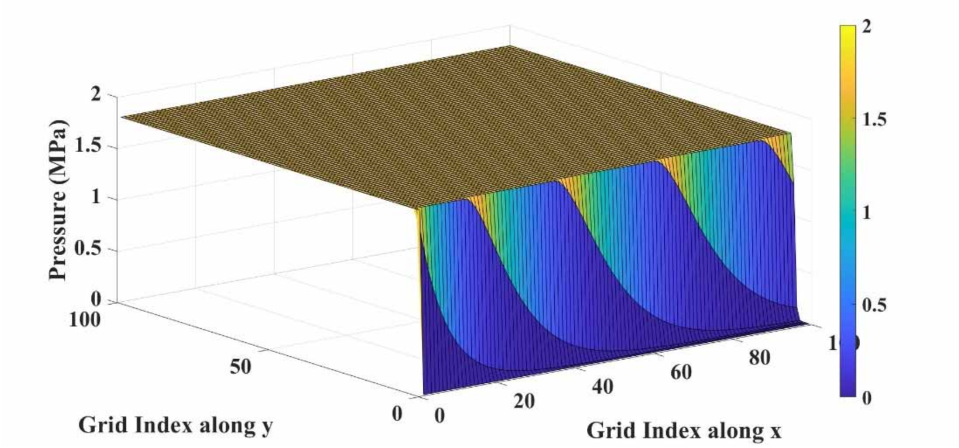

Figure 4-10 Pressure distribution in wedge area at 1 m/min fiber velocity

Respectively, the higher pressures gradients suggest poor contact time for the polypropylene polymer to be able to penetrate successfully into the glass fiber bundles, which would in turn lead to lower quality of impregnation. Results in MATLAB indicate that the impregnation is not better for higher velocities due to the polypropylene polymer not have enough time to coat into the glass fibers.

The nonlinear Darcy flow model offers theoretical support for these observations, specifically through the mathematical framework provided by equations 4-1 and 4-2. According to the model, as velocity increases, a higher pressure gradient is required to maintain polymer flow through the fiber bundles, equation 4-1, captures this relationship, predicting that increasing fiber velocity requires a higher pressure gradient. Equation 4-2, which introduces the nonlinear permeability Function, further explains that this rising pressure gradient reduces permeability, which in turn limits the polymer's ability to infiltrate the glass fiber bundles. The mathematical model thus suggests that lower fiber velocity, such as 0.5 m/min, are optimal because they maintain a moderate pressure gradient, preserving permeability and enabling effective impregnation. As the fiber velocity rises, the model predicts a reduction in impregnation efficiency due to the decreased permeability and reduced contact time between the PP polymer and GF. These predictions align closely with the results observed in both the MATLAB simulations and SEM analysis, validating the model's accuracy.

The combined insight from the MATLAB simulations, SEM analysis, and model predictions leads to a clear conclusion regarding optimal impregnation conditions. The data from all four parameters consistently suggest that 0.5 m/min is the ideal fiber velocity for melt impregnation in thermoplastic composites, at this fiber velocity, the moderate pressure gradient allows sufficient contact time, resulting in dense polymer infiltration and strong glass fiber polypropylene matrix bonding. To balance impregnation quality and production speed intermediate fiber velocities can provide a suitable compromise between sufficient resin impregnation time and higher productivity. Enhancing resin flow properties through additives, slightly higher temperature or resin with improved viscosity characteristics can support better infiltration at higher velocities. Optimizing pressure distribution by redesigning the wedge or die geometry and employing dynamic pressure control system ensures uniform flow and minimizes voids. A multi stage impregnation process with partial infiltration at higher velocity followed by slower final impregnation can further improve quality. Real time monitoring system with sensors and feedback loops can dynamically adjust parameters like fiber velocity, pressure and resin flow for optimal performance. Additionally, modifying fiber bundling and surface treatment can enhance resin infiltration while simulation can fine tune key variables such as porosity and flow rate to achieve efficient production without compromising quality. This result is supported by SEM Figures showing minimal voids and robust bonding, while the mathematical model explains this result theoretically through the balance between pressure gradient and permeability. Higher velocities, such as 0.8 m/min, 1.0 m/min, and 1.5 m/min, result in higher pressure gradients and reduced contact time.

The findings of this study highlight the importance of selecting an optimal fiber velocity to achieve high-quality impregnation in GFPP composites. The nonlinear Darcy flow model, validated by both experimental SEM analysis and MATLAB simulation data, provides a reliable framework for predicting impregnation results based on fiber velocity adjustments. By maintaining a low fiber velocity, manufacturers can ensure that the pressure gradient remains moderate, allowing adequate permeability for the polymer to infiltrate the fiber bundles effectively. This approach enhances the mechanical properties of the final composite material, offering superior strength and durability for various industrial applications. In comparison, high fiber velocity may increase productivity but compromise the structural integrity of the composite due to insufficient impregnation.

While the current model provides reliable predictions for fiber velocity, extending the model to include temperature variation poses challenges such as; handling temperature-dependent viscosity, thermal effects on permeability and porosity and heat transfer equation and localized pressure variation from uneven viscosity.

Table 4-1 MATLAB Simulation Parameter

Parameter	**symbol**	**value**	**unit**
Fiber diameter	d_f	17 µm.	m
fiber monofilament	n_f	4400	-
Porosity	ε_0	0.7	-
PP viscosity	η	97.616	Pa.s
Temperature	T	250	°C
Velocity	U	0.5, 0.8, 1, and 1.5	m/min
Melt film thickness	δ	40 µm.	m
Carman kozeny co efficient	k_0	5	-
Non linearity exponent	n	0.8	

4.4 Summary

The aim of the present study is to analyze the effect of various velocity (0.5 to 1.5 m/min) on pressure in wedge melt impregnation for glass fiber reinforced polypropylene (GFPP) thermoplastic composites using Nonlinear Darcy flow mathematical model.

- The results obtained indicate that the lower fiber velocity (0.5 m/min) produce a moderate pressure in wedge area and which improve the polymer polypropylene (PP) penetration by allowing sufficient contact time between melt PP polymer and glass fiber.
- Compared to lower velocity, higher velocities lead to increases the pressure in wedge area that reduces the impregnation quality due to limited penetration time.
- The MATLAB simulation confirmed that the model predictions, indicating an increase in pressure with velocity, especially at velocities above 1 m/min.
- This comprehension aligns with SEM analysis, which exposed minimal voids at low velocities, confirming that lower fiber velocity optimizes impregnation quality by maintaining permeability and sufficient pressure distribution.
- These findings can guide manufactures in optimizing fiber velocity to balance production efficiency and impregnation quality, particularly in aerospace and automotive applications.

Chapter 5 Impact of Mold Temperature and Fiber velocity on Impregnation in Carbon Fiber Polyamide 66 Composites

5.1 Introduction

The manufacturing of thermoplastic composites has become a critical area of research due to their growing applications in industrial sectors. Thermoplastic matrices such as; Polyamide 66 are extensively used for fiber reinforced plastic composites owing to their excellent mechanical strength, light weight properties, low cost and superior thermal stability [234, 235]. However, the impregnation process for thermoplastic is highly demanding due to their high temperature and pressures required to reduce resin viscosity for effective fiber penetration [236, 237], compared to thermosetting composites thermoplastic matrices often have less favorable impregnation conditions which can lead to void contents, reduced composite quality and weak fiber – matrix bonding [238, 239]. Carbon fiber have emerged as a leading reinforcement material in thermoplastic composites due to their outstanding mechanical and chemical resistance properties coupled with their recyclability [240]. Achieving high quality impregnation requires balancing processing parameters, such as; pressures, temperature and fiber velocity. High temperature reduces resin viscosity enhancing resin flow and fiber wetting, though continued exposure risks thermal degradation of matrix and reinforcement [152, 241], Conversely, high fiber velocities while improving production throughput reduce resin resistance time within fiber bed leading to incomplete impregnation and increased void contents [153, 242].

Modeling resin flow during the impregnation process is critical for optimizing these parameters, Darcy law is the standard approach for describing fluid flow in porous media like fiber bundles, while effective for low fiber velocity, Newtonian fluids it fails to account for the non-Newtonian behavior of thermoplastic melts such as shear thinning and viscoelasticity [156, 238]. Mukherjee et al. [156], shows that viscoelastic effect and temperature gradients could significantly enhance dispersion in thermoplastic melt flow, highlighting the limitations of traditional flow models. Hu et al. [152], studied the relationship between temperature and viscosity in polyether ether ketone (PEEK) composites showing that while moderate temperature enhances resin flow and impregnation, excessively high temperature reduced viscosity due to thermal crosslinking compromising mechanical properties.

Experimental optimization has also focused on the effect of temperature and velocity on impregnation quality. Tipboonsri et al. [153], optimized the pultrusion process for jute and glass fiber reinforced polypropylene composites, finding that increasing temperature improved resin flow and fiber matrix bonding up to a critical threshold, beyond which void content increased due to fiber degradation. Similarly, Steggall et al. [147], emphasized the importance of temperature control to minimize deconsolidation and surface defects during pultrusion of thermoplastic composites.

Despite these advancement, significant challenges persist, existing model often oversimplify the complex interactions between flow dynamics thermal gradients and mechanical deformation during thermoplastic melt impregnation [144]. Most studies fail to integrate non-Newtonian rheological behavior into fluid flow models, which is critical for accurately predicting impregnation quality [144, 173].

The impregnation of thermoplastic composites remains a challenging process due to the complex interplay of temperature, fiber velocity and resin properties, current models, Darcy law inadequately describes the non-Newtonian behavior of thermoplastic melts including shear-thinning and viscoelastic effects. Furthermore, while high temperature improve resin flow they pose a risk of thermal degradation, while high velocities lead to incomplete impregnation and void contents [152, 153, 242].

However, the aim of this study addresses these gaps by developing an advanced multi-physics model that integrates the Carreau-Yasuda viscosity model with Darcy law to accurately predict resin flow, pressure distribution and impregnation quality. Experimental validation using PA66 – carbon fiber composites under varying temperature and velocity conditions refine the mathematical model, by linking the gap between theoretical modeling and practical application, this research seeks to optimize processing parameters ensuring the production of high performances thermoplastic composites with enhanced carbon fiber-polyamide (66) (CFPA66) matrix adhesion and minimal void content.

5.2 Materials and Methods

5.2.1 Materials

In this research work, the thermoplastic polymer polyamide 66 (PA66), specifically the Zytel® 101L NC010 grade from DuPont, was chosen due to its excellent mechanical strength and thermal stability. To improve the stability of the matrix during high-temperature processing, antioxidants Irgafos 168 (0.2%) [243, 244], and Irgafos 198 (0.2%) [245], were added to prevent oxidative degradation. The carbon fiber used was an 800 Tex product from Toray, specifically developed for high-performance applications, the fiber consisted of 12,000 carbon monofilaments, each with a diameter of 8 μm.

5.2.2 Preparation of Carbon Fiber Reinforced Polyamide 66 Samples

The CFPA66 samples were prepared using a twin-screw melt impregnation machine, as shown in Figure 5-1. The temperatures of the impregnation mold were set as 320 °C, and 330 °C, and fiber velocities were set at 1.5 m/min and 2 m/min and the polymer feed rate was set at 0.672 and 0.896 kg/h to ensure stable resin flow and matrix distribution in order to investigate how the processing conditions affect the resin impregnation as shown in Figure 5-2. The constant polyamide 66 resin to carbon fiber weight ratio was kept at 70% and 30%, and the fiber tension was kept at 10 N for all experiments.

In preparation, carbon fibers were fed from a creel into a mold where molten polyamide 66 resin was uniformly distributed into the carbon fiber bundles through a twin-screw extruder. The resulting impregnated tapes were cooled by rollers to stabilize the composite structure for further analysis [144, 173, 174].

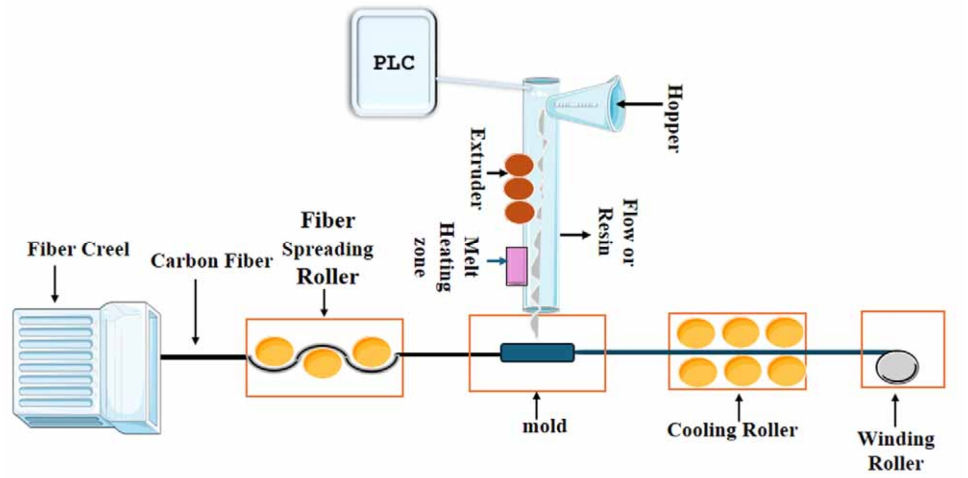

Figure 5-1 Flow diagram of melt impregnation machine

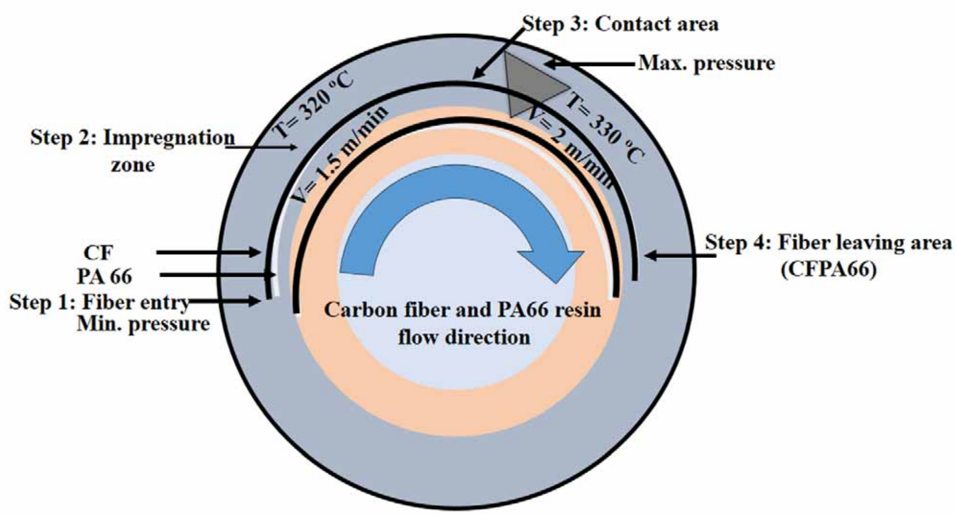

Figure 5-2 Impact of temperature and Velocity in wedge area of melt impregnation

5.2.3 Characterization

In this study various characterization techniques were used to evaluate the quality of CFPA66 composites under different processing parameters.

Scanning electron microscopy (SEM) was used to analyze the morphological structures and impregnation quality of CFPA66 composites. SEM shown void content, resin penetration and carbon fiber-polyamide 66 (CFPA66) adhesion, helping to determine optimal processing conditions.

Differential scanning calorimetry (DSC) was used to examine the thermal properties of carbon fiber-polyamide 66 (CFPA66) including crystallization and melting behavior correlating improved impregnation with crystanllinity.

Thermogravimetric analysis (TGA) calculated thermal stability of carbon fiber-polyamide 66 (CFPA66) showing how impregnation quality enhances resistance to degradation.

5.2.4 Mathematical Model Frame work

The model combines Darcy law with the Carreau-Yasuda viscosity model to provide an accurate predictions of resin flow dynamics, pressure distribution and impregnation efficiency during melt impregnation process. This approach addressees the limitation inherent in traditional flow models by incorporating the non-Newtonian characteristics of thermoplastic melts including shear thinning behavior and viscosity dependence on temperature.

5.2.4.1 Mathematical Model

Assumptions

1. Non-Newtonian fluid behavior: The melt exhibits shear-thinning properties, meaning its viscosity decreases with increasing shear rate.

Thermoplastic melts often behave as non-Newtonian fluids under processing conditions. Accurate modeling of viscosity is crucial for predicting flow behavior within the fiber bed.

2. Incompressible fluid: The melt is considered incompressible.

For most practical purposes in polymer processing, the density of the melt does not change significantly under pressure, simplifying the equations.

3. Laminar flow: The flow of the melt through the fiber bed is laminar.

The Reynolds number in melt impregnation is typically low due to high viscosity and slow flow rates, leading to laminar flow conditions.

4. Isothermal process: Initially, assume a constant temperature to simplify the analysis.

Focusing first on flow mechanics without the complexity of thermal gradients allows for initial model development. Temperature effects can be added later.

5. No slip condition: There is no relative motion between the melt and the fiber surfaces.

This is a standard assumption in fluid mechanics that simplifies boundary conditions and accurately represents the interaction between the fluid and solid surfaces.

6. Flow in radial and axial directions: Flow is considered in both radial and axial directions relative to the fiber bundle.

In real-world applications, melt flow within the fiber bed is not unidirectional. Including both radial and axial components provides a more realistic model.

The assumptions of incompressible fluid and laminar flow simplify the complexity of the model making it feasible to solve accurately for this specific resin-fiber process. These assumptions are valid under current conditions, where the processing pressure is not significantly high and Reynolds numbers are low ensuring laminar flow. However for different resin-fiber conditions that may involve higher shear rates, these assumptions could limit model applicability.

Darcy’s law for fluid flow through a porous medium

Darcy's law is fundamental in modeling fluid flow through porous media like the fiber bed in melt impregnation: and it is necessary to describe how the melt flows through the fibrous structure, which is essential for predicting impregnation efficiency and quality. The radial and axial components account for the multi-directional flow within the fiber bed.

$$\mathrm{V} = -\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\nabla\mathrm{P} \qquad (5-1)$$

Now considering radial (r) and axial (z) components in equation (5-1):

$$v_r = -\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\frac{\partial \mathrm{p}}{\partial r}\ , \qquad v_z = -\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\frac{\partial \mathrm{p}}{\partial z} \qquad (5-2)$$

Whereas; “V” is the fluid velocity vector (m/s), $k(\in)$ is the permeability (m^2) of the porous medium, dependent on the deformation strain “ϵ” of the fiber bed, $\mu(\gamma^{\cdot},T)$ is the non-Newtonian viscosity (Pa·s), which depends on shear rate “ $\gamma^{\cdot}$” and temperature “T”, “∇P” is the pressure gradient (Pa/m).

Continuity equation (Mass conservation)

The continuity equation ensures that mass is conserved within the fluid flow

$$\frac{\partial v_r}{\partial r} + \frac{\partial v_z}{\partial z} + \frac{v_r}{r} = 0 \qquad (5-3)$$

The equation 5-3, is a standard form of the continuity equation in cylindrical coordinates, ensuring mass conservation for an incompressible fluid. The second term, $\frac{\partial v_z}{\partial z}$ is consistent with the continuity equation's requirements for cylindrical coordinates,

Whereas v_r *and* v_z are the velocity components in the radial and axial directions, respectively.

Substituting Darcy’s Law equation (5-2) into the continuity equation (5-3) and get:

$$\frac{\partial}{\partial r}\left(-\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\frac{\partial \mathrm{p}}{\partial r}\right) + \frac{\partial}{\partial z}\left(-\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\frac{\partial \mathrm{p}}{\partial z}\right) + \frac{1}{r}\left(-\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\frac{\partial \mathrm{p}}{\partial r}\right) = 0 \qquad (5-4)$$

Simplifying, equation (5-4) to obtain the governing equation for pressure distribution:

$$\frac{\partial}{\partial r}\left(\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\ \frac{\partial \mathrm{p}}{\partial r}\right)+\frac{1}{r}\ \frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\ \frac{\partial \mathrm{p}}{\partial r}+\frac{\partial}{\partial z}\left(\frac{k(\in)}{\mu\,(\gamma^{\cdot},\mathrm{T})}\ \frac{\partial \mathrm{p}}{\partial z}\right)=0 \qquad (5-5)$$

The continuity equation 5-5, is crucial to ensure that the flow rates predicted by the model do not violate the principle of mass conservation, which is a fundamental requirement in fluid dynamics.

Viscosity modeling for non-Newtonian fluids

The viscosity μ (γ˙, T)) of the thermoplastic melt is modeled using the Carreau-Yasuda model:

$$\mu(\gamma^{\cdot}) = \mu_{\infty} + (\mu_0 - \mu_{\infty})\,[1 + (\lambda\gamma^{\cdot})^{a}]^{\frac{n-1}{a}} \qquad (5-6)$$

Whereas; "μ0" the zero-shear viscosity (Pa·s), "μ∞" is the infinite-shear viscosity (Pa·s), "λ" is a time constant (s), "a" and "n"are dimensionless parameters.

For temperature dependence:

$$\mu\,(T) = \mu_0\, exp\left(\frac{E}{RT}\right) \qquad (5-7)$$

Whereas: "E" is the activation energy (J/mol), "R" is the universal gas constant (J/mol·K), "T" is the temperature (K).

The thermoplastic melt's viscosity changes with shear rate, which significantly impacts the flow behavior. Using a model like Carreau-Yasuda equation 5-7, allows for accurately capturing this non-Newtonian behavior, ensuring that the flow predictions align with real observations. This is critical for understanding how the melt behaves under varying shear conditions within the fiber bed.

Boundary conditions

To solve the governing equations, appropriate boundary conditions must be defined:

Inlet Boundary Condition:

Condition: $P_{inlet} = P_0$

The inlet pressure is the starting point for calculating the pressure distribution within the domain. This boundary condition is essential because it drives the flow through the fiber bed, determining how the melt enters the structure.

Outlet Boundary Condition:

Condition: Pressure gradient $\left.\frac{\partial p}{\partial x}\right|_{outlet} = G_{outlet}$

Given that the outlet pressure is difficult to calculate directly, specifying the pressure gradient allows the model to naturally determine the outlet pressure as part of the solution. This approach is flexible and practical when direct measurement or estimation of outlet pressure is challenging.

Wall boundary conditions:

Condition: No-slip “V=0” at the fiber-melt interface.

The no-slip condition accurately represents the interaction between the melt and the fiber surfaces, ensuring that the flow near the boundaries is correctly modeled.

Initial conditions:

Condition: Initial temperature “T(x, y, z, t = 0)” and pressure P(x, y, z, t=0)

These conditions define the state of the system at the beginning of the process, ensuring that the simulation starts from a physically realistic state.

Pressure distribution in the domain

To solve for the pressure distribution, integrate Darcy’s Law:

$$\frac{dP}{dx} = -\frac{\mu}{k} v_z \qquad (5-8)$$

For constant “k” and “μ”:

$$P(x) = P_0 - \frac{\mu}{k}\int_0^x v_z\ (x')dx' \qquad (5-9)$$

The equation (5-9) represents the pressure at any point “x” along the axial direction, derived by integrating the relationship between the pressure gradient and the velocity v_z.

Here, “P_0” is the inlet pressure, and the integral accounts for the cumulative effect of velocity v_z along the distance “x”.

Assumption for axial flow: To further simplify the analysis, we assume steady, fully-developed axial flow, where the velocity "v_z" is constant along the length of the domain. Under this assumption, the pressure gradient becomes "$\frac{dp}{dx}$" a constant, which denote as “G”.

For an axial flow:

$$P(x) = P_0 - Gx \tag{5-10}$$

"G" represents the uniform pressure gradient along the axial direction. This linear relationship between pressure and distance "x" indicates that the pressure decreases uniformly as move along the fiber bed.

Whereas; "G" is the pressure gradient. At the outlet x=L the pressure is:

$$P(L) = P_0 - GL \tag{5-11}$$

The equation 5-11, provides the pressure at the outlet, which is crucial for understanding how the pressure distribution impacts the flow and impregnation quality within the fiber bed.

Velocity Calculation Using Darcy's Law

Now, calculate the velocity:

$$v_z(\mathrm{x}) = \frac{k}{\mu\,(\dot{\gamma})}\frac{d\mathrm{p}}{dx} \tag{5-12}$$

Substitute the pressure gradient "G":

$$v_z(\mathrm{x}) = -\frac{k}{\mu\,(\dot{\gamma})}G \tag{5-13}$$

The equation 5-13, described the velocity calculation is critical for understanding how the melt flows through the fiber bed.

Tension in the Fiber Bundle

To calculate the tension in the fiber bundle:

$$T = \int_0^R \mathrm{P(r)}\ .A(r)\,dr \tag{5-14}$$

Whereas; "A(r)"is the cross-sectional area of the fiber bundle as a function of radius.

The equation 5-14, is important for ensuring the structural integrity of the fibers during the impregnation process.

5.3 Results and Discussion

5.3.1 Comparison of Experimental results of CFPA66 with Mathematical model

The comparison experimental results of PA66 results with mathematical model with the predictions shows a strong correlations, as shown in Figures 5-3 to 5-7. This results confirms the mathematical model precision in explaining the impregnation in fiber reinforced thermoplastics. The sublabels Figures 5-3 to 5-7 indicate different Figures comparison with temperature including viscosity Figure 5-3, shear stress Figure 5-4, storage modulus Figure 5-5, loss modulus Figure 5-6, and torque Figure 5-7.

As shown in Figure 5-3 demonstrating the viscosity, the experimental data closely validate with the Carreau – Yasuda equation 5-6. The observed decrease in viscosity with increasing temperature reflects the expected shear thinning behavior of PA66 thermoplastic melt, a critical factor for effective PA66 resin flow and carbon fiber wetting. The error ranges between 4 to 5 %, it validating the mathematical model ability to describe the non-Newtonian viscosity behavior and its dependence on temperature shear rate.

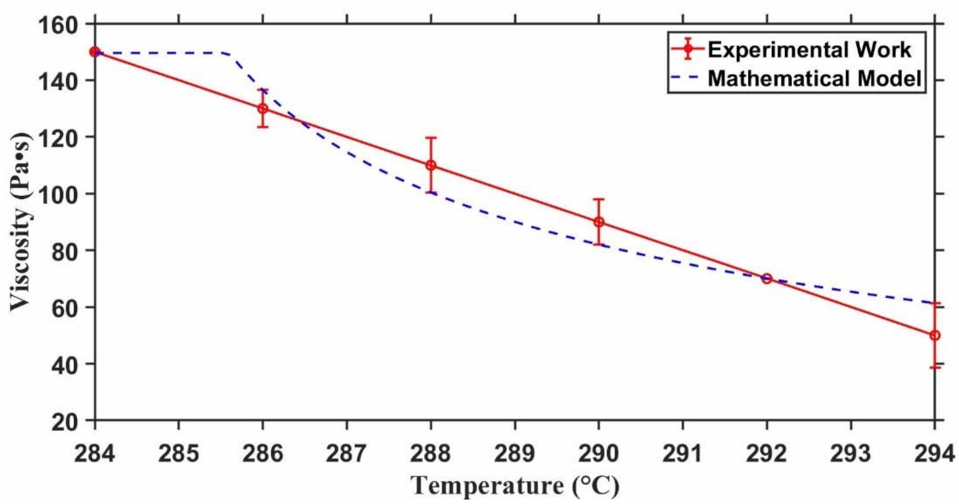

Figure 5-3 Experimental viscosity of PA66 vs. Mathematical model

As shown in Figure 5-4 it indicates the shear stress, which was derived from the pressure gradient and flow velocity as explain in equation 5-13, the experimental and predicated values shows a high degree consistency with error range 2 to 4%, This validate the Darcy law equation 5-2, and pressure distribution equation 5-9, in correctly mathematical modeling PA66 resin flow through carbon fiber under laminar flow conditions.

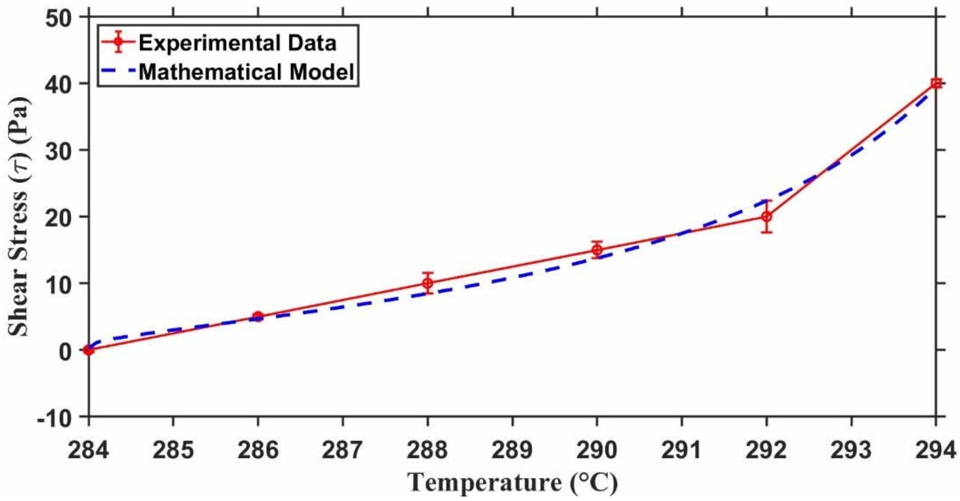

Figure 5-4 Experimental shear stress of PA66 vs. Mathematical model

Figure 5-5 show the storage modulus; it states that the elastic properties of the CFPA66 composite material. This increase in storage modulus with temperature is attributed to the viscoelastic behavior of the PA66 polymer resin which becomes more responsive to external forces during impregnation. The model predictions closely validate with the experimental data with around 3 to 5% error confirming the validity of integrated equations in capturing the elastic response of the CFPA66 composites.

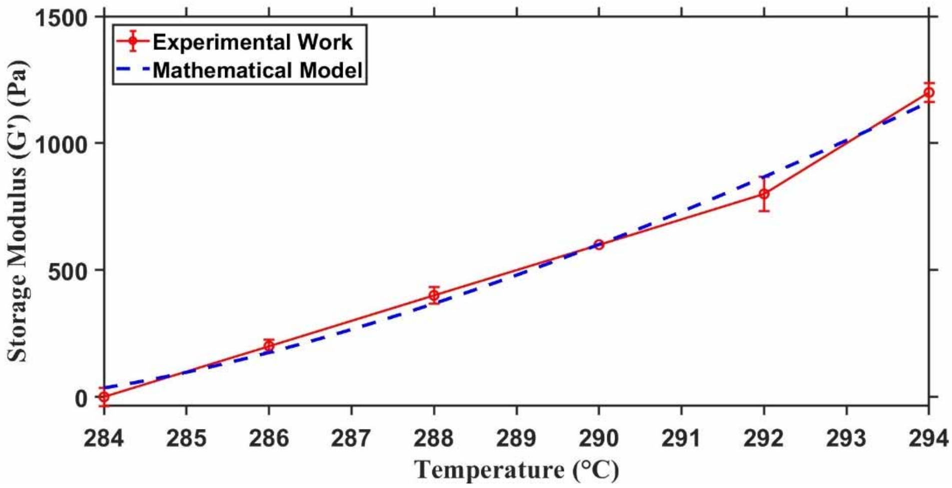

Figure 5-5 Experimental storage modulus of PA66 vs. Mathematical model

The loss modulus as shown in Figure 5-6 which indicates the viscous energy dissipation within the CFPA66 composite was analyzed. The model predictions validate with experimental results with error about 4 to 5%. This validate the mathematical model accuracy in capturing the viscous behavior of the PA66 resin during processing which was critical for understanding the PA66 resin and carbon fiber interaction under dynamics conditions.

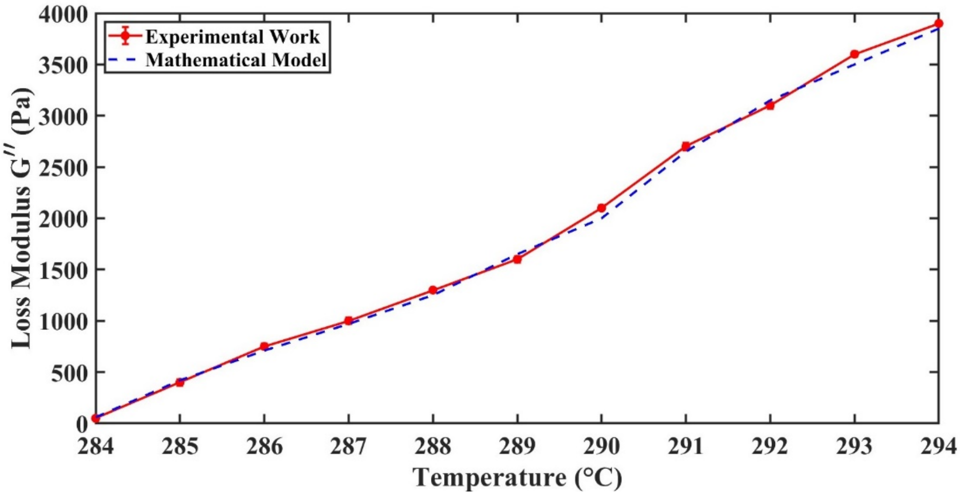

Figure 5-6 Experimental loss modulus of PA66 vs. Mathematical model

As shown in Figure 5-7 which focus on torque, which was a key indicator of the interaction between pressure gradient and PA66 resin flow through carbon fiber. Predications based on Darcy law equation 5-2 and pressure distribution relationship equation 5-9, validate with experimental results the error range between ranges from 2 to 4%. This validates the model's ability to correctly predict flow behavior and pressure dynamics during PA66 thermoplastic impregnations.

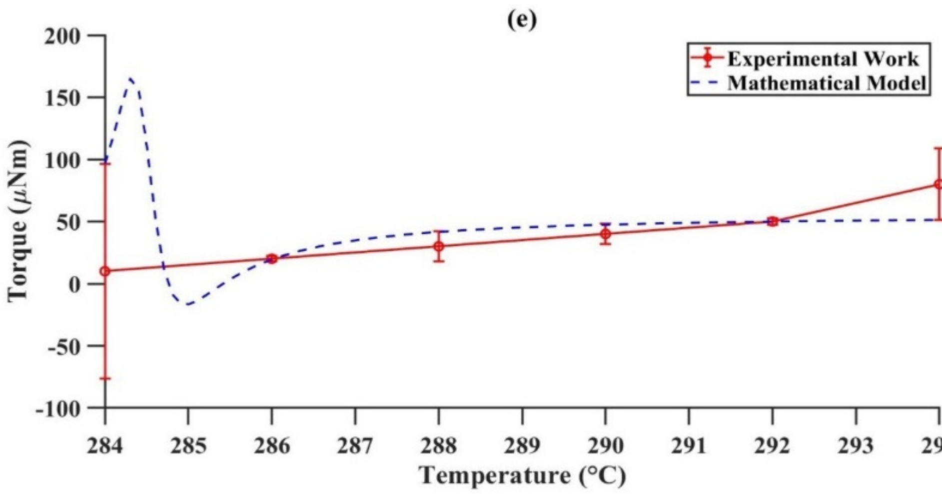

Figure 5-7 Experimental torque of PA66 vs. Mathematical model

The error percentage around 2 to 5% across experimental results with mathematical model as shown in Figures 5-3 to 5-7, confirms the strength of mathematical model in capturing the behavior of PA66 and carbon fiber thermoplastic composites. These above results demonstrate the model's capability to optimize impregnation conditions and improve the quality of CFPA66 reinforced thermoplastics.

The error percentage around 2 -5% across experimental results with mathematical model as shown in Figures 5-3 to 5-7, confirms the strength of mathematical model in capturing the behavior of polyamide 66 and carbon fiber thermoplastic composites. The reported error can be attributed to small fluctuations in temperature control, resin flow dynamics and experimental results variations, which slightly impact resin impregnation behavior. The Carreau-Yasuda viscosity model accurately captures the shear-thinning properties of PA66 melts, though variation such as slight fiber misalignment may also contribute to the deviations observed.

The observed 2-5% error range falls within the acceptable limits for thermoplastic composite manufacturing, this minor deviation reflects inherent experimental uncertainties such as variation in material properties, fiber bed permeability and measurement precision. The consistency of this range across varying conditions highlights the robustness and reliability of the integrated Darcy, Carreau-Yasuda model in predicting resin flow behavior and impregnation efficiency demonstrating its practical applicability for CFPA66 composite optimization.

5.3.2 Impact of Temperature and Fiber velocity on Pressure distribution in the Melt impregnation Wedge area

The Figure 5-2 visually illustrate the impact of mold temperature and fiber velocity in the wedge area of melt impregnation, highlighting the optimal conditions for CFPA66 composites. The Figure 5-8 show the impact of various temperature 320 °C and 330 °C, and fiber velocity 1.5 m/min and 2 m/min on pressure in melt impregnation wedge area. The key parameters used to model and simulate the impregnation process are shown in Table 5-1, it shows that the 330 °C temperature and 1.5 m/min fiber velocity are optimal for achieving better impregnation in CFPA66 composites. The Carreau-Yasuda viscosity model equation 4-6 and its temperature dependence equation 5-7, shows that higher temperature reduced the resin viscosity and improving the PA66 resin flow and carbon fiber wetting. Darcy law equation 5-9, and the pressure distribution equation show that at 330 °C, the reduced viscosity maintain a stable pressure gradient in the wedge of melt impregnation and ensuring uniform PA66 penetration, The Carreau-Yasuda model equation 5-6, accurately explain how viscosity decreases with increasing shear rate and temperature allowing PA66 resin to flow uniformly through the porous carbon fiber, at fiber velocity 1.5 m/min the PA66 resin has sufficient residence time to thoroughly impregnate the carbon fiber bundle and minimal void content, the continuity equation 5-4, confirms mass conservation ensuring uniform PA66 resin distribution across the wedge area, higher velocity 2 m/min reduce the contact time and incomplete impregnation, The Figure 5-8 MATLAB results align with mathematical and confirming that at 330 °C and 1.5 m/min is suitable able parameter for CFPA66 composite due to stable pressure and thorough impregnation for high quality composites.

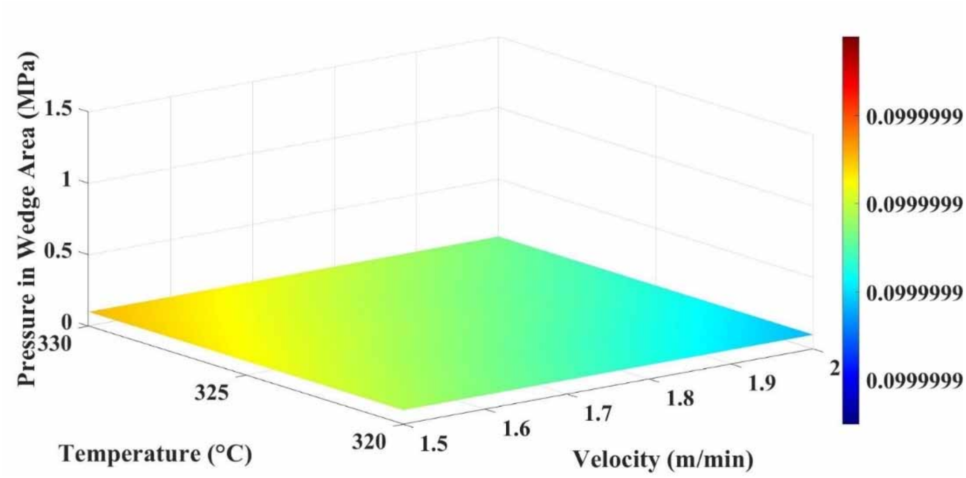

Figure 5-8 Pressure distribution in melt impregnation wedge area at various temperature (320 °C, 330 °C) and fiber velocity (1.5 m/min, 2 m/min)

Table 5-1 MATLAB parameters

Parameters	Symbol	Values	Unit
Fiber diameter	d_f	5 μm	m
Fiber monofilament	n_f	12000	-
Porosity	ε_0	0.7	-
PA66 viscosity at	η	30 ~ 152	Pa.s
Temperature	T	320 ~ 330	^{0}C
Velocity	U	1.5 ~ 2	m/min
Melt film thickness	δ	50 ~ 60 μm	m
Carman kozeny co efficient	k_0	4.06 ~ 5.55	-
Length of wedge area	L	30 ~ 100	mm

5.3.3 Morphological Structure of CFPA66 at various Temperature and Fiber velocity

As shown in Figures 5-9 to 5-10 (a to b), the morphological analysis of CFPA66 composites show different variation in void contents and PA66 penetration under various processing parameters. The molten PA66 polymer shows improved penetration into the carbon fiber, effectively impregnated PA66 into carbon fiber and reduced porosity. In comparison at higher fiber velocity at 2 m/min, the residence time of the polyamide 66 melt in carbon fiber bed reduced, prominent to insufficient PA66 impregnated and higher void content and weak adhesion. Respectively temperature plays an important role in controlling the viscosity of the PA66 melt, as shown in Figure 5-9 and 5-10 a higher temperature of 330 °C reduced the PA66 viscosity and allowing to better flow on carbon fiber and improved impregnation as compared to 320 °C. The optimal composite of CFPA66 was achieved at 330 °C and 1.5 m/min with lower voids and strong carbon fiber and PA66 resin interaction as show in Figure 5-10 (a).

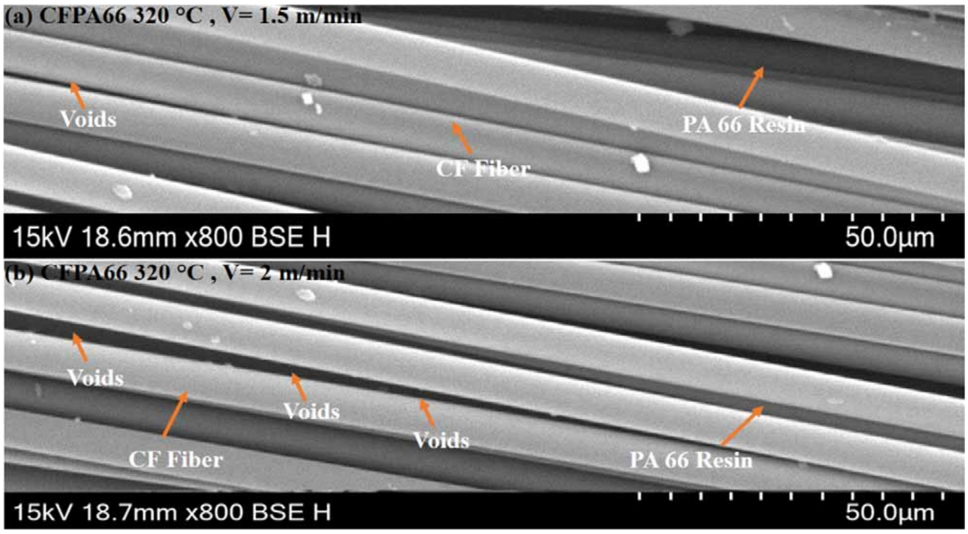

Figure 5-9 Morphological structure of CFPA66 at 320 °C temperature and fiber velocity (a) 1.5 m/min (b) 2 m/min

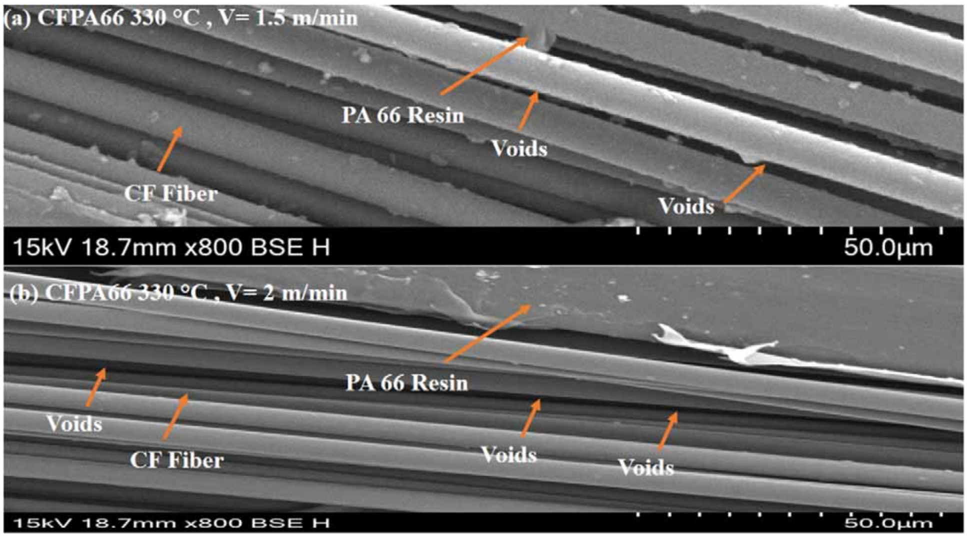

Figure 5-10 Morphological structure of CFPA66 at 330 °C temperature and fiber velocity (a) 1.5 m/min (b) 2 m/min

The SEM analysis shown a significant reduction in void content under optimal conditions, which directly compares to improved mechanical properties, reduced void content enhances carbon fiber-polyamide 66 matrix bonding, leading to increased tensile strength, stiffness and durability of the CFPA66 composites. Moreover, uniform fiber distribution ensures consistent load transfer across the CFPA66 composites and enhancing overall structural integrity. These morphological improvement play a crucial role in achieving high-performance CFPA66 composites suitable for demanding suitable for demanding industrial applications.

Above experimental results finding align closely with the predictions made using the Carreau –Yasuda viscosity model equation 5-7, which show that the shear thinning behavior of thermoplastic melts, the model related for the decrease viscosity under shear forces and combines the effect of temperature on the viscosity of PA66. As observed from experimental results, this decrease in viscosity improved the PA66 resin ability to wet and penetrate into the carbon fiber and reduced void content and improved the interfacial bonding. This alignment confirm that the Carreau-Yasuda model accuracy in capturing the viscosity changes under different processing parameters and provide a reliable framework for optimizing composite of CFPA66. The above experimental results also align with Darcy law equation 5-2, which governs fluid flow through porous media like fiber bundles, at 1.5 m/min the lower fiber velocity provided enough time for the PA66 penetrate to the carbon fiber, ensured the uniform impregnation and reduced void content. The observed reduction in voids at 1.5 m/min confirm that the Darcy law equation 5-2, accurately predict resin flow behavior in the carbon fiber and support its use for optimizing impregnation process. The continuity equation 5-4, promote validate the above experimental results by ensuring mass conservation during the impregnation process, at 330 °C and 1.5 m/min, the even distribution of PA66 resin across the carbon fiber shows that the flow rates adhered to the principles of mass conservation. This consistent PA66 resin penetrate throughout the composite leading to uniform impregnation quality. The continuity equation shows a critical theoretical foundation for understanding how PA66 resin flow rates interact with carbon fiber during impregnation. Moreover, the pressure distribution equation 5-9, derive from Darcy law highlights the gradual pressure drop across the length of carbon fiber , SEM analysis shown that the impregnation quality remained uniform along the carbon fiber length at the optimal conditions indicate the continued PA66 resin flow throughout the process. This alignment with pressure distribution equation 5-9, confirms that a stable pressure gradient is essential for effective impregnation further validating the experimental results.

5.3.4 Differential Scanning Calorimetric (DSC) of CFPA66 at various Temperature

Figures 5-6 (a and b) and Table 5-2 show the results of the differential scanning calorimetric (DSC) analysis conducted to study the crystallization and melting behavior of Polyamide 66 (PA66) and carbon fiber-reinforced polyamide 66 (CFPA66) at various temperature. The analysis was performed using a DSC Q2000 under a nitrogen atmosphere. Samples weighing approximately 7.5 mg were heated from room temperature to 300 °C at a rate of 10 °C/min, held at 300 °C to eliminate any thermal processing, cooled to 50 °C at 10 °C/min, and then reheated to 300 °C at the same rate.

The degree of crystanllinity (Xc) for each CFPA66 composites sample was calculated from the second heating cycle using the equation [246];

$$X_c = \frac{\Delta H_m}{\emptyset \Delta H_m^0} \times 100\% \qquad (5-14)$$

Whereas ΔH_m is the melting enthalpy measured by DSC, ΔH_m^0 is the melting enthalpy of fully crystalline PA66 (255 J/g) [247], and $\emptyset$ represents the weight fraction of PA66 in the composite, which is 0.7 for the CFPA66 samples.

The DSC results shows critical insights into the thermal properties of PA66 and CFPA66 composite. CFPA66 processed at 330 ºC shows the highest crystallization temperature (Tc), and melting temperature as shown in Figures 5-11 (a, and b) and Table 5-2, compared to 320 °C and PA66 it shows that the enhanced nucleation due to carbon fibers. The reduction in crystallization correlates with improved PA66 resin flow and carbon fiber wetting due to reduced PA66 polymer chain rigidity. The DSC results confirms that the CFPA66 proceed at 330 °C offer optimal conditions for thermoplastic melt impregnation, ensuring better resin flow and improved carbon fiber and PA66 adhesion.

The Carreau–Yasuda viscosity model equation 5-6, is validate by the experimental results as it accurately captures the shear thinning behavior of thermoplastic PA66 melts under processing condition. The reduced viscosity at 330 °C improved the PA66 resin flow into the carbon fiber ensured the better impregnation. This reduction in viscosity confirms the better impregnation by reducing resistance to flow. The model equation 5-6, prediction align with improved thermal transitions observed in the DSC analysis validating its applicability in describing the link between temperature, viscosity and shear rate. Moreover, the temperature dependence of viscosity equation 5-7, validate these finding as it relation to the viscosity reduced to the temperature dependent behavior of PA66 resin. The DSC analysis shows the enhanced PA66 flow and penetration at 330 °C temperature confirms the significance of equation 5-7, in understanding and optimizing the melt impregnation process. The Darcy law for fluid flow equation 5-2, align with experimental results as it governs the flow of PA66 resin through the carbon fiber. The DSC analysis which shows at 330 °C better PA66 resin penetration and uniform flow into the carbon fiber, this uniform flow improved the impregnation quality. The equation 5-13, shear stress show that at 330 °C reduced viscosity low shear stress required for PA66 resin flow, the lower shear stress assists smoother penetration of PA66 resin into carbon fiber and confirms better impregnation quality.

Above equations 5-6, 5-7, 5-2 and 5-13, mutually explain the optimal impregnation quality observed at 330 °C. The lower crystanllinity support better flow of PA66 resin and fiber interaction, although reduced viscosity confirms the efficient PA66 resin distribution and penetration.

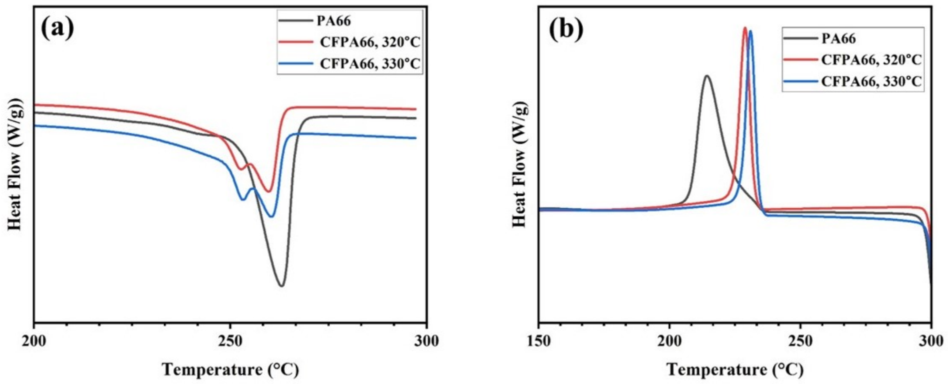

Figure 5-11 DSC curve of CFPA66 at various temperature (a) melting 320 °C to 330 °C (b) Crystallization 320 °C to 330 °C

Table 5-2 Crystallization and Melting of CFPA66 at different temperature

Sample	T_c (°C)	ΔH_c (J/g)	T_m (°C)	ΔH_m (J/g)	X_c (%)
PA66	214.64	54.88	262.82	41.59	31.74
CFPA66 320 °C	228.89	31.57	259.80	14.26	17.68
CFPA66 330 °C	231	28.26	260.57	24.91	15.83

5.3.5 Thermal Degradation behavior of PA66 and CFPA66 at different Temperatures

The TGA results shows in Figure 5-12 and Table 5-3, critical insight into the thermal stability and impregnation quality of thermoplastic fiber composites. Pure PA66, a non-reinforced thermoplastic resin, exhibits the lower thermal stability among the CFPA66 composites samples. The degradation starts at a lower temperature, and the residual weight percentage is minimal, indicating poor thermal resistance due to the absence of reinforcing carbon fibers. This lacks of reinforcement results in weak thermal properties, making PA66 the least effective for high performance composite applications. CFPA66 proceed at 330 °C, shows the improved thermal stability as compared to pure PA66 due to the inclusion of carbon fibers, which enhance the PA66 resin resistance to thermal degradation. Though, the residual weight and thermal stability were still lower than those of CFPA66 at 330 °C indicate partial impregnation. CFPA66 processed at 330 °C shows superior thermal stability with a delayed beginning of degradation and highest residual weight. This performance was attributed to improve the impregnation quality and PA66 and CFPA66 bonding, as higher temperature lower the melt viscosity of PA66, promoting better resin flow into the carbon fiber and reducing porosity.

The experimental results align with the predictions developed by mathematical model, the Carreau –yasuda viscosity equation 5-6, accurately shows the shear thinning behavior of PA66 melt, where higher temperature 330 °C reduced the viscosity permitting better flow into the carbon fiber. This viscosity reduction directly enhanced the wetting and bonding between the PA66 resin and carbon fibers, comparing 330 ºC CFPA66 the superior performance. Darcy law equation 5-2, supports the observed results, explaining the flow of the thermoplastic PA66 resin through the porous medium formed by the carbon fiber. At 330 °C improved the impregnation quality and validate the model predictions about the PA66 resin behavior during processing. Darcy law equation 5-2, promote supports the observed results and shows that the flow of PA66 resin through porous medium formed by the carbon fiber bundle and improved the impregnation quality at 330 °C Validate the law predictions about PA66 resin behavior during processing. The continuity equation 5-4, ensures mass conservation within impregnation process and its validation was evident in the uniform PA66 distribution observed at 330 °C. Correspondingly the pressure distribution equation 5-9, align with experimental results, as stable pressure gradient maintained during processing ensures uniform impregnation along the length of carbon fiber, shear stress equation 5-13, confirms that reduced viscosity at 330 ºC minimize the shear stress required for PA66 resin flow allowing efficient impregnation and reduced porosity

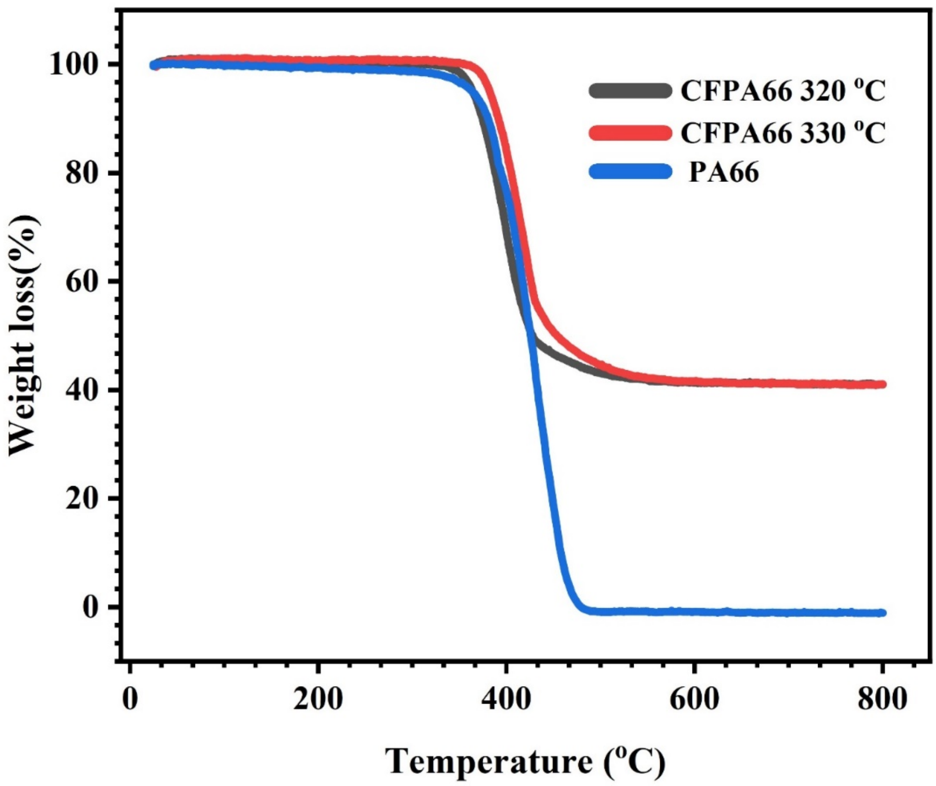

Figure 5-12 TGA curve of CFPA66 at various temperature 320 °C to 330 °C

Table 5-3 Weight retention and loss % of PA66 and CFPA66 composite from TGA

SN	Sample	Weight Retention (%)	Weight Loss (%)
1	PA66	0	100
2	CFPA66 320 °C	41	59
3	CFPA66 330 °C	39	61

5.4 Summary

This study developed and validate an integrated multi-physics mathematical model combining Darcy law and Carreau-Yasuda viscosity model to optimize the melt impregnation process of CFPA66 composites.

- The optimal impregnation quality of CFPA66 was achieved at mold temperature 330 °C and fiber velocity of 1.5 m/min, these conditions resulted in reduced viscosity, minimal void content and improved flow dynamics ensuring strong carbon fiber and polyamide 66 (CFPA66) matrix bonding.

- Experimental results showed strong alignment with mathematical model with error 2 to 5%. This validates the model ability to accurate capture the non-Newtonian behavior of thermoplastic melts and predict flow behavior and impregnation quality under varying conditions.
- Characterization techniques (SEM, DSC and TGA) confirmed that higher temperatures improved resin flow and interfacial adhesion while maintaining thermal stability and crystanllinity confirming the worth of the model.
- The integration of advanced viscosity modeling and fluid dynamics provides a robust framework for understanding and optimizing thermoplastic melt impregnation processes.

Chapter 6 Conclusion

Thermoplastic composites with their excellent mechanical properties, recyclability and potential for lightweight, high-performance applications are increasingly significant in various industries.

This research systematically investigates the effects of critical processing parameters such as mold temperature and fiber velocity on the impregnation behavior and structural performance of basalt fiber-polypropylene (BFPP), glass fiber-polypropylene (GFPP) and carbon fiber-polyamide 66 (CFPA66) composites. By employing advanced mathematical modeling and experimental validation, the study provides a comprehensive framework for optimizing the thermoplastic melt impregnation process.

In the first section, BFPP composites were fabricated at mold temperatures of 240 °C, 250 °C, and 260 °C and fiber velocities of 0.5, 0.8, 1.0, and 1.5 m/min. The optimal impregnation and lowest fiber fracture were achieved at 260 °C and 0.5 m/min, where porosity and fracture rate were minimized to 2.94% and 1.69% respectively. SEM images confirmed improved fiber-matrix adhesion, while TGA showed enhanced thermal stability with 41% weight retention. The integrated model, incorporating Darcy's law, Reynolds equation, Weibull distribution, cohesive zone model, and Griffith's criterion, effectively predicted pressure distribution and fracture behavior, matching experimental trends and validating its use in process optimization.

In the second section, GFPP composites were examined under mold temperatures from 220 °C to 250 °C at a constant fiber velocity of 0.7 m/min. The best results were obtained at 250 °C, where TGA revealed the highest thermal resistance with 48% weight retention, and DSC showed the highest crystallinity at 40.57%. The temperature-dependent model used the Arrhenius viscosity equation and Oldroyd-B stress predictions, with MATLAB simulations confirming pressure and viscosity relationships. The model's predictions closely aligned with experimental results with an error margin of 2–5%, confirming its accuracy in predicting the influence of mold temperature on fiber tension, viscosity, and resin penetration.

The third section, analyzed the effect of fiber velocity on impregnation quality in GFPP using a nonlinear Darcy flow model. Experimental runs at 250 °C with velocities of 0.5 to 1.5 m/min demonstrated that 0.5 m/min offered the best balance between pressure and impregnation efficiency. Pressure readings ranged from approximately 1.2 MPa at 0.8 m/min to over 2.0 MPa at 1.5 m/min, with corresponding declines in impregnation quality and increases in void content. SEM analysis confirmed that low velocities enabled better wetting and bonding. The model accurately captured the nonlinear relationship between velocity, pressure, and permeability, with simulation results aligning closely with physical observations.

Finally, the study of CFPA66 composites involved developing a multi-physics model combining Darcy's law with the Carreau–Yasuda viscosity model to simulate the non-Newtonian behavior of PA66. Optimal impregnation was achieved at 330 °C and 1.5 m/min, where reduced viscosity enabled better resin flow and lower void content. TGA confirmed thermal stability with 39% weight retention, and DSC analysis showed a crystallinity of 15.83%. The model's predictions on viscosity, shear stress, storage modulus, and torque closely matched experimental data within an error range of 2–5%. SEM images showed uniform resin distribution and strong interfacial bonding under these optimal conditions.

This research highlights the importance of optimizing processing parameters for thermoplastic composites to achieve superior resin impregnation and fiber-matrix bonding. The findings for BFPP, GFPP and CFPA66 composites offer practical guidelines for their industrial application in sectors such as aerospace and automotive. Furthermore, the development of advanced mathematical models provides a robust predictive framework for addressing the complexities of resin flow, fiber-matrix interactions and process optimization.

6.1 Innovation Points

Thermoplastic composites with their lightweight, durable and recyclable properties are essential materials in industries such as aerospace and automotive. Despite their advantages optimizing resin impregnation, fiber bonding and processing parameters such as fiber velocity and mold temperature has remained a persistent challenge. This research addresses these gaps by developing mathematical model that enhanced the understanding and control of thermoplastic melt impregnation processes, establishing a foundation for the development of advanced high-performance composites.

1 Integrated Multi-Physics Models

Combined Darcy's law, Reynolds equation, Oldroyd-B, Carreau-Yasuda, Weibull distribution, and Griffith's criterion to model resin flow, pressure, fiber tension, viscosity behavior, and fracture. This level of multi-domain integration is not present in existing models and is clearly novel.

2 New Understanding of Processing Parameters

Revealed how mold temperature, fiber velocity, and film thickness interact to affect resin viscosity, pressure distribution, and fiber-matrix bonding. These insights were derived theoretically and experimentally, filling a critical gap in process optimization.

3 Simultaneous Prediction of Impregnation & Fracture

Developed the models to predict both resin impregnation quality and fiber fracture in BFPP composites, using a cohesive and validated framework.

This dual prediction approach is unique and impactful in composite manufacturing.

This research represents a significant advancement in the field of thermoplastic melt impregnation by developing mathematical model that optimize resin flow, fiber bonding and processing parameters such as mold temperature and fiber velocities. These innovations ensure superior impregnation quality with minimal void content and provide a robust framework for manufacturing high-performance composites, applicable to aerospace and automotive industries. Furthermore this research establishes a foundation for the development of lightweight and high–strength materials, addressing critical performance requirements for future composite application.

6.2 Future Recommendation

Future work should focus on enhancing the current mathematical models by incorporating fiber deformation, thermal degradation, and dynamic interfacial behavior to better represent real processing conditions. Additionally, the integration of in-situ pressure and temperature monitoring can improve model validation and support the development of adaptive control strategies.

Further studies should also explore non-conventional thermoplastic matrices (e.g., bio-based or high-temperature resins) and hybrid fiber systems to broaden the applicability of the model. Lastly, expanding the framework to 3D and transient simulations will improve prediction accuracy for complex geometries and continuous processing environments.

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